

BRIGHT WEB CRAFTERS

SCHOOL- PARENTAL SYSTEM

ITC 327W

EST.2024

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1. PROBLEM STATEMENT

Most schools currently lacks a centralized, accessible, and easy-to-use system for tracking and managing Learner records, attendance, grades, and parent communication. Learner-related data is scattered across paper registers, spreadsheets, email records, and manual filing systems, making it difficult to maintain accuracy, track Learner progress, or plan academic activities effectively.
Manual tracking of attendance, grades, and parent communication is inefficient, error-prone, and lacks real- time insights, leading to poor decision-making, slower communication, difficult to fix your child ways before is too late and incomplete or outdated Learner profiles.

The problem affects school administrators who spend excessive time on manual administrative tasks, teachers who are burdened with record-keeping instead of focusing on teaching, parents who lack real-time visibility into their children's academic progress and child who drop out, suspended from school and parents don't have knowledge of that until is too late, and Learners who experience delays in receiving feedback from their parents / teachers.

Addressing this problem will improve school operations, ensure data accuracy, and enable administrators to respond to academic needs proactively. School administrators will have access to real-time reports and dashboards for better decision-making and strategic planning. Teachers will be able to add, manage, and update attendance and grades digitally. Parents will have access to their children's information through a mobile/web application.

Our School-Parental Portal system will help fulfil long-term organizational goals while providing accurately updated Learner data. The system will support multiple schools, allowing a Super Admin to manage all schools from a single dashboard, while School Admins manage only their respective schools.

This problem affects all departments of the school, including administrators, teachers, parents, and Learners. The overall scope of this system emphasizes Learner data collection, attendance tracking, grade management and parent communication.

The desired outcome is a secure, fully centralized, up-to-date, and accessible repository of school information that supports CRUD functionalities where administrators can easily manage Learners, teachers, classes, attendance, and grades in real time. The School Portal System will benefit the school through improved operational efficiency, better parent communication, and optimized academic management

2. INTRODUCTION

2.1 Purpose

This Software Requirements Specification (SRS) document outlines the requirements for the School Portal System, a web-based and mobile-enabled platform developed for educational institutions. The system aims to centralize school management operations, streamline administrative tasks, improve parent-school communication, and provide real-time access to Learner information for all stakeholders.

The document serves as a formal agreement between the development team and the stakeholder and will guide the development process throughout the semester.

2.2 Scope

The School Portal System enables:

- **Super Admin** to manage multiple schools from a single dashboard
- **School Admin** to manage Learners, teachers, classes, attendance, and grades for their school
- **Teachers** to mark attendance, record grades, and view their assigned classes
- **Parents** to view their children's attendance, grades, and school announcements

The system emphasizes data security, role-based access control, and compliance with data protection regulations. It will be accessible via web browsers on desktops, tablets, and smartphones, and through a mobile application developed with Flutter, ensuring responsive and intuitive user experience.

2.2.1 Deliverables

The project deliverables include:

- Software Requirements Specification
- System design documentation
- UML diagrams
- Entity Relationship Diagram
- Flutter mobile application prototype
- ASP.NET Core Web API
- Supabase PostgreSQL Database
- Authentication and Role-Based Access Control Implementation
- Testing Documentation
- Final Project Presentation

2.2.2 Acceptance Criteria

The School Portal System prototype will be considered acceptable when:

- Users can successfully authenticate and access the system.
- Role-Based Access Control functions correctly.
- School Administrators can manage Learners, teachers and classes.
- Teachers can record attendance and grades.
- Parents can view authorized Learner information.
- Communication features function correctly.
- Flutter and ASP.NET Core successfully integrate with Supabase.
- System security requirements are satisfied

2.2.3 Exclusions

The initial prototype will not include:

- School fee payment processing
- Payroll management
- Library management
- Transport tracking
- Online classroom functionality
- Integration with external government education systems
- Full production deployment

2.2.4 Constraints

- The project must be completed within the academic semester.
- Development must use Flutter, ASP.NET Core and Supabase.
- The project is developed by a Learner team with limited resources.
- Internet connectivity is required for system operation.
- The system is developed as an academic prototype rather than a production system.

2.2.5 Assumptions

- Stakeholders will provide the information required for analysis and design.
- Users will have access to internet-enabled devices.
- Schools maintain accurate Learner and staff records.
- Parents are linked only to their own children.
- Users possess basic computer or smartphone literacy

2.3 Project Aim

The aim of the School Portal System is to design and develop a centralized web and mobile platform that improves the management of Learner information, attendance, grades and communication between school administrators, teachers and parents. The system will provide secure role-based access to information while improving operational efficiency, data accuracy and stakeholder communication.

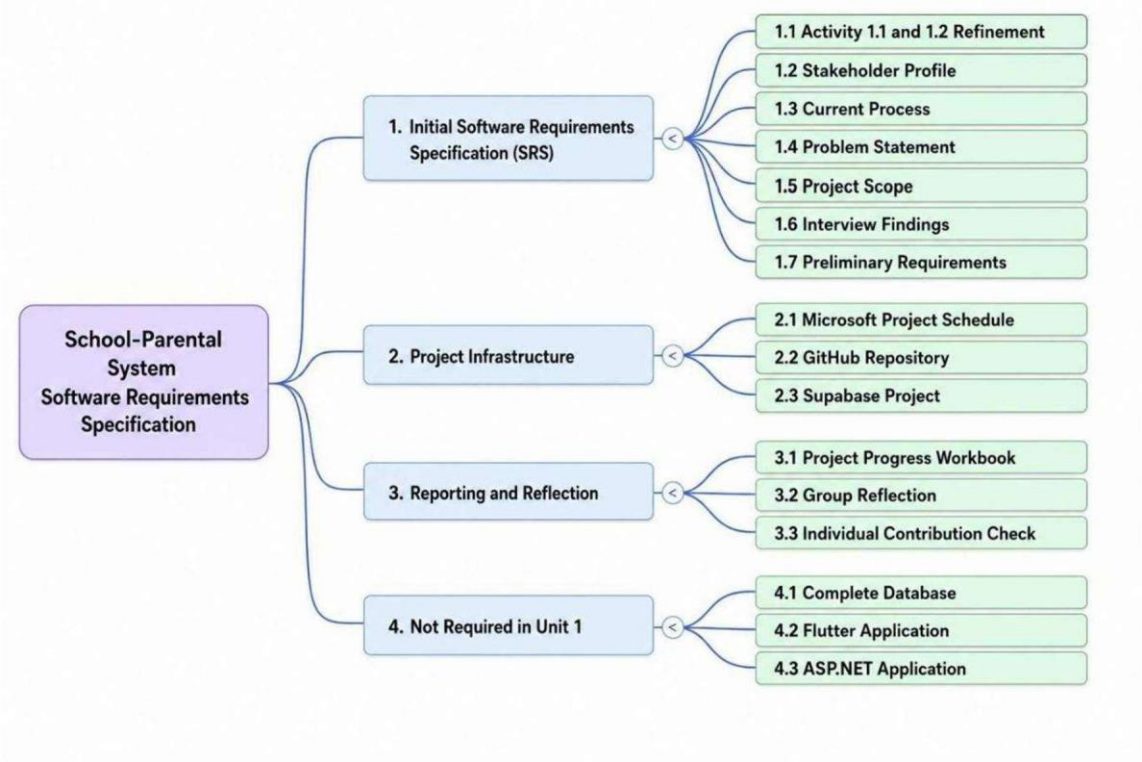
2.4 Project Objectives

- Centralize Learner, teacher, attendance and grade information within a single system.
- Enable School Administrators to manage Learners, teachers and classes digitally.
- Allow teachers to record attendance and Learner grades electronically.
- Provide parents with access to their children's attendance, grades and school communication.
- Improve communication between schools and parents through announcements, notifications and events.
- Implement secure role-based access control for Super Admins, School Admins, Teachers and Parents.
- Support multi-school management through a centralized platform.
- Integrate Flutter, ASP.NET Core and Supabase into a single solution.
- Improve the accuracy, accessibility and availability of school information.

2.5.1 Stakeholders Profile and Current Process

1 **Participating School -** Primary stakeholder and source of system requirements **Super**
2 **Admin -** Manages multiple schools and system-wide information
3 **School Admin -** Manages Learners, teachers, classes and school information
4 **Teacher -** Records attendance, grades and manages assigned classes
5 **Parent -** Views children's information and school communication
6 **Learners -** Beneficiaries of improved school administration and communication
7 **Development Team -** Responsible for analysis, design, development and testing

Current Process



Schools currently manage Learner information through a combination of paper registers, spreadsheets, email communication and manual filling systems. Attendance records, Learner grades and parent communication are often managed using separate processes and tools, resulting in duplication of work, delays in accessing information and increased administrative effort. School administrators spend significant time collecting and maintaining Learner records, while parents have limited access to real-time information regarding their children's academic progress and school activities. The lack of a centralized system makes it difficult to maintain accurate record, monitor Learner performance and facilitate effective communication between schools and parents.

2.5.2 Stakeholder Needs

Stakeholder engagement activities, including interviews, discussions and requirements gathering with administrators and educators from **Reitzpark Primary School, Lenakeng Technical School and Thabong Primary School**, identified the following needs for the School Portal System.

1 School Administrators

School Administrators require a centralized platform that can assist with managing Learner records, teacher information, classes, attendance records and academic results more efficiently.

Stakeholder feedback showed that some schools still depend on paper registers and separate systems when managing learner information. Problems identified include incomplete mark lists, delays in receiving attendance registers, duplication of manually captured information and difficulty locating information that has been captured manually.

School Administrators also require accurate and up-to-date parent or guardian contact information to improve communication and follow-up with parents.

2 Teachers

Teachers require a system that allows them to view assigned classes, record attendance, capture Learner grades and access learner information without depending heavily on paper-based records.

Stakeholder feedback identified manual data capturing, paper registers, duplicated information and difficulty locating manually captured information as challenges experienced by educators. Teachers also require more reliable communication with parents. Current communication methods include printed letters, telephone calls, WhatsApp groups and other messaging methods. However, educators reported challenges such as parents not responding, incorrect or changing contact details, parents not attending meetings and learners not always delivering letters to their parents.

3 Parents

Stakeholder feedback indicates that parents require convenient and timely access to information concerning their children, including attendance records, academic performance, school announcements, notifications, meetings and events.
Parents should not have to depend only on learners bringing printed letters or reports home, because stakeholders confirmed that some learners do not always give these documents to their parents. Parents also require communication that allows them to receive important school information directly and, where necessary, confirm that the information has been received. This supports the proposed **Read/Unread Tracking and Parent Acknowledgement** functions.

The system should also support the updating of parent or guardian contact information because stakeholders identified outdated, incorrect or changed telephone numbers as a common communication problem.

4 School Management

School Management requires improved visibility of school operations, accurate Learner information and improved communication between administrators, teachers and parents.

Stakeholder interviews showed that participating schools currently use a combination of paper registers, printed letters, telephone calls, WhatsApp, Facebook and systems such as SA-SAMS. These separate methods can make information management and communication difficult to monitor consistently. School Management therefore requires a more centralized approach that can reduce dependence on manual processes and provide better access to current school information.

5 **Development Team**

The Development Team requires a clearly defined and prioritized set of functional and non-functional requirements to guide system design, implementation, integration and testing activities.

The stakeholder findings provide evidence for the main proposed functions, including Learner Management, Teacher Management, Attendance Management, Grade Management and Communication Management. The Development Team must also consider limitations identified during stakeholder engagement, including unreliable internet access, limited mobile data, outdated contact information and differences in parents' access to smartphones or digital communication platforms. Reitzpark Primary School, for example, identified parents who may not use WhatsApp or Facebook and may have limited access to mobile data or smartphones.

6 Stakeholder Confirmation

The identified requirements are supported by stakeholder engagement conducted with administrators and educators from participating schools in Welkom, Free State.

The interviews confirmed recurring challenges relating to:

- manual and paper-based record keeping;
- learners not always delivering important documents to parents;

- difficulty confirming whether parents received or read communication;
- parents not responding to messages or attending meetings;
- outdated or incorrect parent contact information;
- duplication and difficulty locating manually captured information; and
- dependence on several separate communication and information-management methods.

2.5.3. Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
RLS	Row Level Security - Database security feature that restricts data access
CRUD	Create, Read, Update, Delete
API	Application Programming Interface
UI	User Interface
RESTful API	Representational State Transfer Application Programming Interface
ASP.NET Core	Microsoft's open-source framework for building web apps and APIs
Flutter	Google's open-source toolkit for building mobile and web apps
Supabase	Open-source Firebase alternative with PostgreSQL database
HTTPS	Hypertext Transfer Protocol Secure
JWT	JSON Web Token - used for secure authentication
RBAC	Role-Based Access Control
RLS	Row Level Security
p	Parent
SA	Super Admin & School Admin

3. Overall Description

3.1 Product Perspective

The School Portal System is an integrated platform designed to allow educational institutions to manage Learner records, attendance, grades, and parent communication efficiently. The system supports multiple schools from a single platform with role-based access control.

3.2 Product Functions

- User Registration and Authentication (Email/Password)
- Role-Based Access Control (Super Admin, School Admin, Teacher, Parent)
- Multi-School Management (Super Admin only)
- Learner Profile Management
- Teacher Management
- Class Management
- Attendance Tracking
- Grade Management
- Communication Management
- Announcements
- Notifications
- Events
- Read/Unread Tracking
- Parent Acknowledgement
- School Selector (Super Admin only)
- Report Generation

3.3 System Environment

The system will integrate with:

- **ASP.NET Core 8** for the Web API backend
- **Flutter** for the mobile application (Android C iOS)
- **Supabase** for PostgreSQL database with Row Level Security
- **Supabase Auth** for authentication
- **Supabase Storage** for profile pictures and documents

4. Business Rules

4.1 User Authentication

- Only registered and verified users can access the system's functionalities
- Authentication is handled through Supabase Auth with email/password
- JWT tokens are used for session management

4.2 Role-Based Access Control

- **Super Admin**: Can view, add, edit, and delete any school's data
- **School Admin**: Can view, add, edit, and delete data for their school only
- **Teacher**: Can view and manage their assigned classes only
- **Parent**: Can view only their children's information

4.3 School Context Rules

- All data must be filtered based on the selected school context
- Super Admin must select a school to view data
- School Admin, Teacher, and Parent are automatically assigned to their school
- School selector is only visible to Super Admin

4.4 Data Validation Rules

- All Learner records must include: full name, grade, and school
- All teachers must be assigned to a school
- Attendance status must be one of: Present, Absent
- Grades must be between 0 and 100
- Announcements must have a title and content

4.5 Attendance Rules

- Attendance can only be marked by teachers for their assigned classes
- Attendance records cannot be changed after 7 days
- Parents can only view attendance for their children

4.6 Grade rules

- Grades can only be recorded by teachers for their assigned classes
- Parents can only view grades for their children
- Grade history must be maintained

4.7 Communication Rules

- Communication must be associated with a school.
- Users shall only receive communication intended for their school and role.
- Announcements may target the whole school, specific roles, grades, classes, or specific users.
- Announcements may have a priority of **Normal, Important, or Urgent**.
- Important or Urgent announcements may require Parent Acknowledgement.
- School Admins may create communication for their own school.
- Super Admins may create communication for permitted schools.
- Teachers may create class-related communication for their assigned classes where permitted.
- Parents may only view communication intended for them or related to their children.
- Read status and acknowledgement status shall be stored separately.
- A parent shall not acknowledge the same announcement more than once.

5.1 Specific Requirements

5.1.1 Platform Development

- The mobile application must be built using Flutter, ensuring compatibility with Android and iOS
- The web application must be developed using ASP.NET Core 8 Web API
- The database must be hosted on Supabase using PostgreSQL

5.1.2 Authentication Security

- Supabase Authentication with support for email/password
- All data must be encrypted in transit using HTTPS
- Role-Based Access Control must restrict user functions according to user role
- Row Level Security (RLS) must be enforced at the database level

5.1.3 User Profile Management

- Users must be able to log in
- Users must have a role assigned upon registration
- Administrators can manage user accounts
- Users can update their profile information

5.1.4 School Management (Super Admin Only)

- Super Admin can add new schools
- Super Admin can edit school details

- Super Admin can delete schools
- Super Admin can view all schools

5.1.5 Learner Management

- School Admin can add Learners to their school
- School Admin can edit Learner details
- School Admin can view all Learners in their school
- Teachers can view Learners in their classes
- Parents can view their children only

5.1.6 Teacher Management

- School Admin can add and remove teachers
- School Admin can edit teacher details
- School Admin can assign teachers to classes
- Super Admin can view teachers across all schools

5.1.7 Class Management

- School Admin can create classes
- School Admin can edit class details
- School Admin can assign Learners to classes
- Teachers can view their assigned classes

5.1.8 Attendance Management

- Teachers can mark attendance for their classes
- Teachers can view attendance records
- Parents can view their children's attendance
- School Admin can view attendance reports

5.1.9 Grade Management

- Teachers can record grades for Learners
- Teachers can edit grades
- Parents can view their children's grades
- School Admin can view and upload grade reports

5.1.10 Announcement Management

The Communication Management Module shall provide a central platform for school communication. The module shall support:

- Announcements
- Notifications
- Events
- Read / Unread Tracking
- Parent Acknowledgement

5.1.10.1 Announcement Management

The system shall allow authorized users to create and manage announcements. An announcement shall contain:

- Title
- Message/content
- School
- Created by
- Date created

- Target audience
- Priority

Announcement priority may be:

- Normal
- Important
- Urgent

The system shall:

- Allow Super Admins to create announcements for permitted schools.
- Allow School Admins to create announcements for their school.
- Allow Teachers to create announcements for their assigned classes where permitted.
- Allow authorized users to edit and delete announcements they are permitted to manage.
- Allow announcements to target the whole school, specific roles, grades, classes, or specific users.
- Display announcements to the intended users.
- Display the newest announcements first.
- Clearly distinguish Important and Urgent announcements from Normal announcements.
- Allow Important or Urgent announcements to require Parent Acknowledgement.

5.1.11 Notification Management

Notifications shall alert users when relevant information becomes available.

Notifications may be generated for:

- New announcements
- Upcoming events
- Grade updates
- Attendance updates
- Other relevant system activities

Each notification shall contain:

- Title
- Message
- Notification type
- Recipient
- Date and time
- Read status

The system shall:

- Display notifications to the intended user.
- Distinguish between read and unread notifications.
- Allow users to mark notifications as read.
- Store the date and time a notification was read.

5.1.12 Important Notice Management

The system shall allow authorized users to create and manage school events.

Events may include:

- Parent meetings
- Sports days
- Examinations
- School holidays
- Award ceremonies
- School trips
- Extra classes

Each event shall contain:

- Event name
- Description
- Event date
- Start time
- End time
- Venue
- School
- Created by
- Target audience

The system shall:

- Allow authorized users to create events
- Allow authorized users to edit events
- Allow authorized users to delete events
- Display upcoming events to relevant users
- Allow events to target the whole school, specific roles, grades or classes.

5.1.13 Read / Unread Tracking

The system shall track whether communication has been read by the intended recipient.

- New communication shall initially have an Unread status.
- When a recipient opens a communication item, the status shall change to Read.
- The system shall store the date and time the communication was read.
- Users shall be able to identify unread communication.
- Authorized staff shall be able to view communication read statistics where required.

5.1.14 Parent Acknowledgement

Parent Acknowledgement shall allow parents to actively confirm that they have received an announcement requiring acknowledgement.

The system shall:

- Allow Important or Urgent announcements to require Parent Acknowledgement.
- Allow parents to acknowledge announcements requiring acknowledgement.

- Store which parent acknowledged the announcement.
- Store the acknowledgement date and time.
- Prevent the same parent from acknowledging the same announcement more than once.
- Allow authorized school users to view acknowledgement status.

The communication status can therefore be:
Unread → Read → Acknowledged

5.1.15 Communication Tracking and reporting

Authorized school users shall be able to view communication statistics, including:

- Number of intended recipients
- Number of recipients who have read an announcement
- Number of recipient who have not read announcement
- Number of parents who acknowledged an announcement where acknowledgement is required
- Number of parents who have not acknowledged the announcement

5.1.16 Administrative Functions

- Admin should be able to view all employee profiles
- Reports must be viewable in the system
- Dashboard must show key metrics

5.2 Functional requirements

5.2.1 Super Admin Interface

#	Requirement	Priority
SA1	Super Admin can register and login using their credentials	High
SA2	Super Admin can view all schools in the system	High
SA3	Super Admin can add new schools	High
SA4	Super Admin can edit school details	High
SA5	Super Admin can delete schools	High
SA6	Super Admin can view all users across all schools	High
SA7	Super Admin can view reports and dashboards displaying data from all schools	High
SA8	Super Admin can select a school to view specific data	High
SA9	Super Admin can create announcements for permitted schools	Medium
SA10	Super Admin can create and manage events	Medium
SA11	Super Admin can view communication statistics	Medium

5.2.2 School Admin Interface

#	Requirement	Priority
SA12	School Admin can register and login using their credentials	High
SA13	School Admin can view only their school's data	High
SA14	School Admin can add Learners to their school	High

SA15	School Admin can edit Learner details	High

SA16	School Admin can add teachers to their school	High
SA17	School Admin can edit teacher details	High
SA18	School Admin can create classes	High
SA19	School Admin can assign Learners to classes	High
SA20	School Admin can view attendance reports	Medium
SA21	School Admin can view grade reports	Medium
SA22	School Admin can create and manage announcements for their school	High
SA23	School Admin can create and manage school events	Medium
SA24	School Admin can require Parent Acknowledgement for Important or Urgent announcements	Medium
SA25	School Admin can view read/unread and acknowledgement statistics	Medium
SA26	School Admin can add, edit and remove subjects to their schools	high

5.2.3 Teacher Interface

#	Requirement	Priority
T1	Teacher can login using their credentials	High
T2	Teacher can view only their school's data	High
T3	Teacher can view their assigned classes	High
T4	Teacher can view Learners in their classes	High
T5	Teacher can mark attendance for their classes	High
T6	Teacher can view attendance records	High
T7	Teacher can record grades for Learners\	High
T8	Teacher can edit grades	High
T9	Teacher can view grade reports for their classes	Medium
T10	Teacher can create announcements for assigned classes where permitted	Medium
T11	Teacher can view communication related to assigned classes	Medium
T12	Teacher can create class-related events where permitted	Medium

5.2.4. Parent Interface

#	Requirement	Priority
P1	Parent can login using their credentials	High
P2	Parent can view only their school's data	High
P3	Parent can view only their children's information	High
P4	Parent can view their children's attendance	High
P5	Parent can view their children's grades	High
P6	Parent can view school announcements	High
P7	Parent can view their children's class information	Medium
P8	Parent can receive and view notifications	High
P9	Parent can view upcoming events	Medium
P10	Parent can identify read and unread communication	Medium
P11	Parent can acknowledge announcements where acknowledgement is required	High

6 Non-Functional Requirements

6.1 Performance Requirements

#	Requirement	Priority
NFR1	Page load times shall be under 3 seconds	High
NFR2	The system shall support at least 100 concurrent users	High
NFR3	Data synchronization shall occur within 5 seconds	High
NFR4	API response time shall be under 500ms	Medium

6.2 Security Requirements

#	Requirement	Priority
NFR5	All data transmission shall use HTTPS	High
NFR6	Passwords shall be encrypted using hashing algorithms	High
NFR7	JWT tokens shall expire after 24 hours	High
NFR8	Row Level Security (RLS) shall be enforced at database level	High
NFRG	User input shall be validated to prevent SQL injection	High

6.3 Usability Requirements

#	Requirement	Priority
NFR10	The interface shall follow Material Design guidelines	Medium
NFR11	The system shall be responsive across different screen sizes	Medium
NFR12	Error messages shall be clear and user-friendly	Medium
NFR13	The system shall provide loading indicators during operations	Medium

6.4. Reliability Requirements

#	Requirement	Priority
NFR14	The system shall have 99% uptime	High
NFR15	The system shall handle errors gracefully	High
NFR16	The system shall maintain data integrity	High

6.5. Maintainability Requirements

#	Requirement	Priority
NFR17	Code shall follow industry best practices	Medium
NFR18	Code shall be properly documented	Medium
NFR19	The system shall use version control (GitHub)	High

6.6. Communication Requirements

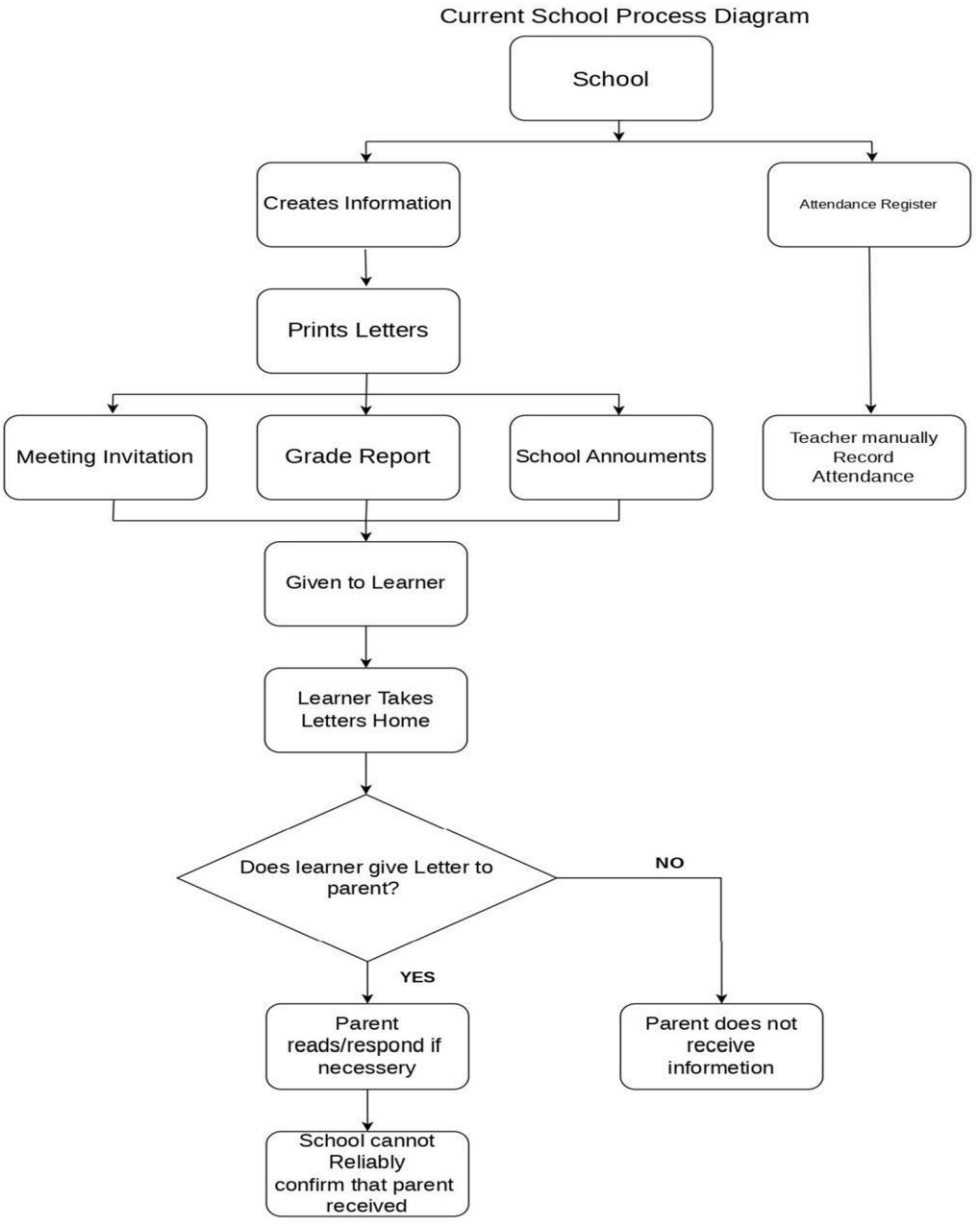
#	Requirement	Priority
NFR20	Communication shall only be accessible to authorized users according to their school, role, class and relationship.	Medium
NFR21	Unread communication shall be visually distinguished from read communication.	Medium
NFR22	Important and Urgent announcements shall be visually distinguished from Normal announcements.	M
NFR23	The system shall provide clear feedback when PARENT Acknowledgement is successfully recorded.	High

7. Data Requirements

6.1 Data Entities

#	Entity	Fields
1	**Schools**	id, name, address, phone, email, logo_url, created_at
2	**Users**	id, email, full_name, role, school_id, phone, profile_picture, is_active, created_at
3	**Learners**	id, admission_number, full_name, date_of_birth, gender, school_id, grade, parent_id, address, is_active, created_at
4	**Teachers**	id, employee_id, user_id, school_id, specialization, joining_date, is_active, created_at
5	**Classes**	id, name, code, school_id, grade, section, teacher_id, room_number, capacity, is_active, created_at
6	**Enrollments**	id, Learner_id, class_id, enrollment_date, academic_year, is_active, created_at
7	**Attendance**	id, Learner_id, class_id, date, status, check_in_time, remarks, marked_by, created_at
8	**Grades**	id, Learner_id, subject, assessment_name, assessment_type, score, grade_letter, school_id, academic_term, created_at
9	**Announcements**	id PK, school_id FK, created_by FK, title, content, priority, target_type, target_grade, target_class_id FK, acknowledgement_required, expires_at, created_at, updated_at • priority(Normal, Important, Urgent) • target_type(Whole School, Role, Grade, Class, Specific Users)
10	**Announcement Recipient**	id PK, announcement_id FK, user_id FK, is_read, read_at, is_acknowledged, acknowledged_at, created_at
11	**Notifications**	id PK, user_id FK, announcement_id FK, title, message, notification_type, is_read, read_at, created_at
12	**Events**	id PK, school_id FK, created_by FK, event_name, description, event_date, start_time, end_time, venue, target_type, target_grade, target_class_id FK, created_at, updated_at

6.2 Data Relationship



6.3 Storage Requirements

- User profile pictures stored in Supabase Storage
- Learner documents (if applicable) stored in Supabase Storage
- Database hosted on Supabase PostgreSQL
- Regular backups maintained by Supabase

7 External Requirements

7.1 User Interfaces

- The system shall provide a responsive web interface accessible via modern browsers
- The system shall offer a mobile app interface developed using Flutter
- Both interfaces shall enable users to securely log in and access role-specific dashboards
- The admin interface shall provide dashboards with data visualizations
- All user interfaces shall follow usability and accessibility best practices

7.2 Software Interfaces

- The mobile app and web interface shall communicate with a centralized backend RESTful API developed with ASP.NET Core Web API
- The backend API shall provide endpoints for authentication, profile management, and data operations
- The backend shall interact with Supabase PostgreSQL for persistent storage
- The system shall use Supabase Auth for authentication

7.3 Communication Interfaces

- The system shall use HTTPS for all communication between clients and the backend API
- Authentication shall use secure token-based methods (JWT) for session management
- File uploads and downloads shall be conducted over HTTP POST and GET requests
- The mobile app and web client shall synchronize data with the backend API in real-time

9 Risk Analysis

9.1 Data Breach /Unauthorised Access

- **Mitigation**: Supabase RLS policies, JWT token validation, HTTPS encryption
- **Contingency Plan**: Revoke keys, force password resets, incident response, notify stakeholders

9.2 Multi-School Data Leakage

- **Mitigation**: Strict RLS policies enforcing school id filtering, server-side validation
- **Contingency Plan**: Audit logs review, patch security rules, re-audit exposed data

9.3 Incorrect Role Enforcement

- **Mitigation**: Automated RBAC tests, server-side validation, role-based middleware
- **Contingency Plan**: Rollback, patch rules, and re-audit access permissions

9.4 Integration Failures

- **Mitigation**: Retry logic, test environments, and error handling
- **Contingency Plan**: Manual data entry fallback, rollback to last known good state

9.5 Network Dependency

- **Mitigation**: Offline caching for essential data, status indicators
- **Contingency Plan**: Notify users of sync status and retry in background

9.6 Browser or Device Compatibility Issues

- **Mitigation**: Cross-platform testing across major browsers and devices
- **Contingency Plan**: Deploy quick patches or provide workarounds

9.7 Outdated or Incorrect Parent/Guardian Contact information

- **Mitigation:** Allow authorized school staff to update parent/guardian contact information and regularly verify contact details
- **Contingency Plan:** Verify incorrect contact information with the learner or guardian and use alternative communication methods until the information is updated

9.8 Parents Not Reading or Acknowledging Important Communication

- **Mitigation:** Use notifications, Read/Unread Tracking, and Parent Acknowledgement for Important or Urgent announcements
- **Contingency Plan:** Identify parents who have not read or acknowledged important communication and follow up using alternative communication methods such as telephone calls or printed notices

10 Feasibility Study

10.1 Technical Feasibility

The proposed School-Parental System requires the development of a Flutter mobile application, an ASP.NET Core web application/API and a shared Supabase backend. The project also requires secure authentication, role-based access control, database security and integration between all components.

The required technologies are available to the development team through Flutter, ASP.NET Core, Supabase PostgreSQL, Supabase Authentication, Supabase Storage and GitHub. The system architecture is technically possible because both Flutter and ASP.NET can communicate with the shared backend.

The more technically challenging areas include:

- Role-based access control;
- Supabase row level security;
- Separating data between multiple schools;
- Integrating flutter and asp.net with supabase;
- Tracking individual read/unread and parent acknowledgement status; and
- Ensuring that parents only access information related to their own children.

10.2 Operational Feasibility

The proposed system addresses problems identified during stakeholder engagement, including manual record keeping and difficulties in communicating important information to parents. The ASP.NET application will mainly support administrative and teacher functions, while the Flutter application will provide parents with convenient access to information regarding their children. The proposed solution fits

the normal activities of schools because administrators already manage Learners, teachers, classes, attendance and academic information, while teachers already record attendance and grades. The proposed system digitises and centralises these existing activities. Stakeholder interviews also identified limitations. Some parents may have outdated contact information, may not respond to communication and may have limited access to smartphones or mobile data. Therefore, the system should not assume that the mobile application will completely replace existing communication methods.

Users with limited technology experience may require basic guidance or training.

- Operational Feasibility Decision
- Feasible with changes.

10.3 Economic Feasibility

Resource	Available?	Action Required
Internet/data	Yes	No action required
Transport to stakeholders	Yes	No action required
Development computers	Yes	No action required
Test mobile devices	Yes	No action required
Flutter	Yes	No action required
ASP.NET Core	Yes	No action required
GitHub	Yes	Maintain the repository and keep documentation current
Supabase	Yes	Use available resources for the prototype
Microsoft Project	Yes	Keep the project schedule updated
Presentation resources	Yes	Prepare presentation materials before assessment

10.4 Schedule Feasibility

Area	Finding
System design	Required before development begins
Flutter development	Time must be allocated for parent/mobile functionality
ASP.NET development	Time required for administration, teacher functions and APIs
Supabase	Database, authentication, storage and RLS require dedicated work
Integration	Must be tested throughout development rather than only at the end
Testing	Time must remain for manual testing, integration testing and corrections
Stakeholder feedback	Must be included before final approval
Documentation	Must continue throughout development

10.5 Legal Feasibility

The proposed system will handle information relating to schools, learners, parents and staff. Personal information must therefore be handled carefully.

The following controls will be applied:

- Only necessary information will be collected;
- Users will only access information permitted by their role;
- Parents will only access information relating to their linked children;
- School admins will only access their own school's information;
- Confidential stakeholder information will not be uploaded to public github repositories;
- Passwords, private keys and api credentials will not be stored in github;
- Supabase rls and rbac will be used to restrict access;
- Https will be used for data transmission;
- Test information should be used during prototype development and demonstration.

10.6 Ethical Feasibility

The prototype must be developed and used responsibly because it involves learners, parents, teachers and school information.

The group will:

- Explain that the system is an educational prototype;
- Collect only information necessary for the project;
- Avoid exposing confidential learner or parent information;
- Ensure that users are treated fairly;
- Restrict information according to user roles;
- Use stakeholder information only for its intended project purpose;
- Keep stakeholder evidence secure; and
- Report system limitations honestly.

The system is intended to improve school administration and communication and should not be used for unauthorized monitoring or disclosure of learner information.

10.7 Feasibility Summary

Feasibility Area	Decision	Main Reason	Change Required
Technical	Feasible with changes	Required technologies can support the prototype, but security and integration are complex	Test integration/RLS early and prioritise core functions
Operational	Feasible with changes	Solves confirmed school problems	Keep alternative communication methods for parents with limited access

Economic	Feasible	Main software tools are accessible	Monitor transport/data/device requirements
Schedule	Feasible with changes	Project is large but manageable through priorities	Develop High-priority functions first
Legal	Feasible with changes	Personal learner/parent data requires protection	Use RBAC, RLS, HTTPS and test/anonymised data
Ethical	Feasible	Solution benefits stakeholders when used responsibly	Collect only necessary information and protect users

10.8 Overall Feasibility Conclusion

The proposed School-Parental System is feasible with changes. The main strengths of the project are that it addresses problems identified through stakeholder engagement, uses technologies available for the project and provides an integrated solution using Flutter, ASP.NET Core and Supabase. The main limitations identified are the large project scope, complexity of system integration and security, dependence on internet connectivity and the possibility that some parents may have limited access to smartphones or mobile data. To ensure that the project can be completed within the semester, the group will prioritise essential requirements, develop the system in manageable modules, test Flutter, ASP.NET and Supabase integration early, maintain suitable security controls and simplify lower-priority functionality where necessary.

11. SWOT Analysis

The SWOT Analysis evaluates the internal strengths and weaknesses of the proposed School-Parental System and the external opportunities and threats that may affect the project.

Strengths	Weakness
The system addresses problems confirmed through stakeholder engagement.	The project has a large scope with many modules and user roles.
Provides one centralized platform for managing Learners, teachers, classes, attendance, grades and communication.	Integration between Flutter, ASP.NET Core and Supabase may be complex
Uses Flutter, ASP.NET Core and Supabase as one integrated solution	The system depends on internet connectivity for many functions.
Improves parent-school communication through announcements, notifications, events, Read/Unread Tracking and Parent Acknowledgement	Some parents may have limited access to smartphones, mobile data or reliable internet
Parents can access authorized information concerning their children	The prototype is an academic system and not a full production-ready school management system
Opportunities	**Threats**
The system can reduce the school's dependence on paper-based communication and manual processes	Data breaches or unauthorized access could expose confidential learner, parent or school information.

Schools can improve communication with parents and have better confirmation that important information has been received.	Incorrect role or school filtering could expose one school's information to another school.
The system can improve access to learner attendance and academic information.	Poor or unreliable internet connectivity could prevent users from accessing or synchronizing information.
The system could later be expanded with additional modules once the core prototype is completed.	Some parents may not adopt the mobile application because of limited smartphones, data or technology familiarity.
The multi-school design creates an opportunity for the system to support more than one school.	Outdated or incorrect parent contact information may reduce communication effectiveness.
Communication statistics can help schools identify parents who have not received or acknowledged important information.	The large number of proposed functions could cause development delays and reduce testing time.
Digital attendance, grades and communication can reduce paperwork and administrative effort.	Parents may still fail to read or acknowledge important communication

11.2. SWOT Analysis Summary

The SWOT Analysis indicates that the School-Parental System has strong potential because it addresses real stakeholder needs and provides an integrated solution for school administration and parent communication. Its main strengths include centralized information management, role-based access control, multi-school support and improved communication between schools and parents.

However, the project also has weaknesses relating to its large scope, technical integration complexity, limited development time and dependence on internet connectivity. Opportunities exist to reduce manual school processes, improve parent access to information and expand the system in future phases.

The main threats include security risks, incorrect access control, poor network connectivity, limited parent access to smartphones or mobile data, outdated contact information and delays caused by the size of the project. These threats will be managed through the project's risk register, security controls, requirement prioritization and contingency planning.

12. Project Risk Register

Risk Register for School-Parental System

Prepared Bright Web Crafters **Date:** 28 August 2026

Risk ID	Risk Statement	Category	Likelihood	Impact	Score	Level	Mitigation	Contingency	Risk Owner	Status
R01	Unauthorized users may access confidential school or learner information.	Security	2	3	6	Medium	Enforce Supabase RLS, RBAC, HTTPS and JWT validation; test access permissions before release.	Revoke affected access or keys, reset credentials, investigate the exposure and correct security policies.	Security/Backend Lead – assign member	Monitoring
R02	Incorrect school filtering may expose one school's data to another school.	Security / Data	2	3	6	Medium	Enforce school_id filtering through RLS and server-side validation; test each school context.	Temporarily disable affected access, correct RLS/filtering rules and re-audit exposed data.	Database/RLS Lead – assign member	Monitoring
R03	Incorrect role enforcement may allow users to perform unauthorized functions.	Security / Access	2	3	6	Medium	Test every user role and permission; apply role-based middleware and server-side validation.	Correct the role or policy configuration, rollback affected permissions and re-test access.	Authentication Lead – assign member	Monitoring
R04	Flutter, ASP.NET and Supabase integration may take longer than planned and delay development.	Technical / Schedule	3	3	9	High	Build and test a small end-to-end integration early; divide integration tasks and review progress weekly.	Simplify affected functionality, reassign development work and use the last stable integration approach.	Integration Lead – assign member	Open
R05	Internet or network problems may interrupt access to the system or delay data synchronization.	Technical / Resource	2	2	4	Medium	Provide connection and sync-status messages, reduce unnecessary network requests and test unstable connections.	Retry synchronization when the connection returns and use temporary manual communication/processes where necessary.	Backend/Mobile Lead – assign member	Monitoring
R06	Browser or device differences may cause screens or functions to behave incorrectly.	Technical / Quality	2	2	4	Medium	Test the web application on common browsers and the mobile application on available devices and screen sizes.	Correct affected screens/functions, release a patch or provide a temporary workaround.	Testing Lead – assign member	Open
R07	Parent or guardian contact information may be incorrect or outdated, reducing communication reliability.	Operational / Data	3	2	6	Medium	Allow authorized school staff to maintain contact information and regularly verify parent/guardian details.	Verify the information with the learner or guardian and use an alternative communication method until updated.	Data Management Lead – assign member	Monitoring
R08	Parents may not read or acknowledge important communication on time.	Operational / Communication	3	2	6	Medium	Use notifications, Read/Unread Tracking and Parent Acknowledgement for Important or Urgent announcements.	Identify non-responsive parents and follow up using telephone calls, printed notices or another approved method.	Communication Module Lead – assign member	Monitoring
R09	Some parents may have limited access to smartphones, mobile data or reliable internet, reducing mobile-app adoption.	Operational / Resource	2	3	6	Medium	Keep the mobile app simple and data-conscious; avoid unnecessary data usage and support commonly available devices.	Continue using alternative communication methods such as telephone calls or printed letters for affected parents.	Mobile App Lead – assign member	Monitoring
R10	The large number of proposed functions may delay prototype completion and leave insufficient time for testing.	Scope / Schedule	3	3	9	High	Prioritise core requirements, monitor the Microsoft Project schedule weekly and develop modules incrementally.	Simplify or postpone lower-priority functions so that core integration and testing can be completed.	Project Manager – assign member	Open

13. Requirements Change Log

ID	Original Requirement/Idea	Change Made	Reason
CH01	Basic announcement functionality	Expanded to Communication Management	Stakeholder engagement confirmed problems with parent communication
CH02	Announcements only	Added Notifications and Events	Schools use multiple forms of communication and require timely information
CH03	No communication confirmation	Added Read/Unread Tracking	Stakeholders indicated difficulty confirming whether parents received communication
CH04	No acknowledgement mechanism	Added Parent Acknowledgement for Important/Urgent announcements	Allows school to confirm receipt of important communication
CH05	Important Notices considered as separate communication type	Combined with Announcement priority: Normal, Important or Urgent	Reduces duplicate functionality and simplifies the communication design
CH06	Assumption that mobile communication can reach all parents	Added connectivity/device constraint and risk	Stakeholder feedback showed that some parents may lack smartphones/data
CH07	Contact information assumed to be current	Added requirement to maintain/update guardian contact details	Stakeholder feedback identified incorrect/outdated contact details as a problem
CH08	Broad proposed system	Added requirement prioritisation	Feasibility and schedule analysis showed that core functions must be completed before lower-priority features

14. Refined Project Scope

Following the feasibility study and project risk analysis, the group has confirmed that the School-Parental System will continue as an integrated Flutter, ASP.NET Core and Supabase prototype. The refined project scope prioritises authentication, role-based access control, school management, Learner management, teacher management, class management, attendance, grades, parent access and school announcements as core functionality. Notifications, events, Parent Acknowledgement, communication statistics and reporting will remain within the project but will be implemented according to priority after the core integrated functions have been completed. Advanced reporting and dashboard functionality may be simplified if required to protect the project schedule and ensure sufficient time for system integration, testing and stakeholder evaluation. Functions already excluded from the prototype, including payment processing, payroll, library management, transport tracking, online classroom functionality and government-system integration, remain outside the refined scope.

15. Conclusion

This School Portal System project aims to provide an efficient, secure, and user-friendly platform for school administrators, teachers, and parents to manage and access Learner information. By combining a responsive web interface and a Flutter-based mobile application, the system ensures accessibility and convenience for users across multiple devices. The design prioritizes data security through role-based access control and Row Level Security, ensuring that users can only access data they are authorized to view. The multi-school architecture allows a Super Admin to manage multiple schools from a single dashboard, while School Admins manage only their respective schools. The project's success will empower schools to make informed decisions regarding resource allocation, academic planning, and parent communication, ultimately contributing to improved educational outcomes and operational efficiency.

16. Reference

- Flutter. (2024). *Flutter Documentation*. https://docs.flutter.dev/
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- IEEE. (1998). *IEEE Std 830-1SS8: IEEE Recommended Practice for Software Requirements Specifications*
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17. AI USE

Artificial Intelligence (AI) tools were used during the preparation of this document to assist with:

a. **Structuring the SRS according to industry standard formats**
b. **Rewording sentences for clarity, conciseness, and formal technical writing style**
c. **Suggesting appropriate terminology, acronyms, and definitions**
d. **Organizing content based on the reference document structure**

All AI-generated content was reviewed, validated, and modified by all group members to ensure accuracy, relevance, and compliance with preparations.

18. Implementation and Final Prototype

18.1 Overview of Implemented Components

The final prototype for the School Portal System was successfully developed as an integrated, three-tier solution. The Supabase backend was implemented as the centralized PostgreSQL database, housing the required tables (such as Users, Learners, Attendance, Grades, and Announcements), along with configured Row Level Security (RLS) policies and Supabase Authentication to ensure strict Role-Based Access Control (RBAC). The ASP.NET Core Web Application was developed to serve as the primary administrative and teaching interface, allowing School Admins to manage learner and teacher records, and enabling teachers to record attendance and capture grades. Finally, the Flutter Mobile Application was developed to provide a user-friendly, mobile-optimized interface primarily for Parents, allowing them to securely log in, view their children's attendance records, check academic grades, and receive school announcements. Both the Flutter and ASP.NET applications were successfully integrated to communicate with the same shared Supabase project, ensuring that data captured by teachers on the web application is immediately accessible to parents via the mobile application.

18.2 Requirement-to-Function Status

This section maps important approved functional requirements in the existing SRS to corresponding implemented application functions. Status categories are Working (verified by execution), Partially Implemented (only part of the requirement is demonstrated), Incomplete (required behaviour is not available), and Not Yet Verified (code or

interface identified but execution has not been confirmed). The current statuses are provisional pending live checks.

18.2.1 Super Administrator Requirements

Req ID	Approved Requirement	Application Function	Status
SA1	Super Admin can register and login	ASP.NET authentication and Supabase Auth	Not Yet Verified
SA2	View all schools	Super Admin schools view	Not Yet Verified
SA3	Add schools	School registration module	Not Yet Verified
SA4	Edit school details	School edit interface	Not Yet Verified
SA5	Delete schools	School management actions	Not Yet Verified
SA6	View all users	Super Admin user management	Not Yet Verified
SA7	View reports and dashboards	Super Admin dashboard and reports	Not Yet Verified
SA8	Select a school	School-context selection	Not Yet Verified
SA9	Create announcements	Super Admin announcements	Not Yet Verified
SA10	Create and manage events	Super Admin events	Not Yet Verified
SA11	View communication statistics	Communication statistics/reporting	Not Yet Verified

18.2.2 School Administrator Requirements

Req ID	Approved Requirement	Application Function	Status
SA12	Register and login	ASP.NET authentication	Not Yet Verified
SA13	View only assigned school data	School-specific filtering and permissions	Not Yet Verified
SA14–SA15	Add and edit learners	Learner management	Not Yet Verified
SA16–SA17	Add and edit teachers	Teacher management	Not Yet Verified
SA18–SA19	Create classes and assign learners	Class management	Not Yet Verified
SA20	View attendance reports	Attendance management	Not Yet Verified
SA21	View grade reports	Grade management	Not Yet Verified
SA22	Manage announcements	School communication module	Not Yet Verified
SA23	Manage events	Events module	Not Yet Verified
SA24	Require parent acknowledgement	Announcement acknowledgement option	Not Yet Verified
SA25	View communication statistics	Communication reporting	Not Yet Verified
SA26	Manage subjects	Subject management — verify full create/edit/remove support	Not Yet Verified

18.2.3 Teacher Requirements

Req ID	Approved Requirement	Application Function	Status
T1	Teacher login	ASP.NET authentication	Not Yet Verified
T2	View only school data	School-scoped access	Not Yet Verified
T3–T4	View assigned classes and learners	Assigned classes	Not Yet Verified
T5–T6	Mark and view attendance	Teacher attendance	Not Yet Verified
T7–T8	Record and edit grades	Teacher grades	Not Yet Verified
T9	View grade reports	Grades/reporting view	Not Yet Verified
T10–T11	Create and view class communication	Teacher communication	Not Yet Verified
T12	Create class-related events	Teacher events	Not Yet Verified

18.2.4 Parent Requirements

Req ID	Approved Requirement	Application Function	Status
P1	Parent login	Flutter authentication	Not Yet Verified

P2–P3	View authorised school and child information	Linked learner retrieval and role restrictions	Not Yet Verified
P4	View children's attendance	Flutter attendance	Not Yet Verified
P5	View children's grades	Flutter grades	Not Yet Verified
P6	View announcements	Flutter announcements	Not Yet Verified
P7	View class information	Learner/class details — verify screen content	Not Yet Verified
P8	Receive and view notifications	Flutter notification functionality	Not Yet Verified
P9	View upcoming events	Flutter events	Not Yet Verified
P10	Identify read/unread communication	Read/unread communication support	Not Yet Verified
P11	Acknowledge important announcements	Flutter parent acknowledgement	Not Yet Verified

Note: The approved SRS uses the SA prefix for both Super Admin and School Admin requirements. The prefixes and IDs have been retained here to preserve traceability to the approved baseline. Status values must be updated only after verifying the corresponding workflow and authorised database operation.

18.3.1 Flutter Mobile Application

Figure 18.1: Parent Login Interface

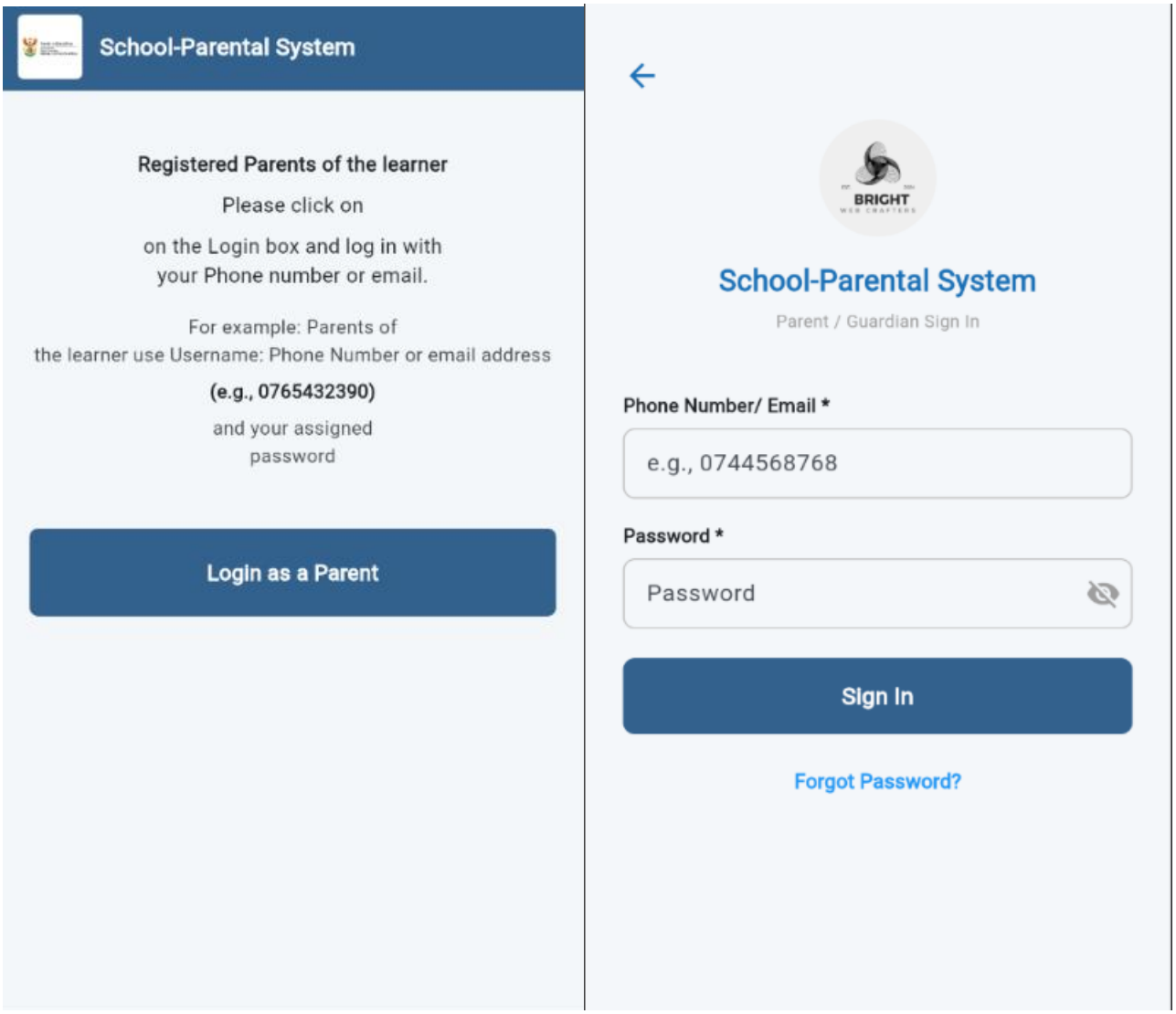


Figure 18.2: Parent Dashboard

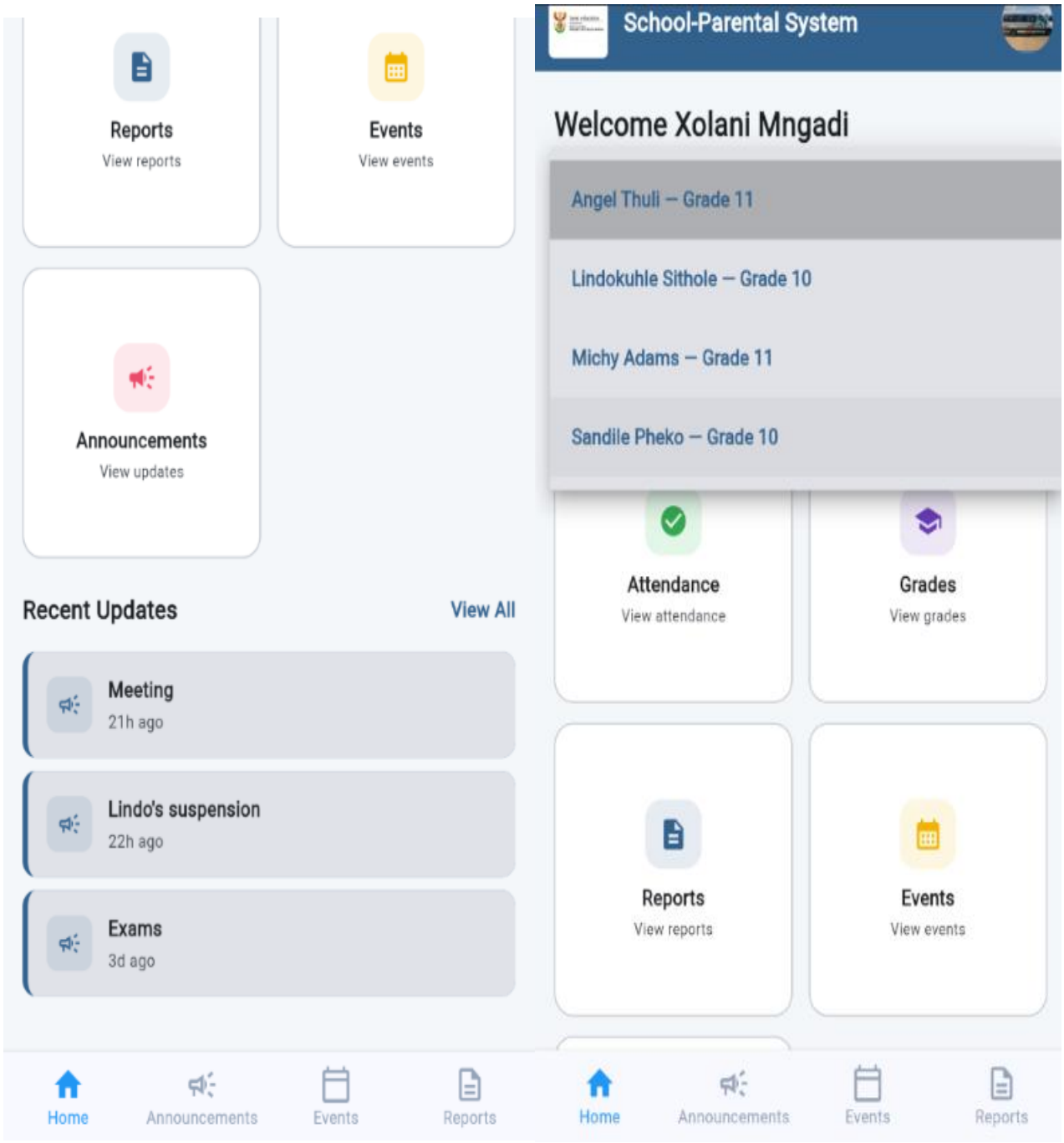


Figure 18.3: Learner Attendance Interface

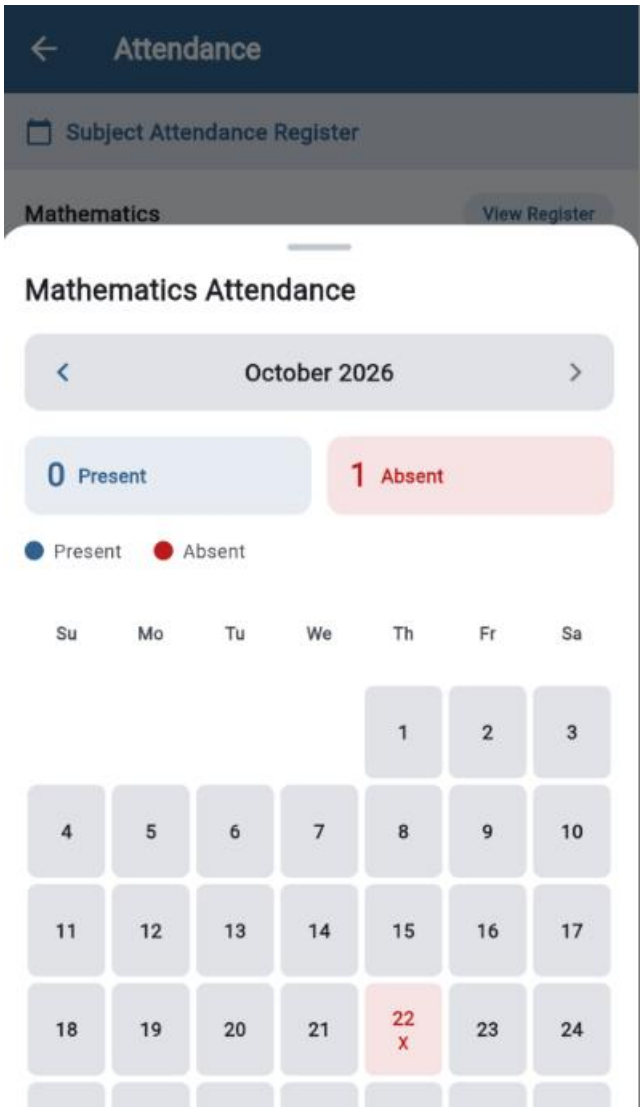


Figure 18.4: Learner Academic Results

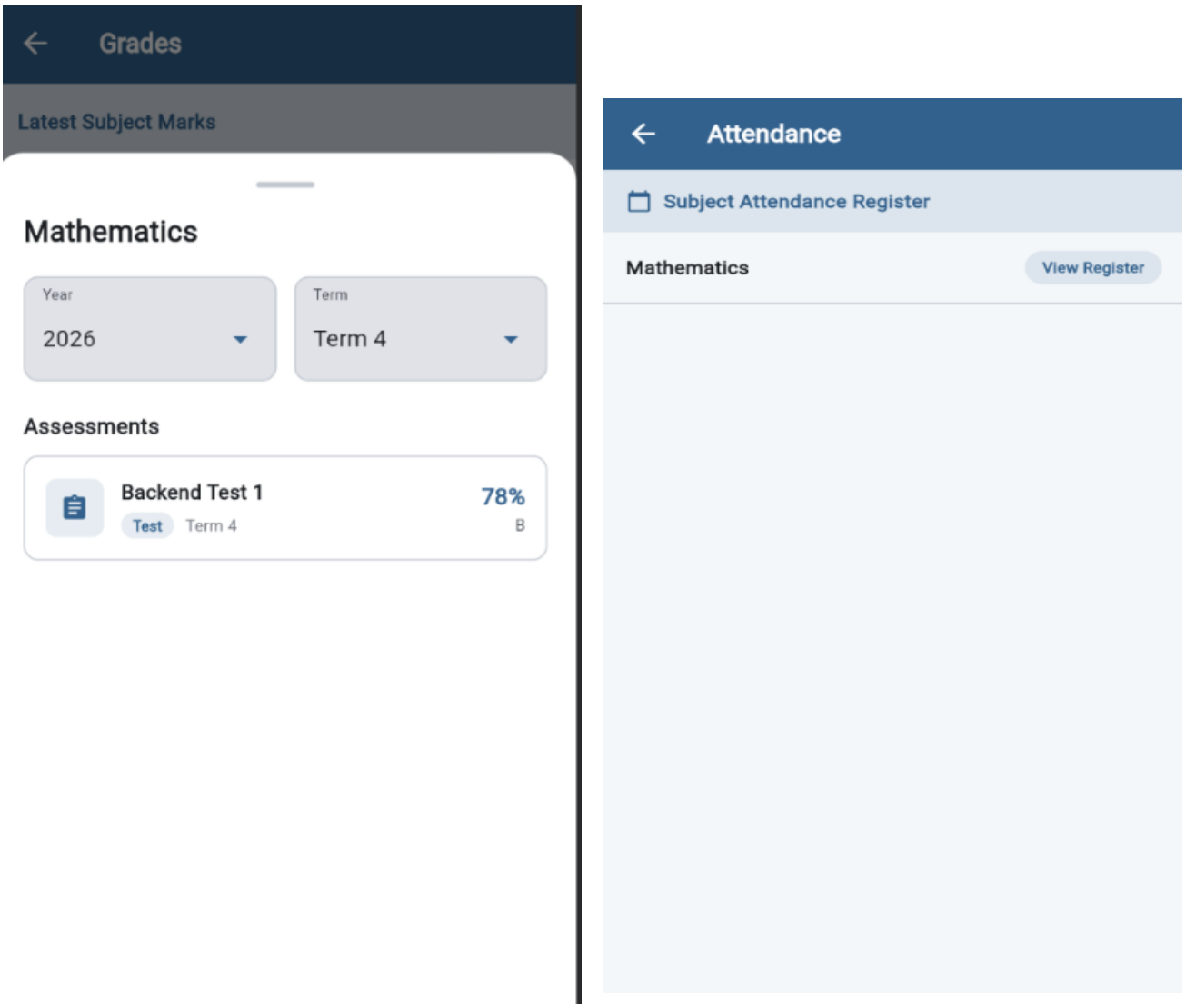


Figure 18.5: School Announcements

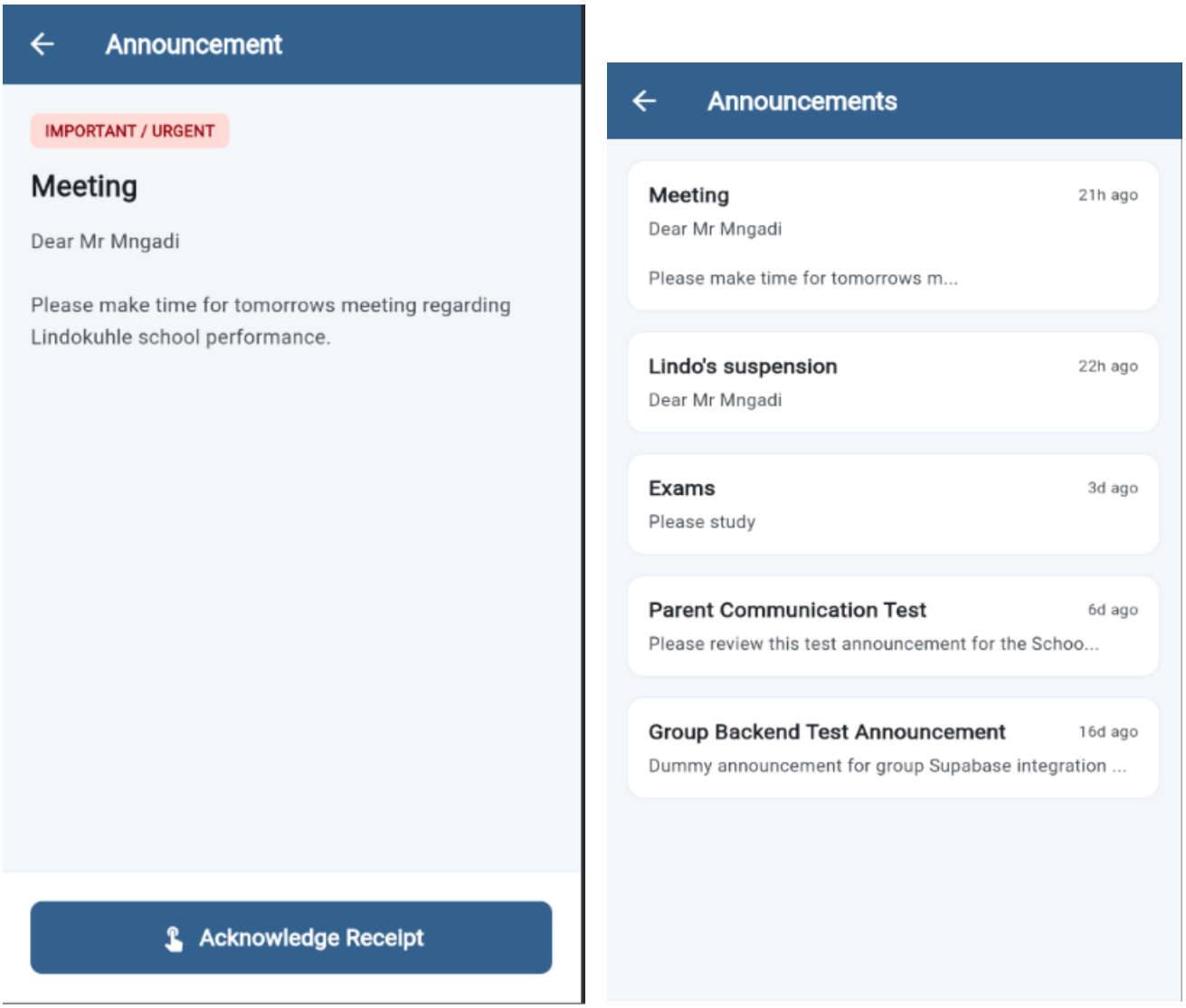
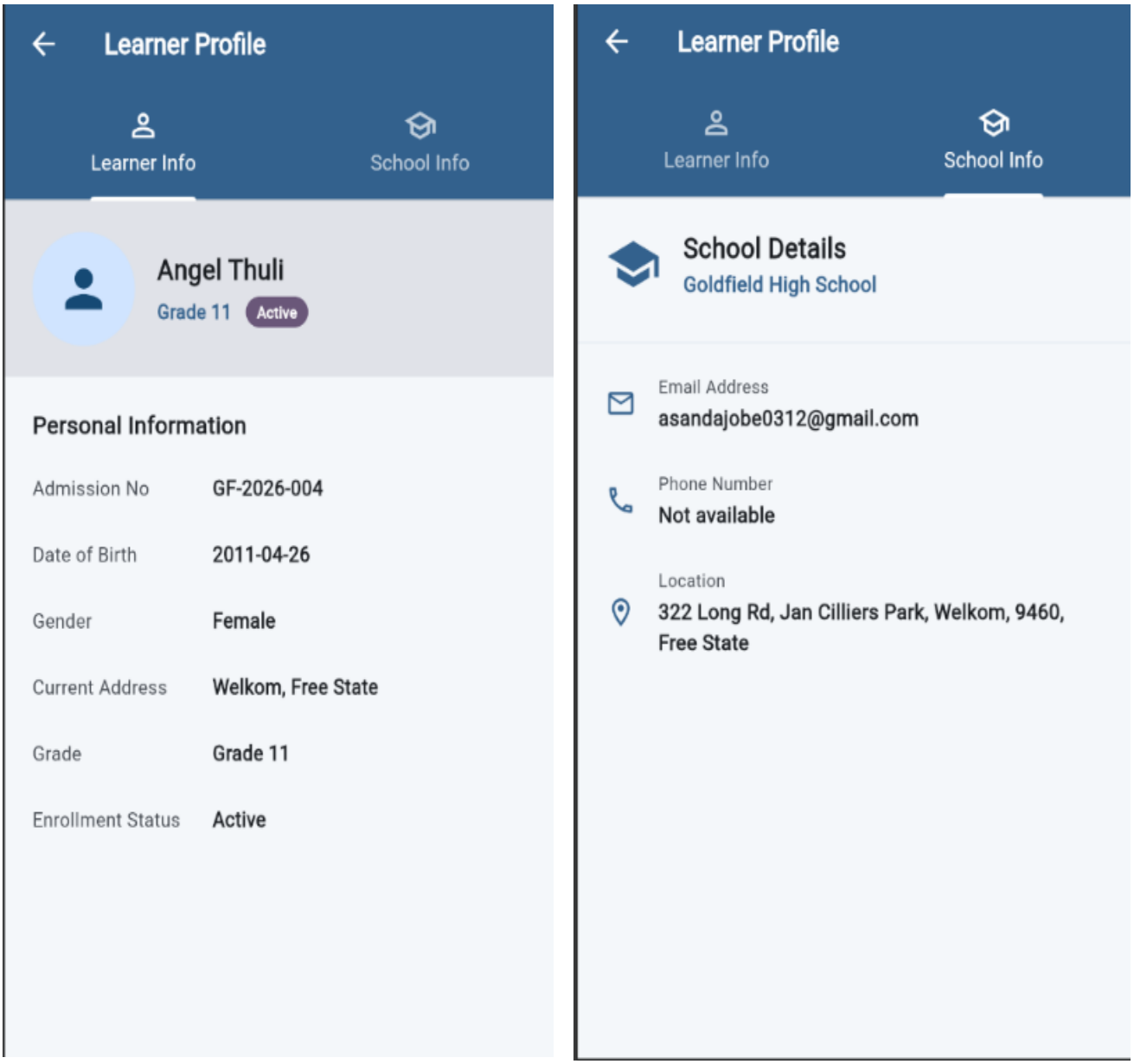


Figure 18.6: Profile

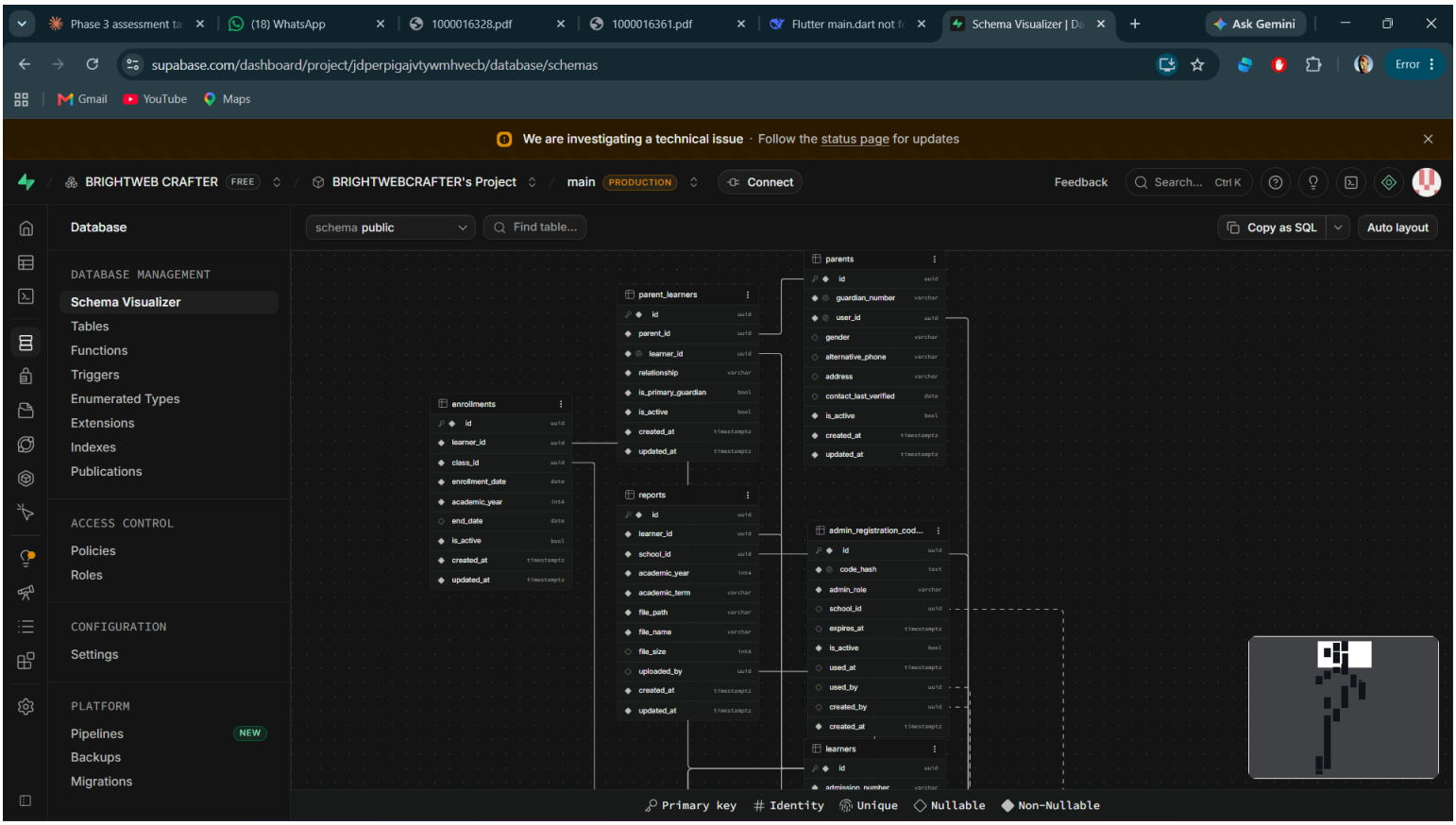


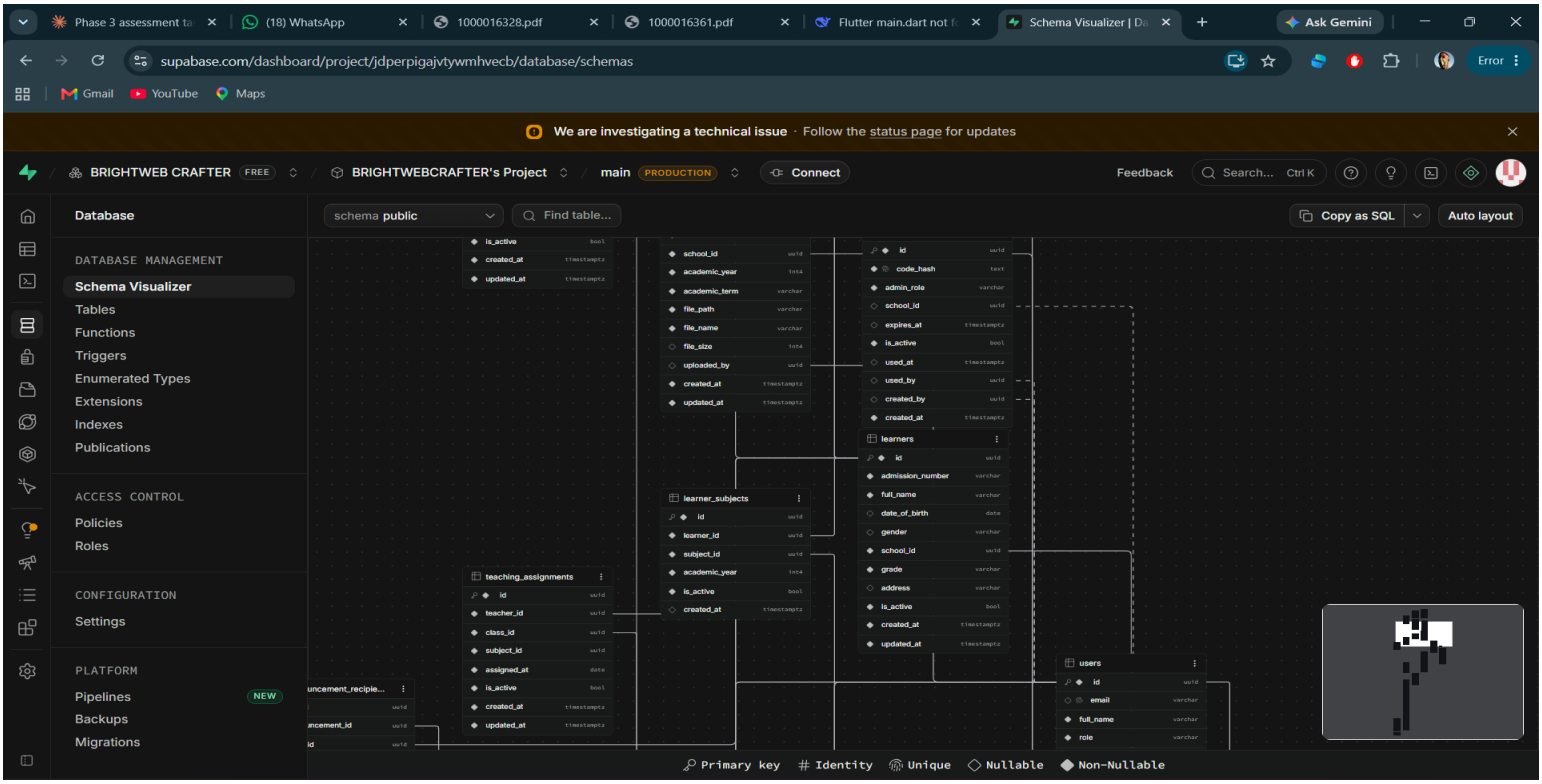
18.3.2 Supabase Backend

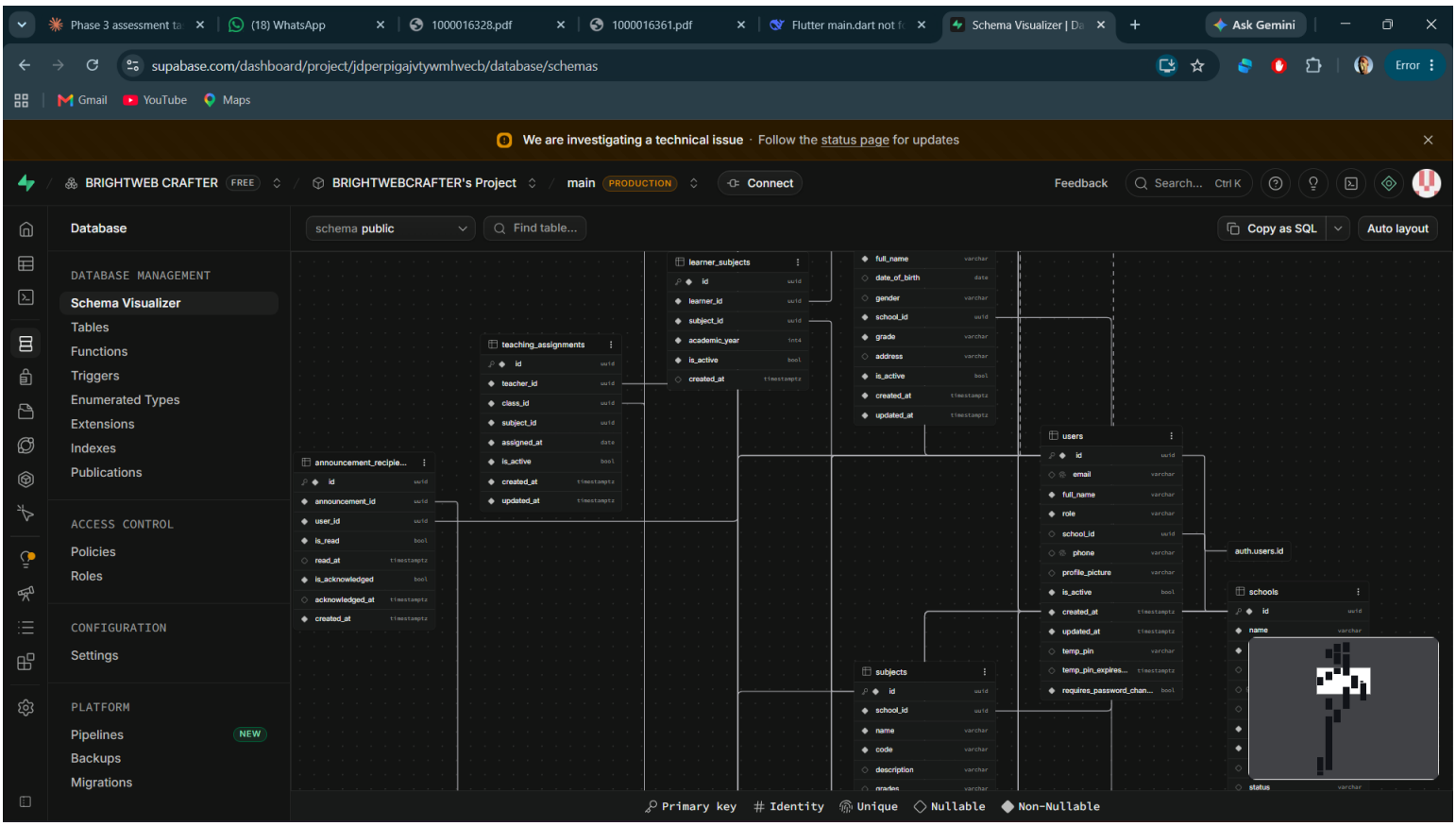
The School-Parental System uses Supabase as its shared backend platform. Supabase provides a PostgreSQL database and authentication services for the ASP.NET Core Web Application and Flutter Mobile Application. The backend is designed to store and manage school information, user profiles, learners, teachers, classes, attendance records, academic grades, announcements, and other relevant data. The application architecture requires both clients to connect to the same Supabase project so that authorized users can access consistent information.

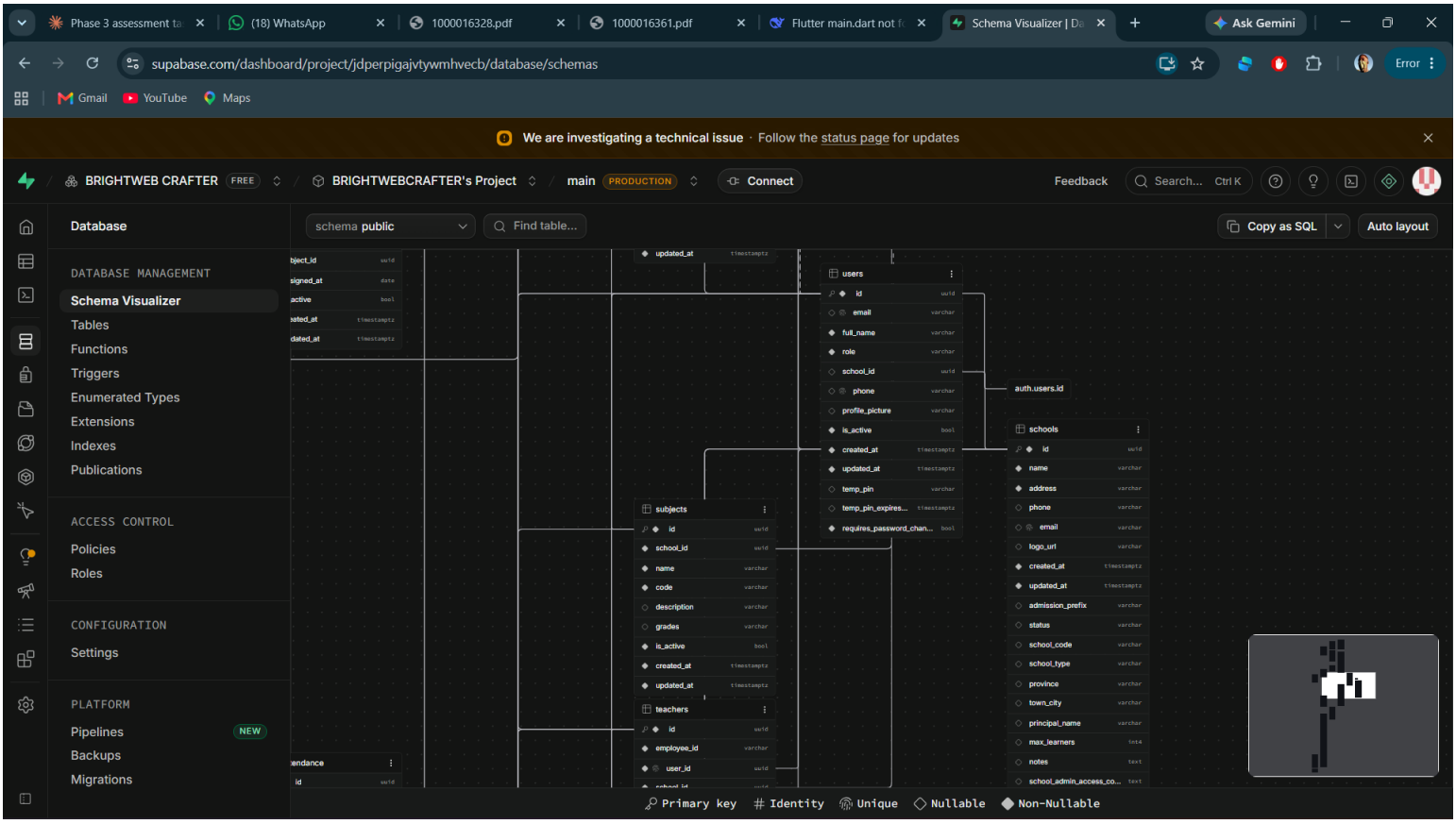
Database relationships and access permissions are important to the system's multi-school design. Role-Based Access Control (RBAC) and Row Level Security (RLS) are intended to restrict access according to the authenticated user's role, assigned school, and permitted learner relationships. The following screenshots provide implementation evidence of the Supabase backend configuration, database structure, and authentication functionality. Successful operation of the access policies and shared data processes must be confirmed during integration testing and demonstration.

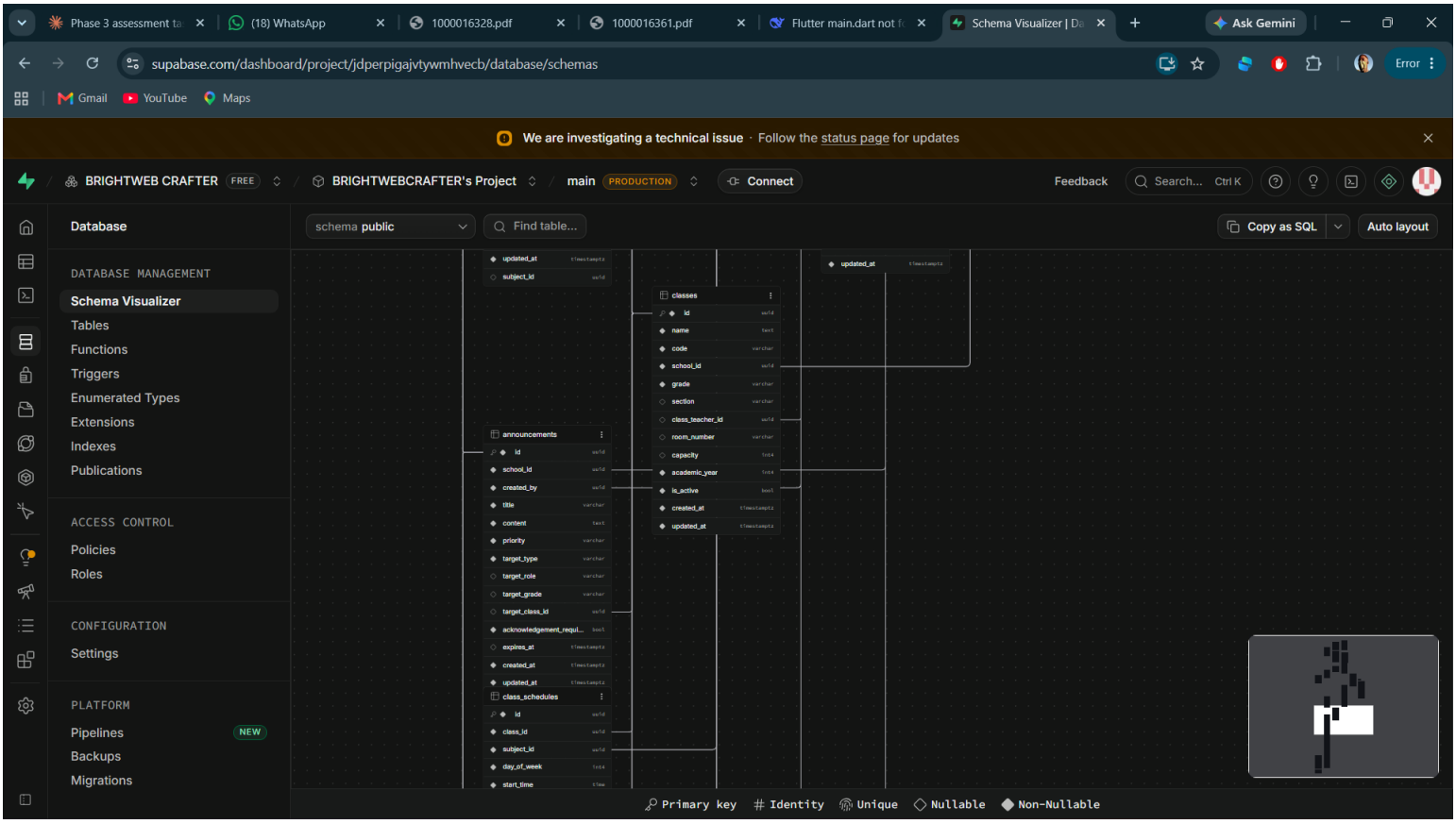
Figure 18.7: Supabase Database Tables

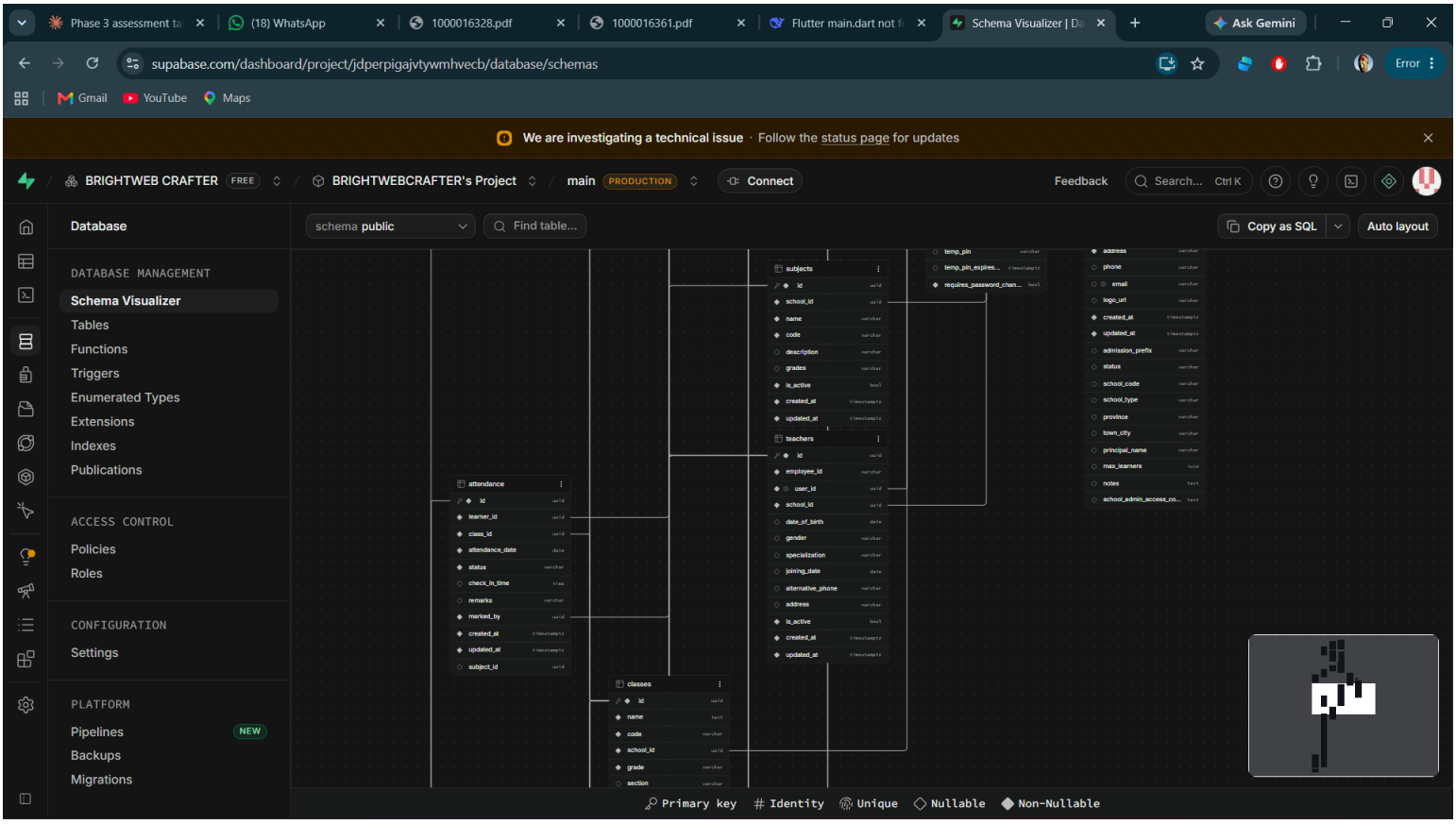


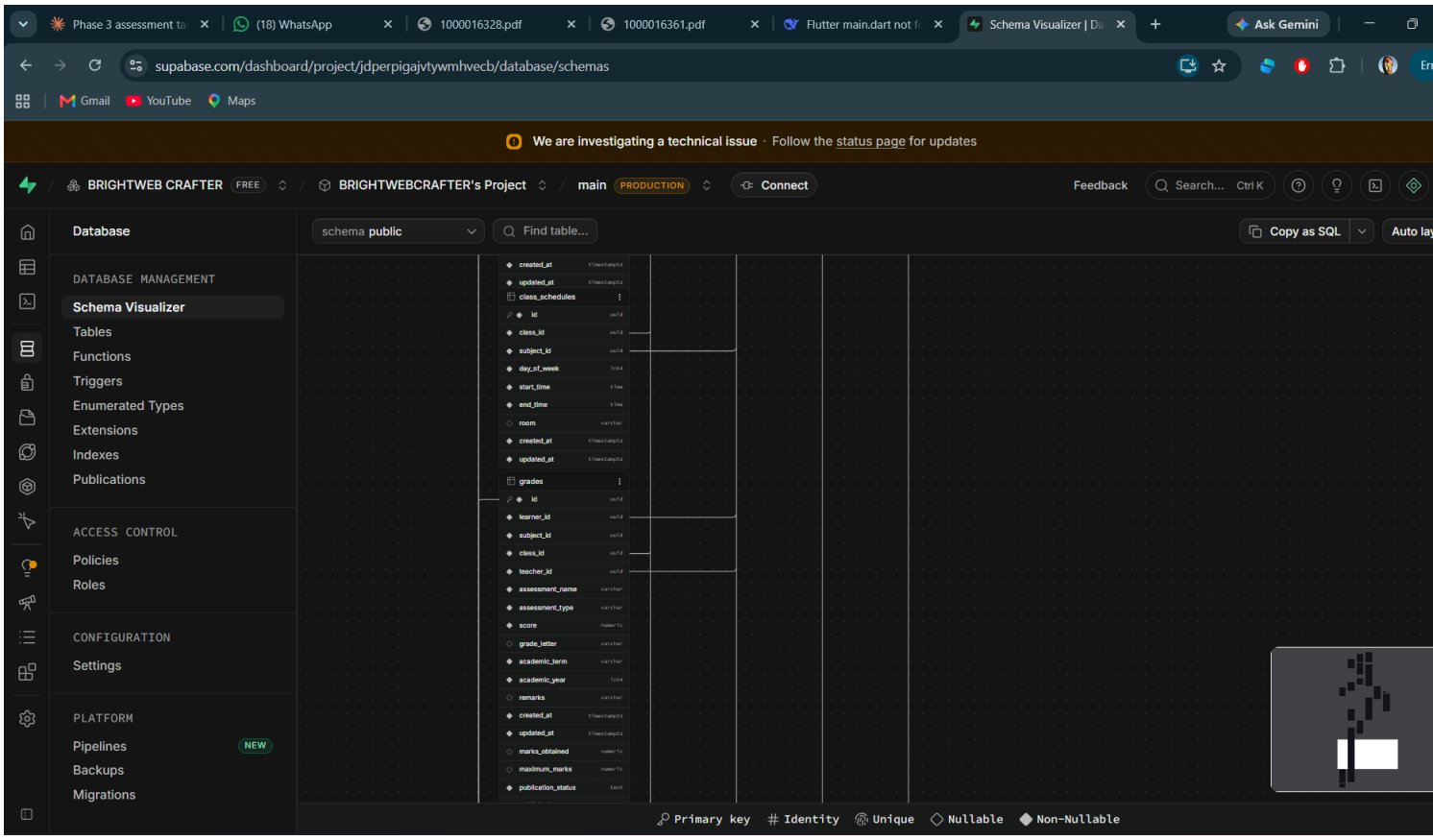

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grades
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Unique
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Non-Nullable

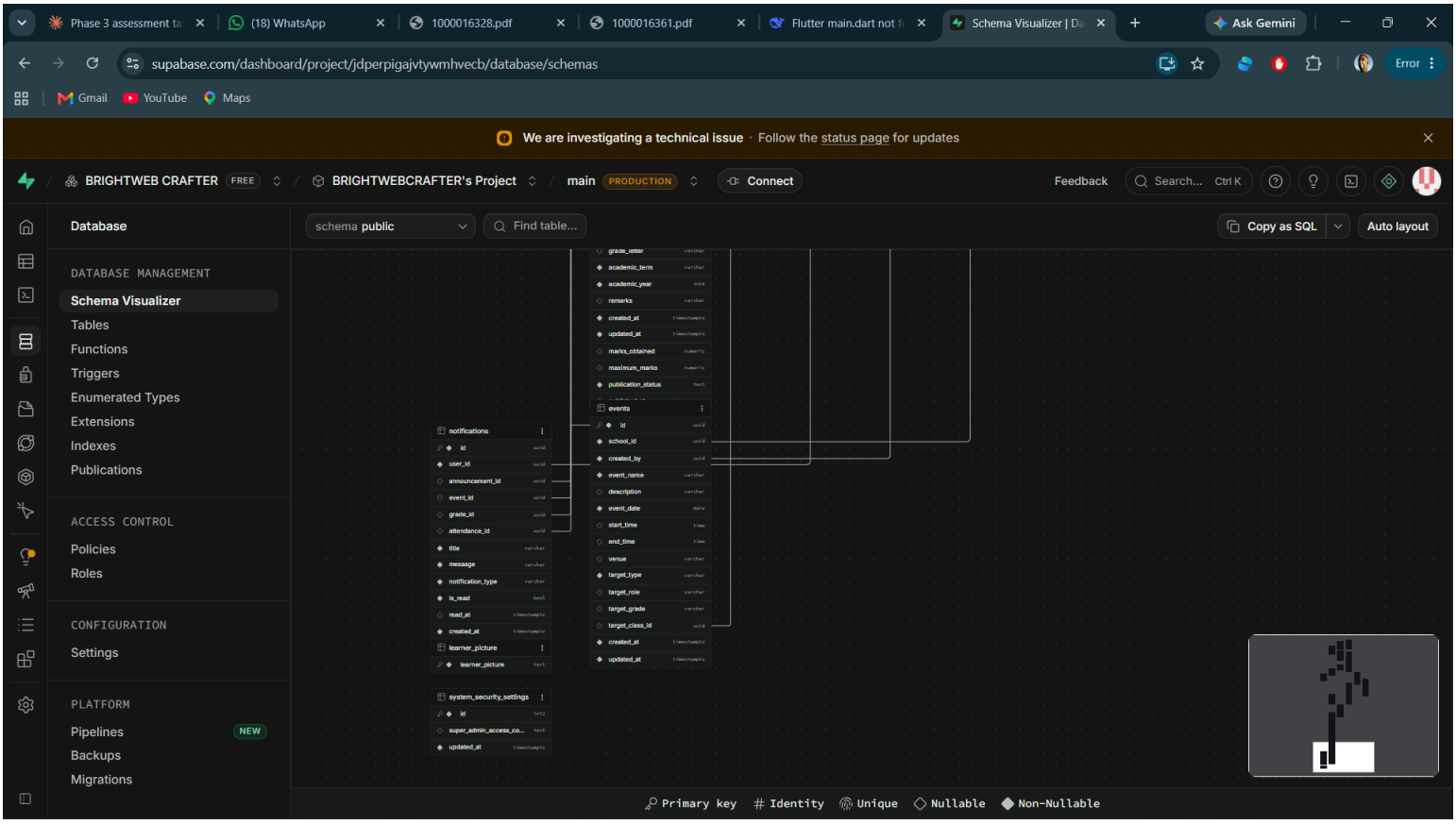


Figure 18.8: Supabase Authentication

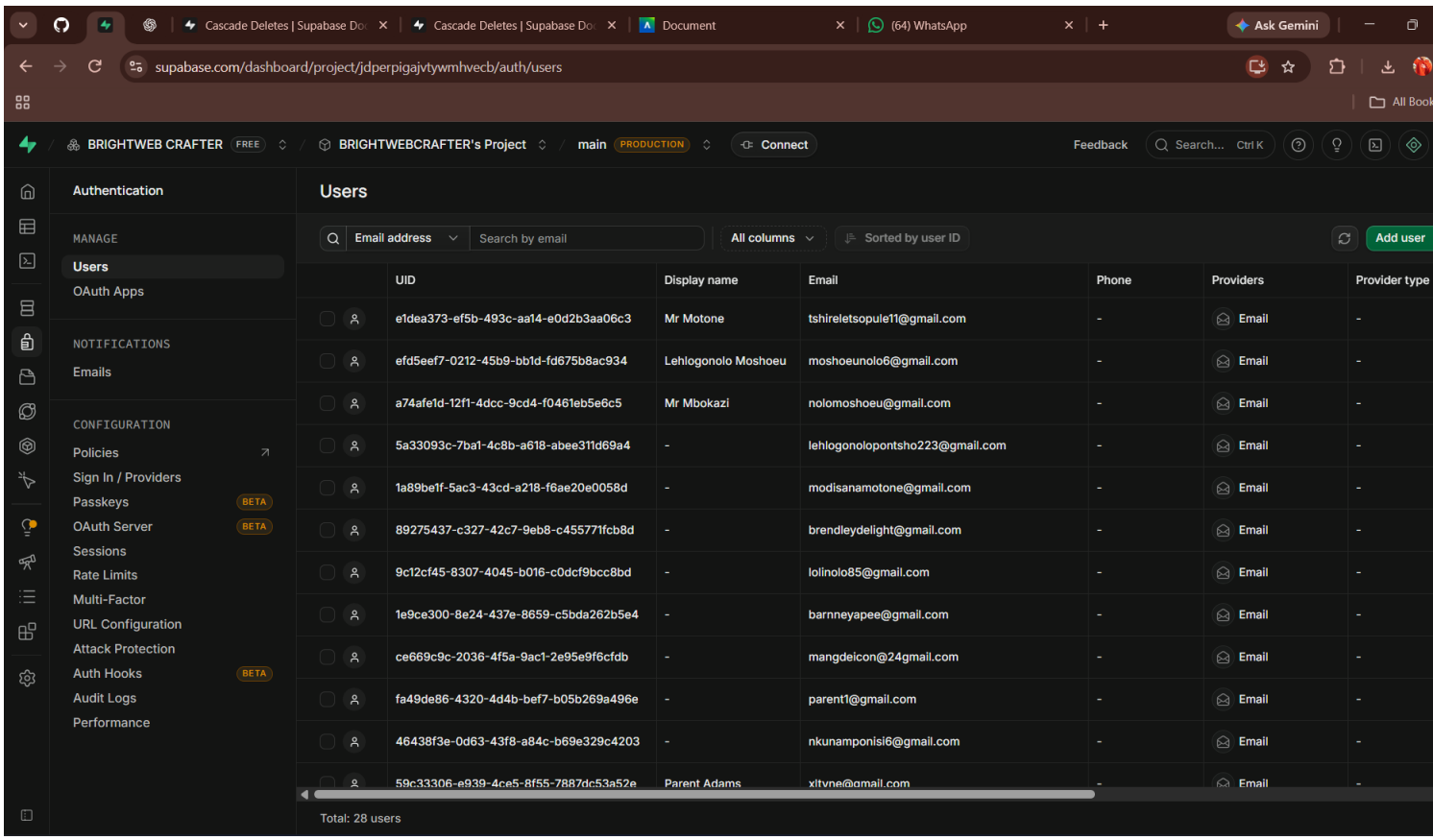
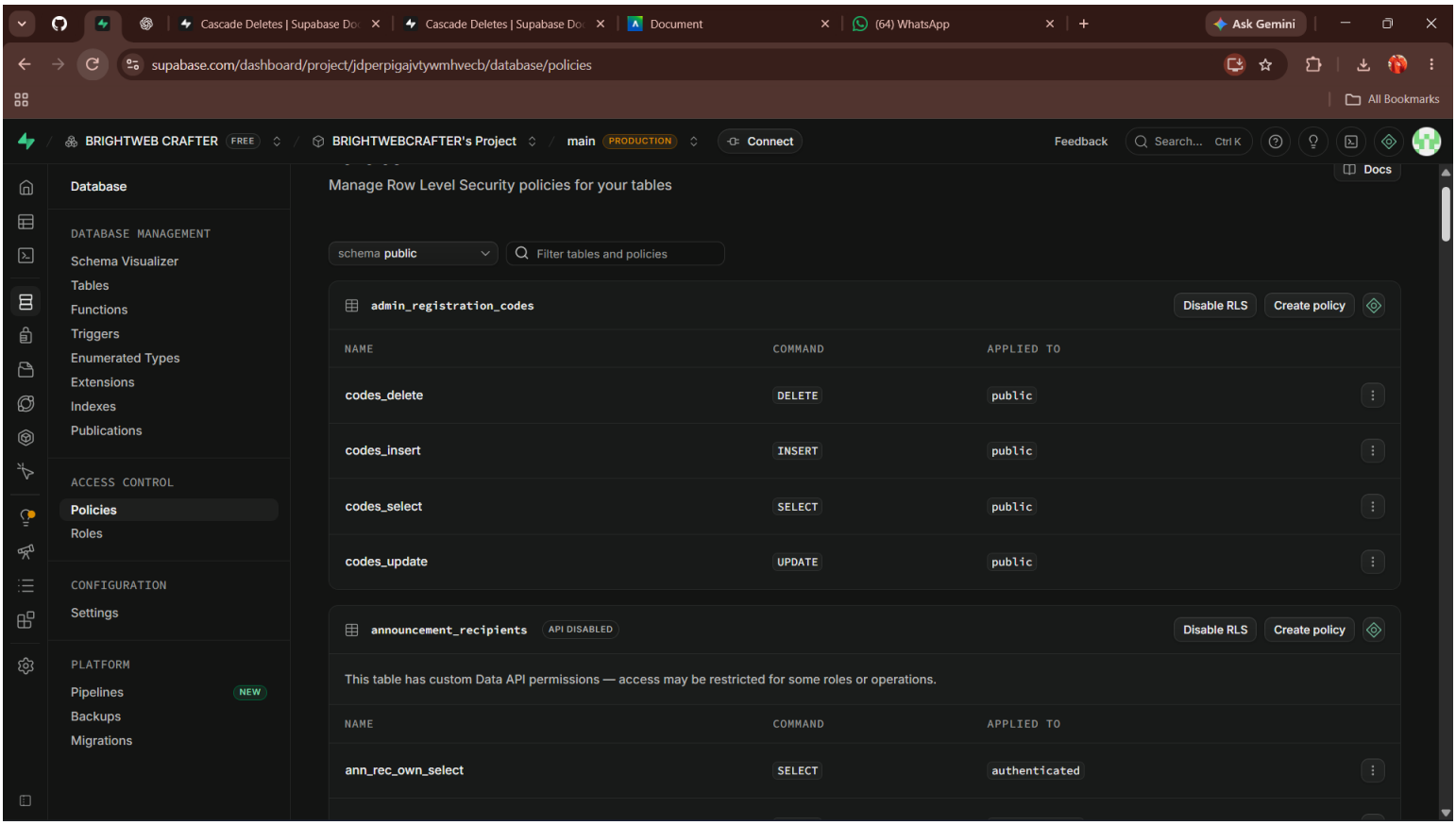
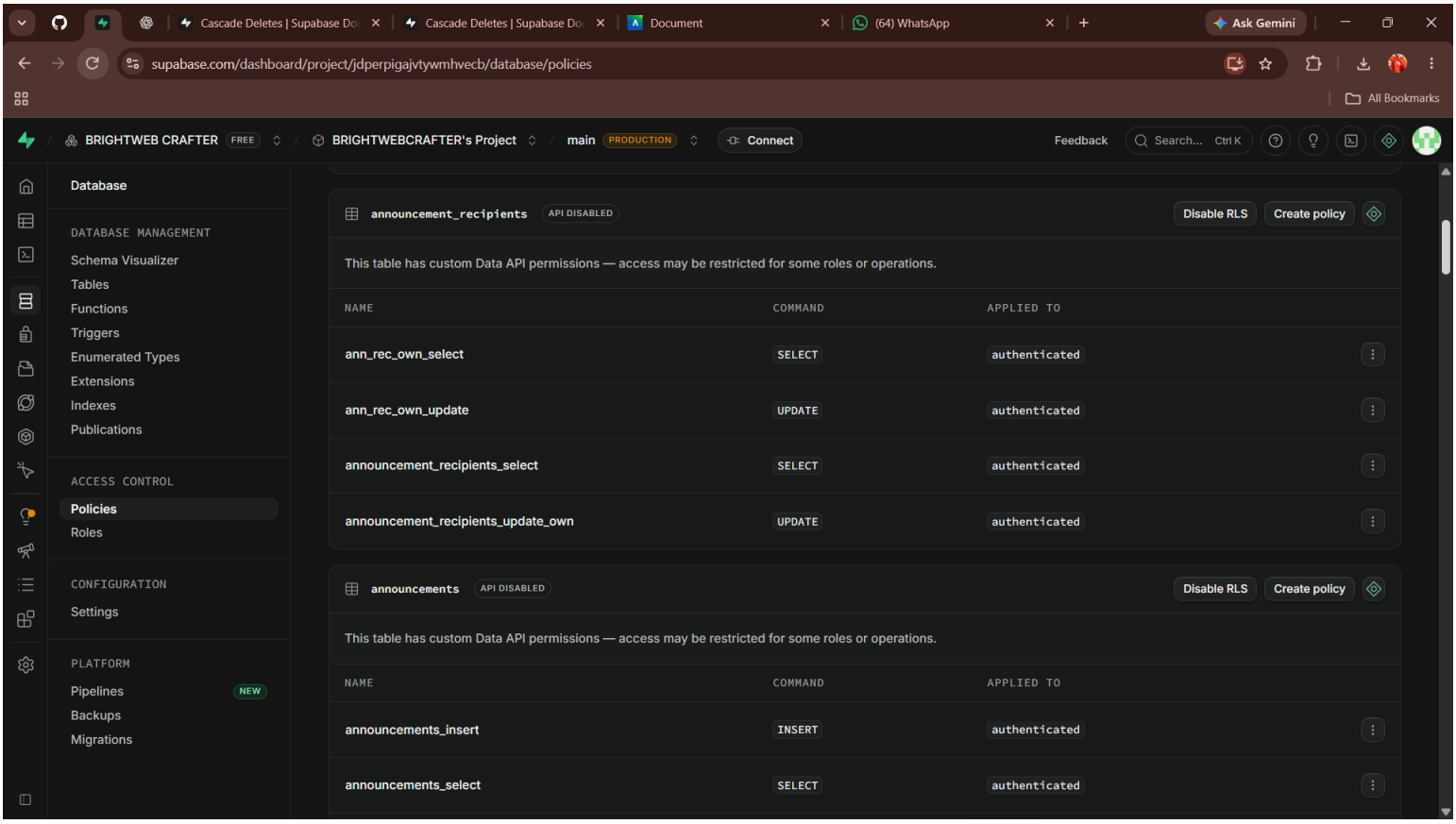
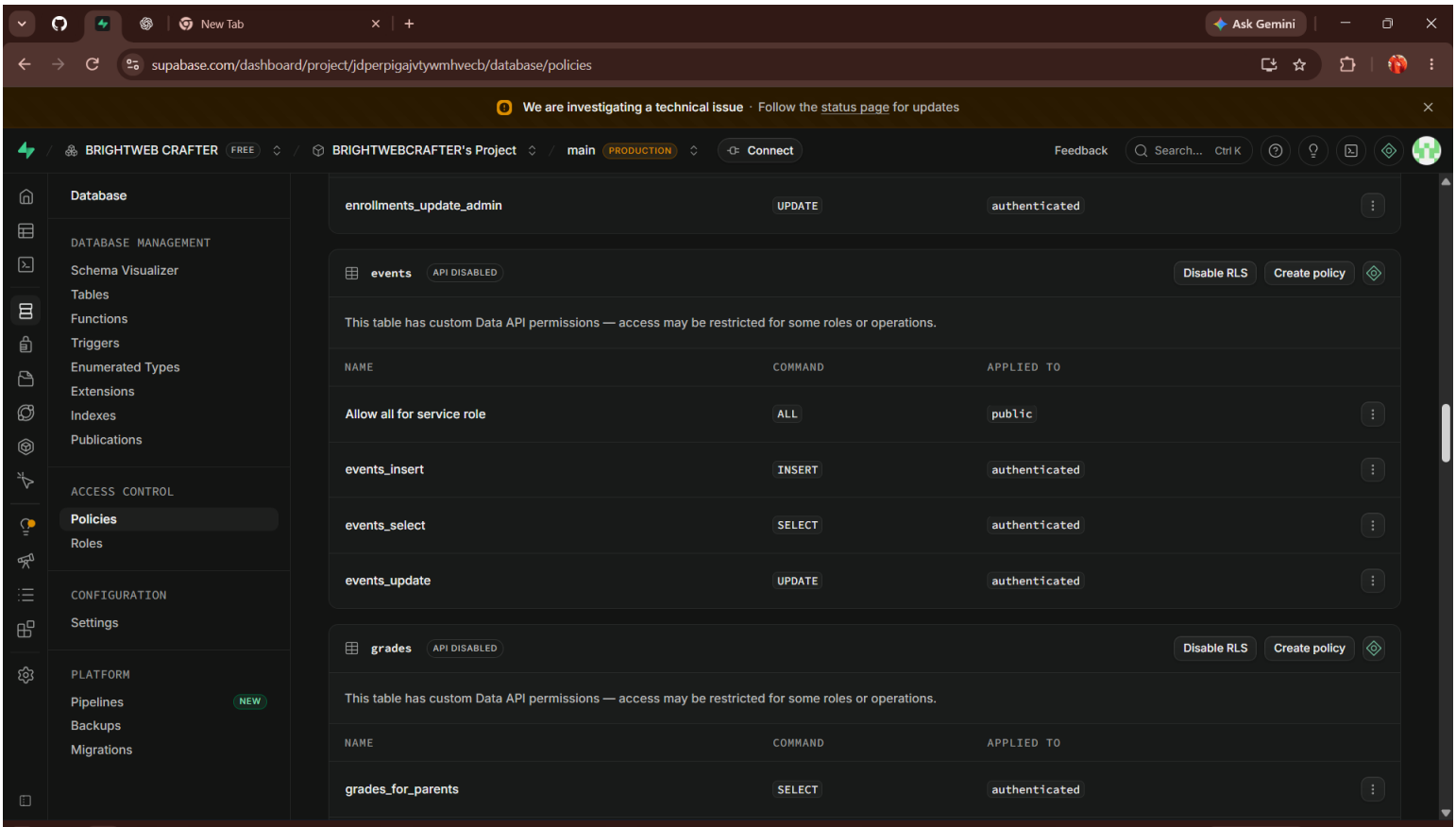
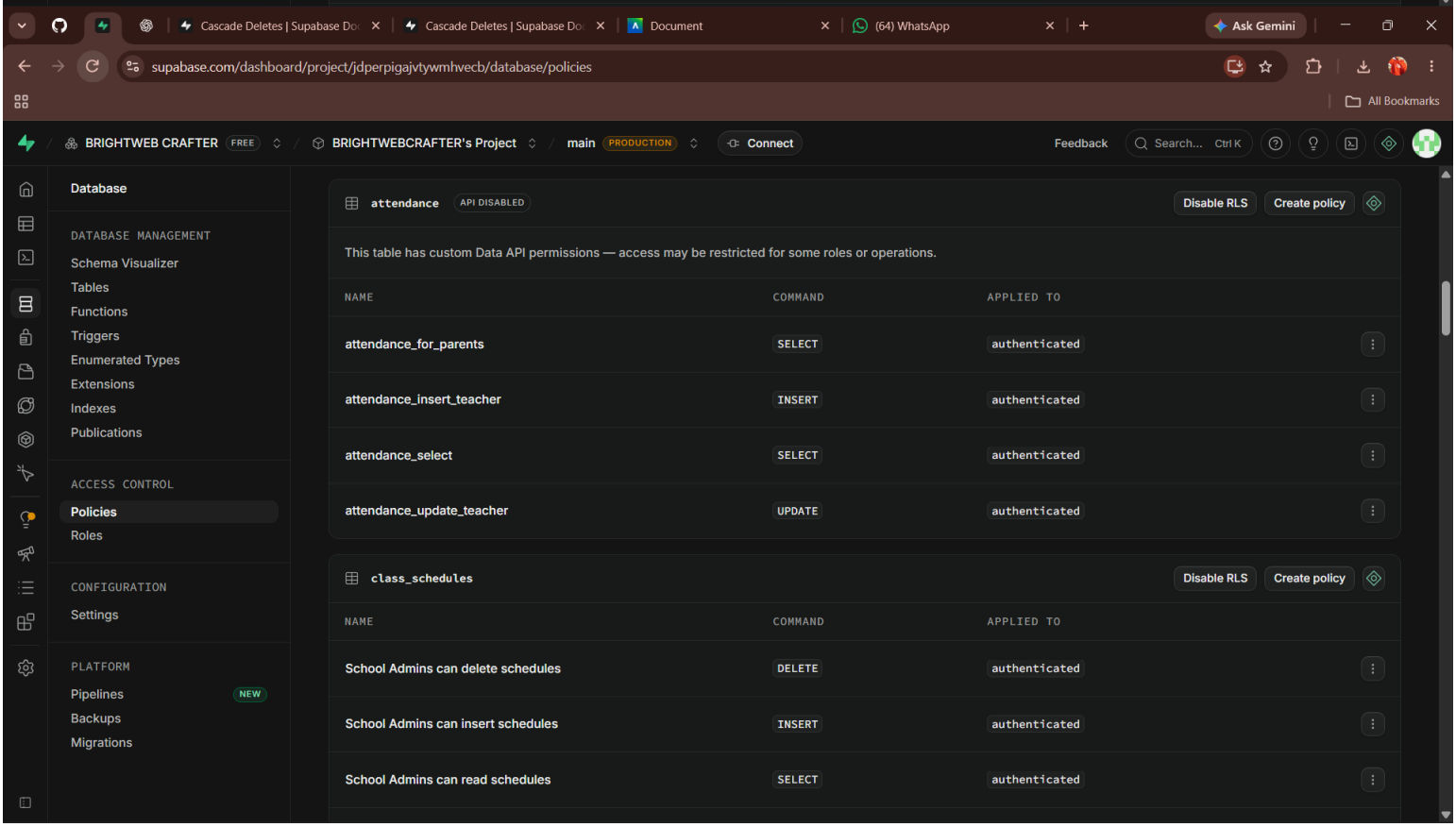


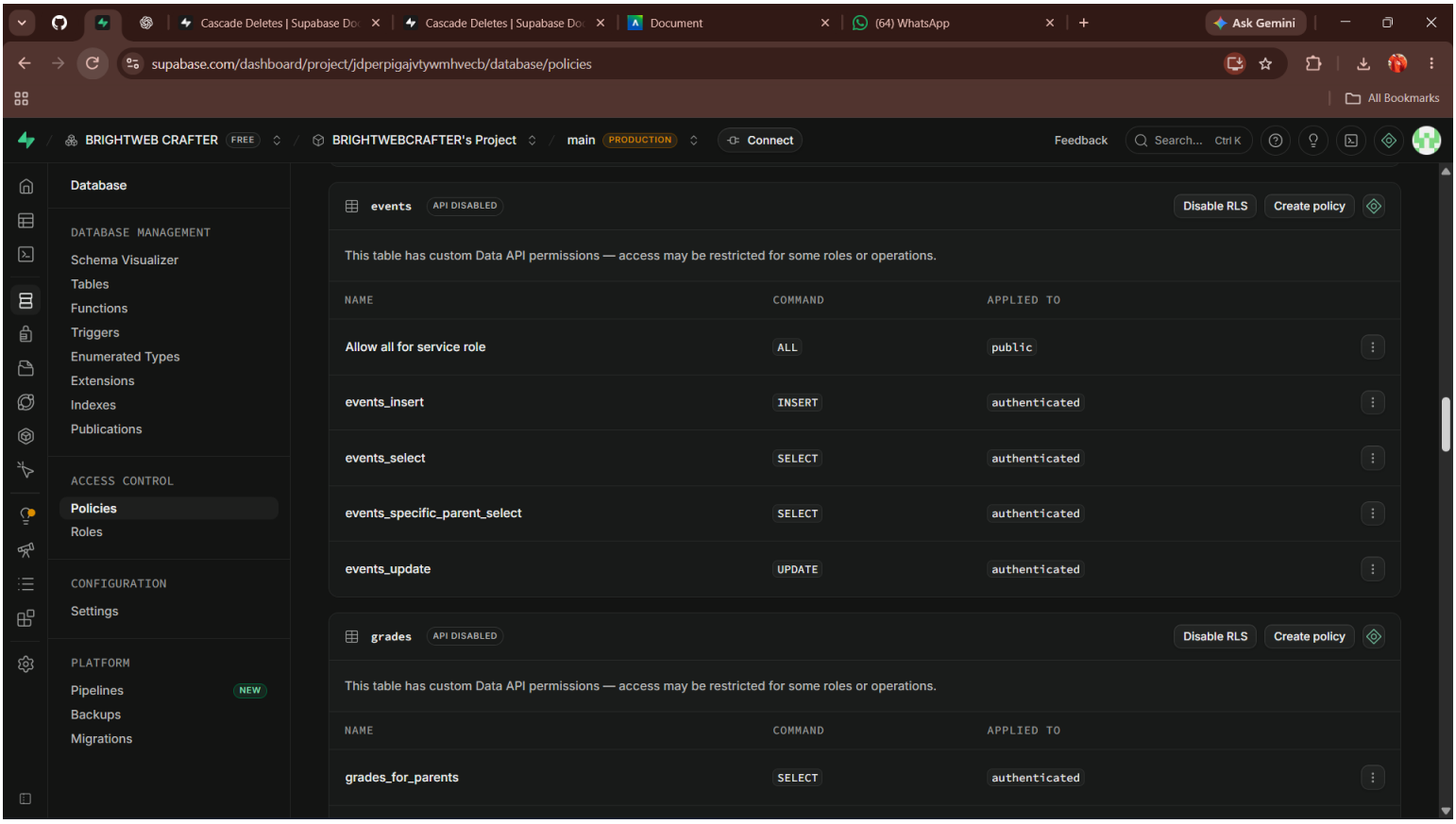
Figure 18.9: Row Level Security Policies

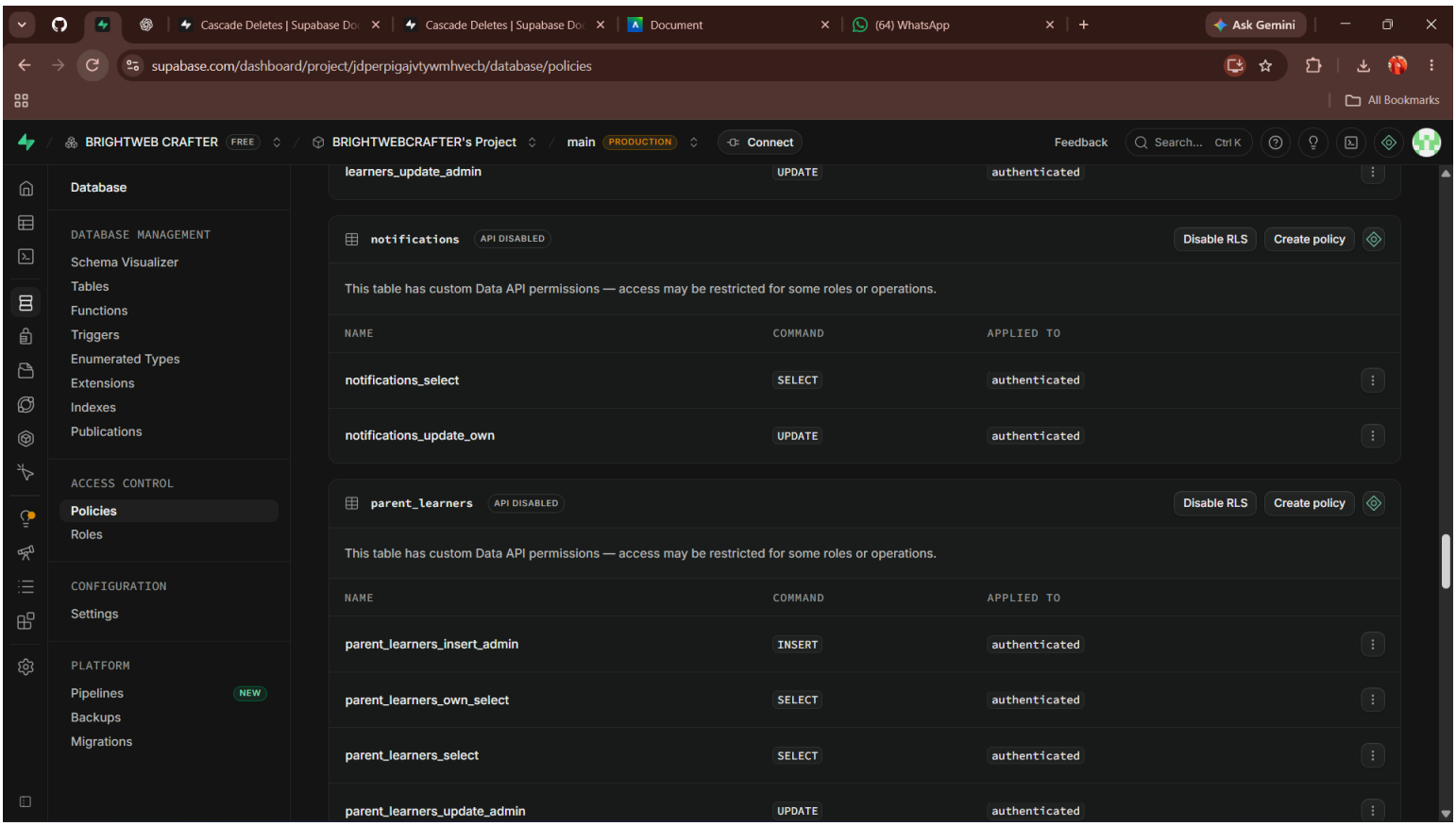


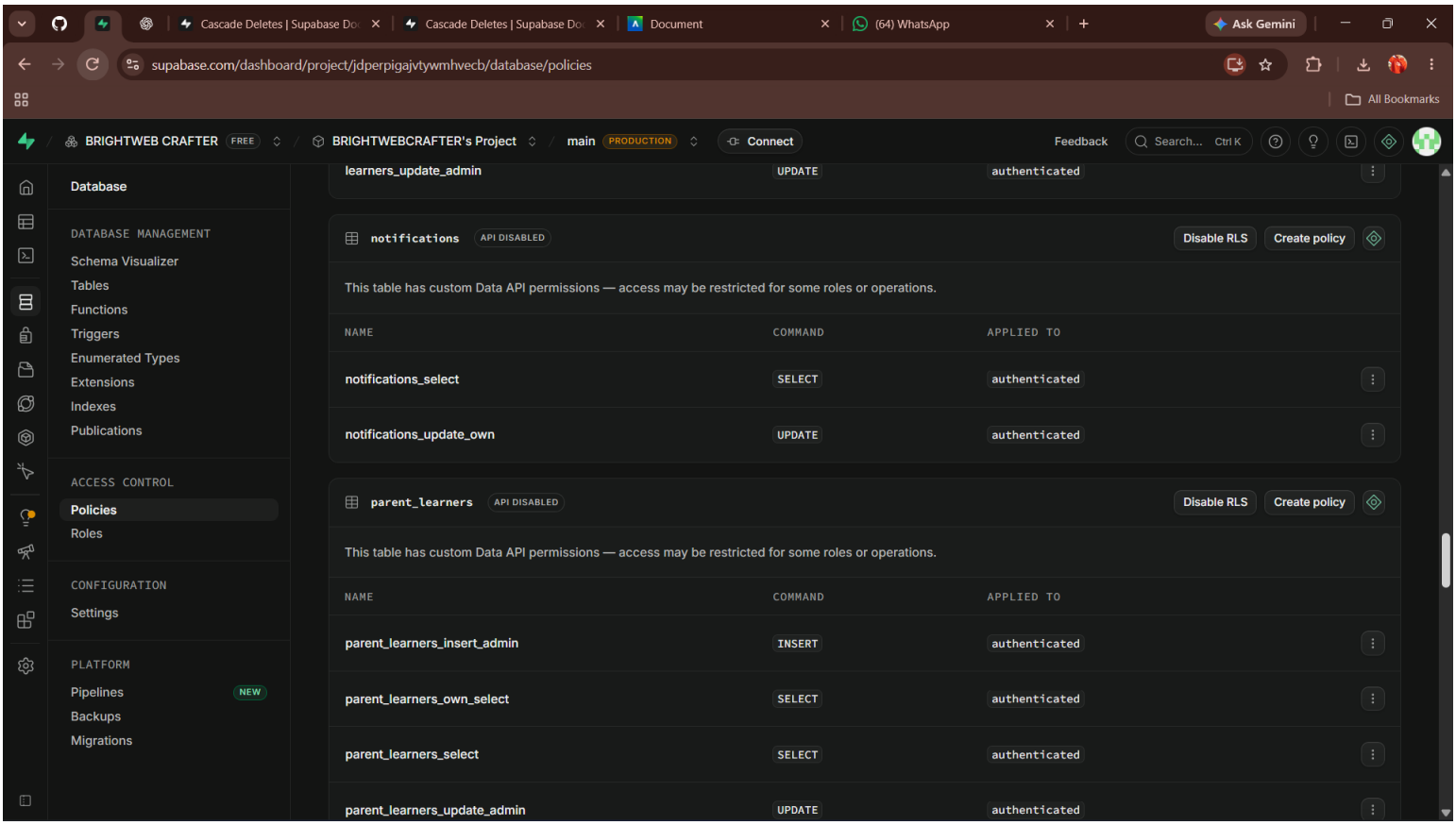


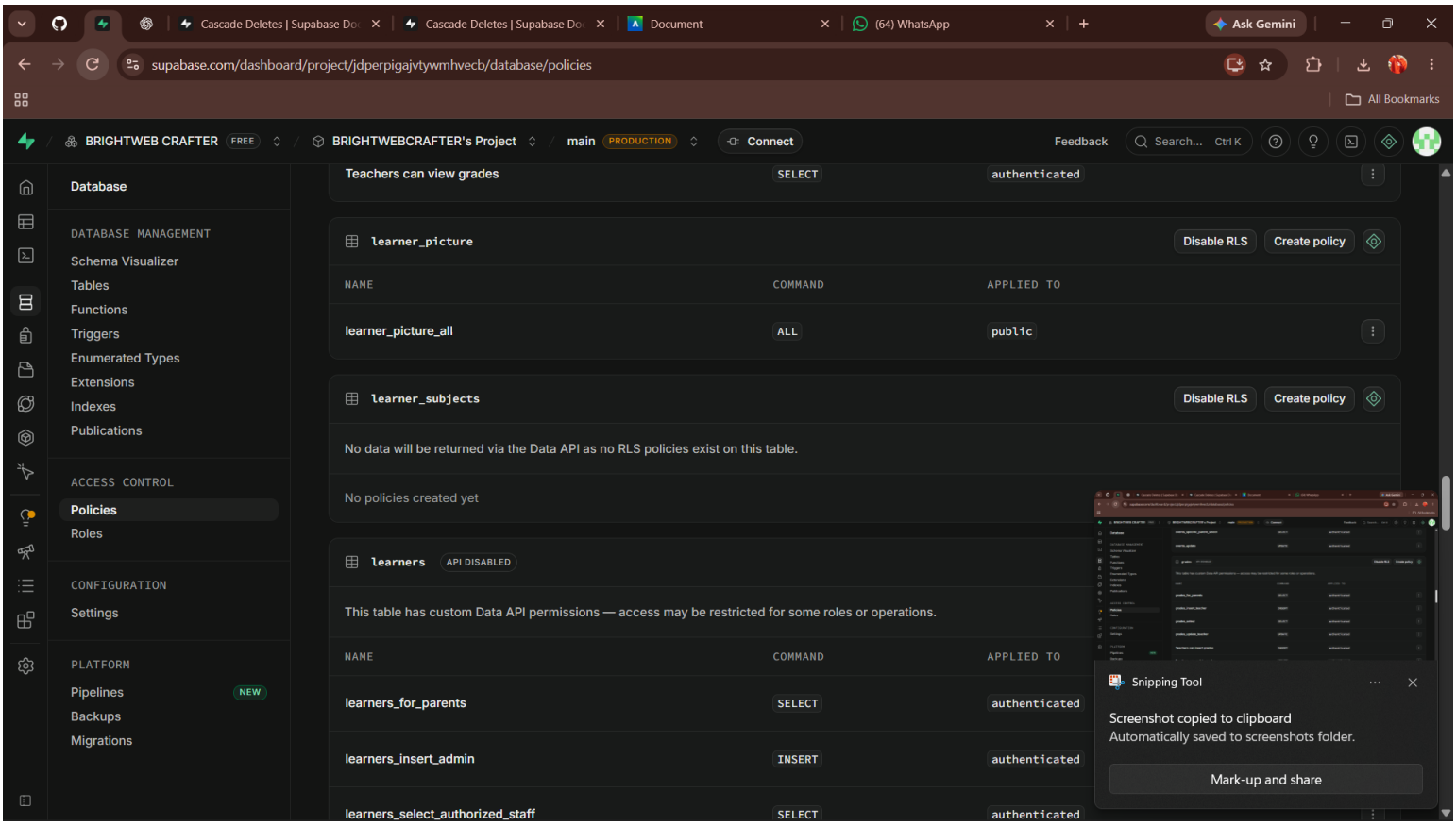

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enrollments_update_admin UPDATE authenticated
events API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME COMMAND APPLIED TO
Allow all for service role ALL public
events_insert INSERT authenticated
events_select SELECT authenticated
events_update UPDATE authenticated
grades API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME COMMAND APPLIED TO
grades_for_parents SELECT authenticated

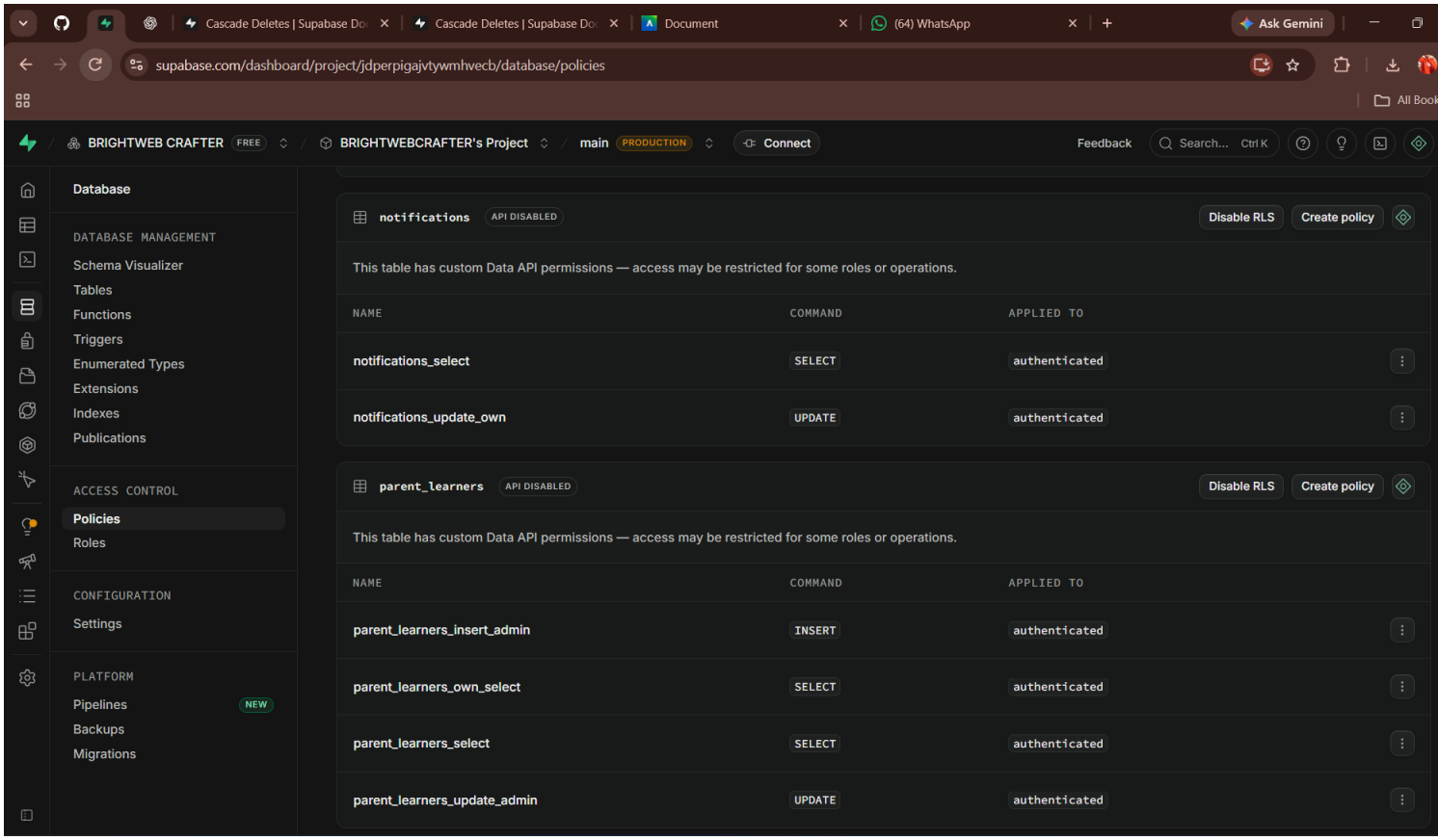

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attendance API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME COMMAND APPLIED TO
attendance_for_parents SELECT authenticated
attendance_insert_teacher INSERT authenticated
attendance_select SELECT authenticated
attendance_update_teacher UPDATE authenticated
class_schedules
Disable RLS
Create policy
NAME COMMAND APPLIED TO
School Admins can delete schedules DELETE authenticated
School Admins can insert schedules INSERT authenticated
School Admins can read schedules SELECT authenticated

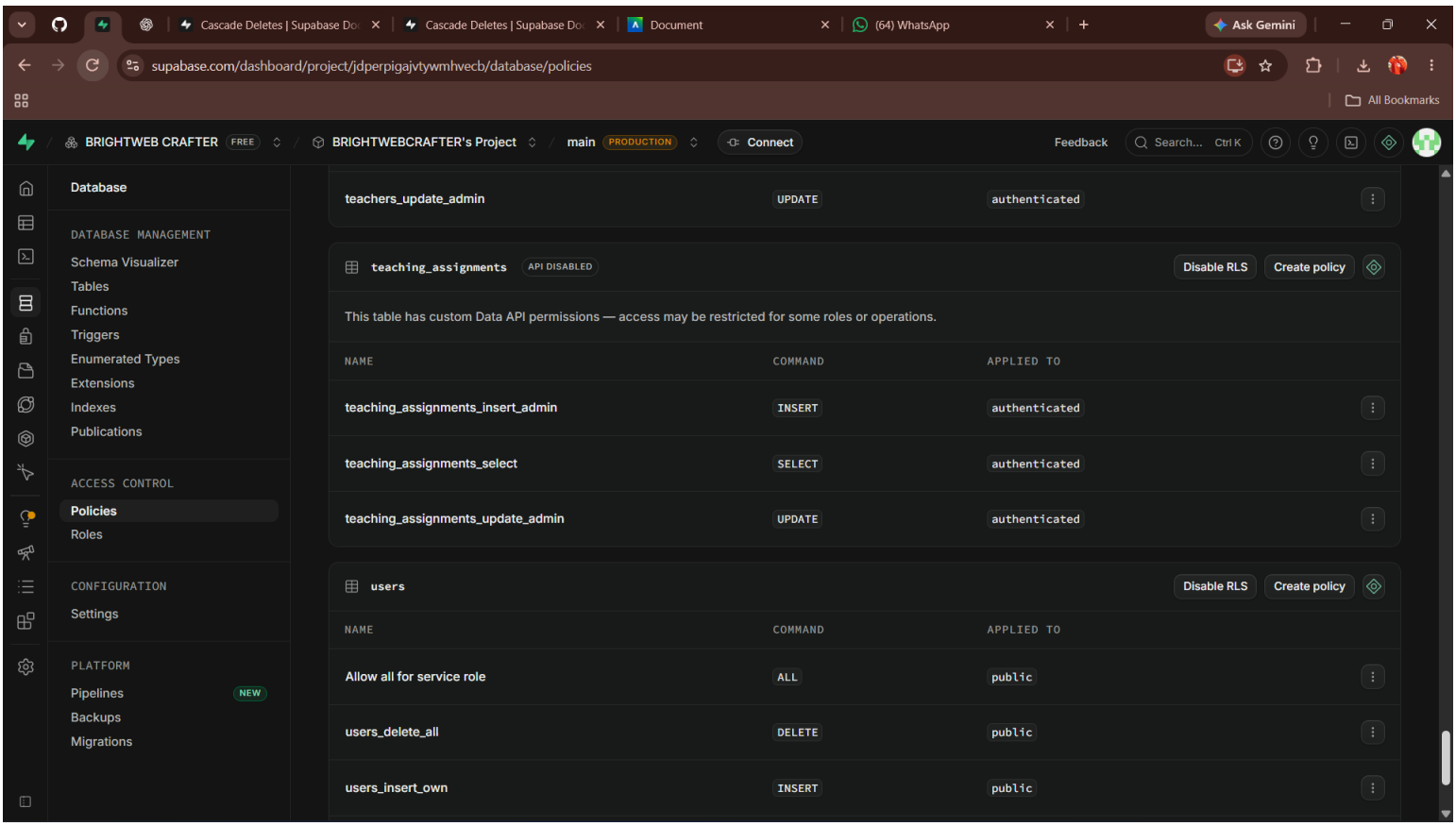

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events API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME
COMMAND
APPLIED TO
Allow all for service role
ALL
public
events_insert
INSERT
authenticated
events_select
SELECT
authenticated
events_specific_parent_select
SELECT
authenticated
events_update
UPDATE
authenticated
grades API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME
COMMAND
APPLIED TO
grades_for_parents
SELECT
authenticated


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learners_update_admin
UPDATE
authenticated
notifications API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME
COMMAND
APPLIED TO
notifications_select
SELECT
authenticated
notifications_update_own
UPDATE
authenticated
parent_learners API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME
COMMAND
APPLIED TO
parent_learners_insert_admin
INSERT
authenticated
parent_learners_own_select
SELECT
authenticated
parent_learners_select
SELECT
authenticated
parent_learners_update_admin
UPDATE
authenticated


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learners_update_admin UPDATE authenticated
notifications API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME
COMMAND
APPLIED TO
notifications_select SELECT authenticated
notifications_update_own UPDATE authenticated
parent_learners API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME
COMMAND
APPLIED TO
parent_learners_insert_admin INSERT authenticated
parent_learners_own_select SELECT authenticated
parent_learners_select SELECT authenticated
parent_learners_update_admin UPDATE authenticated

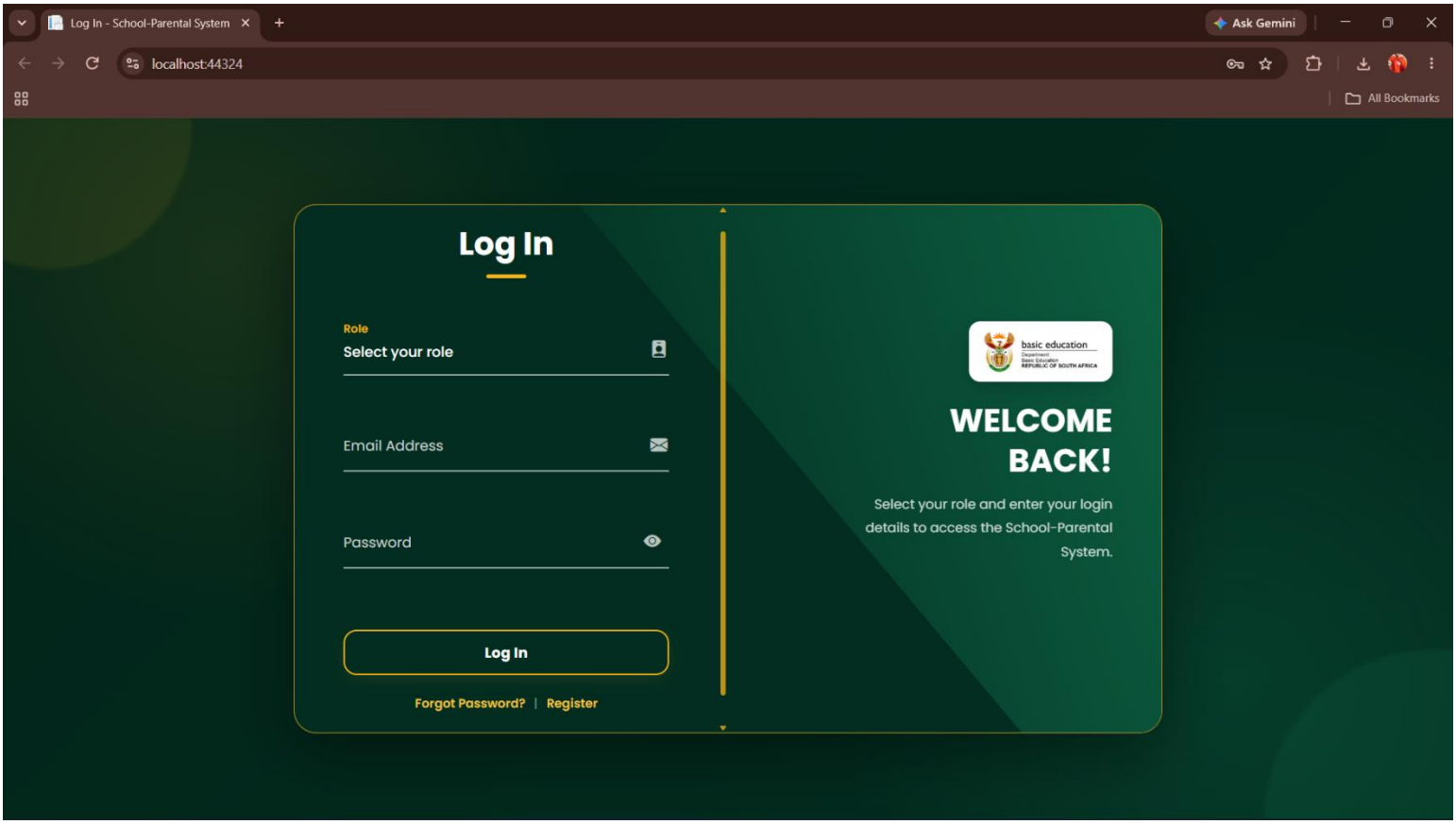

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Teachers can view grades SELECT authenticated
learner_picture
Disable RLS
Create policy
NAME
COMMAND
APPLIED TO
learner_picture_all ALL public
learner_subjects
Disable RLS
Create policy
No data will be returned via the Data API as no RLS policies exist on this table.
No policies created yet
learners API DISABLED
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME
COMMAND
APPLIED TO
learners_for_parents SELECT authenticated
learners_insert_admin INSERT authenticated
learners_select_authorized_staff SELECT authenticated
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share

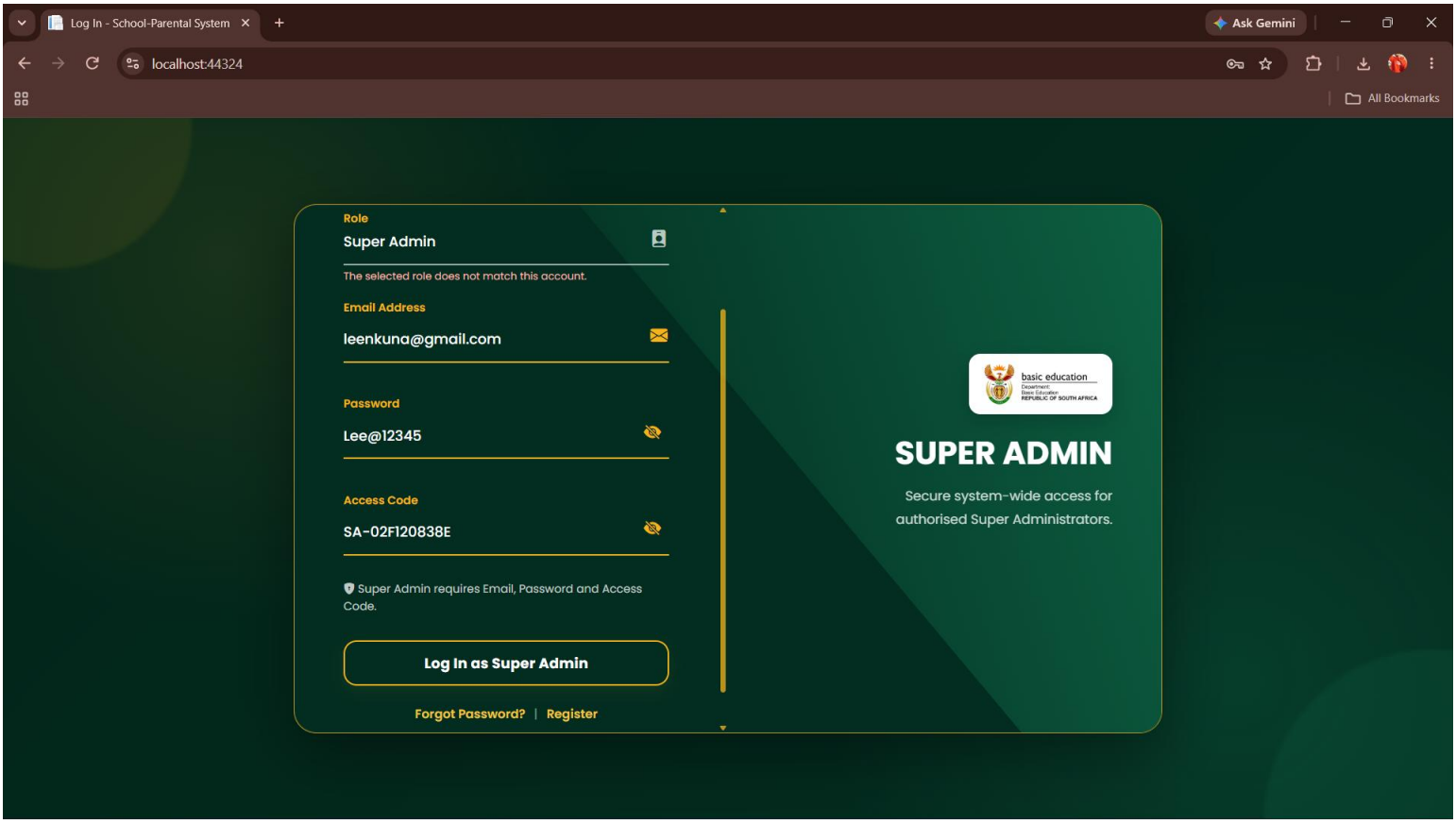

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notifications API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME
COMMAND
APPLIED TO
notifications_select
SELECT
authenticated
notifications_update_own
UPDATE
authenticated
parent_learners API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME
COMMAND
APPLIED TO
parent_learners_insert_admin
INSERT
authenticated
parent_learners_own_select
SELECT
authenticated
parent_learners_select
SELECT
authenticated
parent_learners_update_admin
UPDATE
authenticated

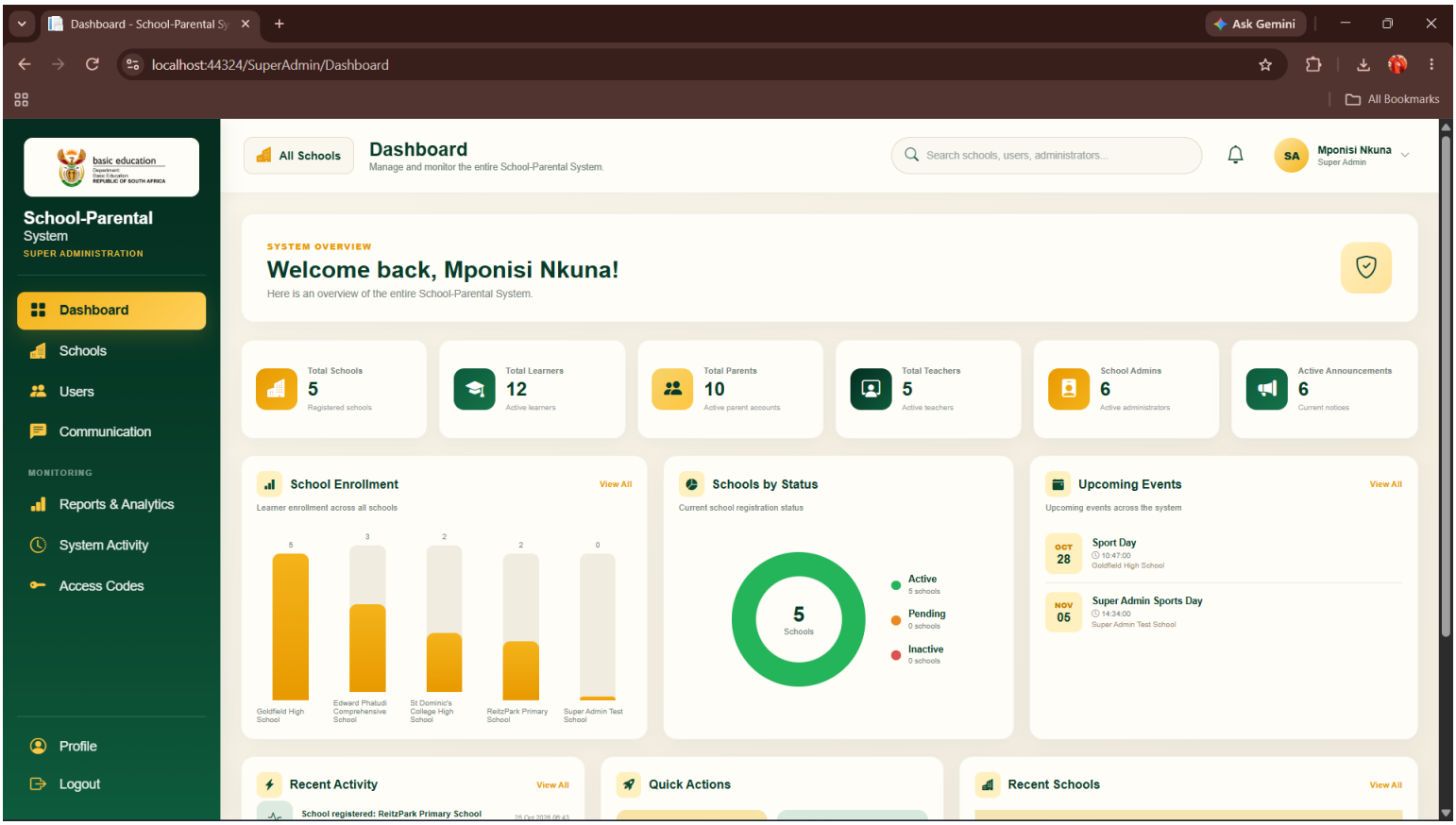

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teachers_update_admin
UPDATE
authenticated
teaching_assignments API DISABLED
Disable RLS
Create policy
This table has custom Data API permissions — access may be restricted for some roles or operations.
NAME
COMMAND
APPLIED TO
teaching_assignments_insert_admin
INSERT
authenticated
teaching_assignments_select
SELECT
authenticated
teaching_assignments_update_admin
UPDATE
authenticated
users
Disable RLS
Create policy
NAME
COMMAND
APPLIED TO
Allow all for service role
ALL
public
users_delete_all
DELETE
public
users_insert_own
INSERT
public

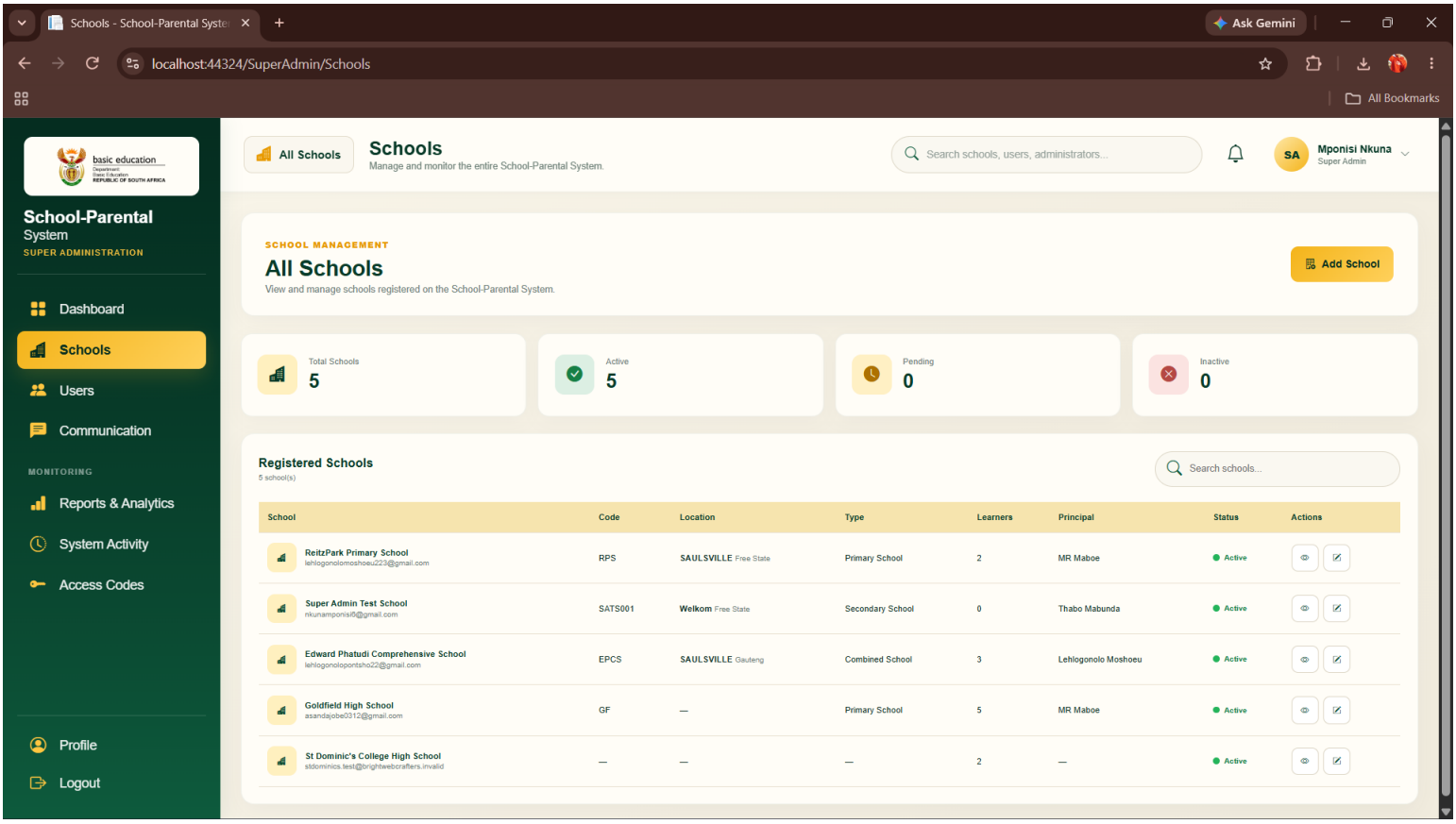
18.3.3 ASP.NET Core Web Application

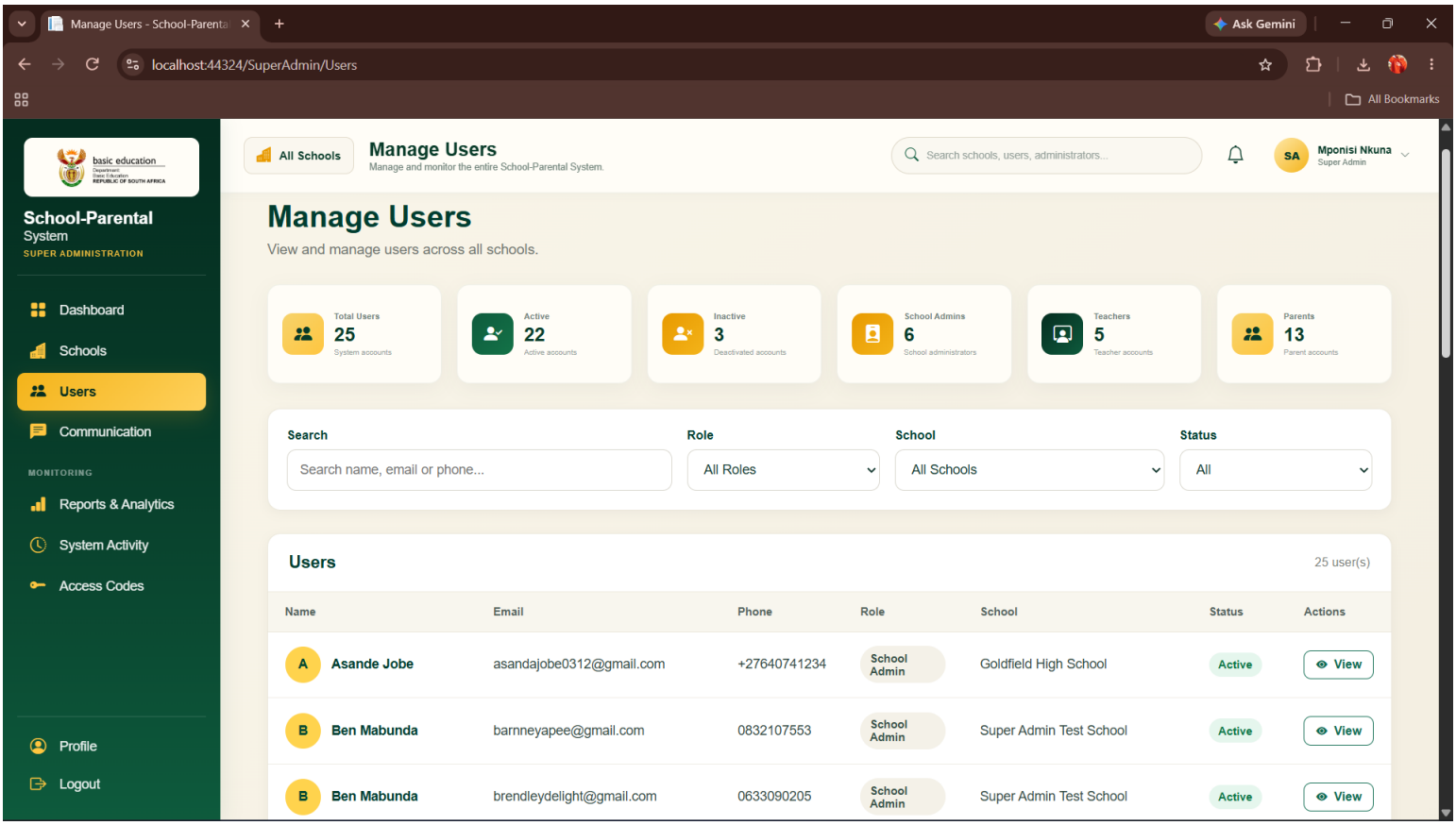
18.3.1 Super ADMIN

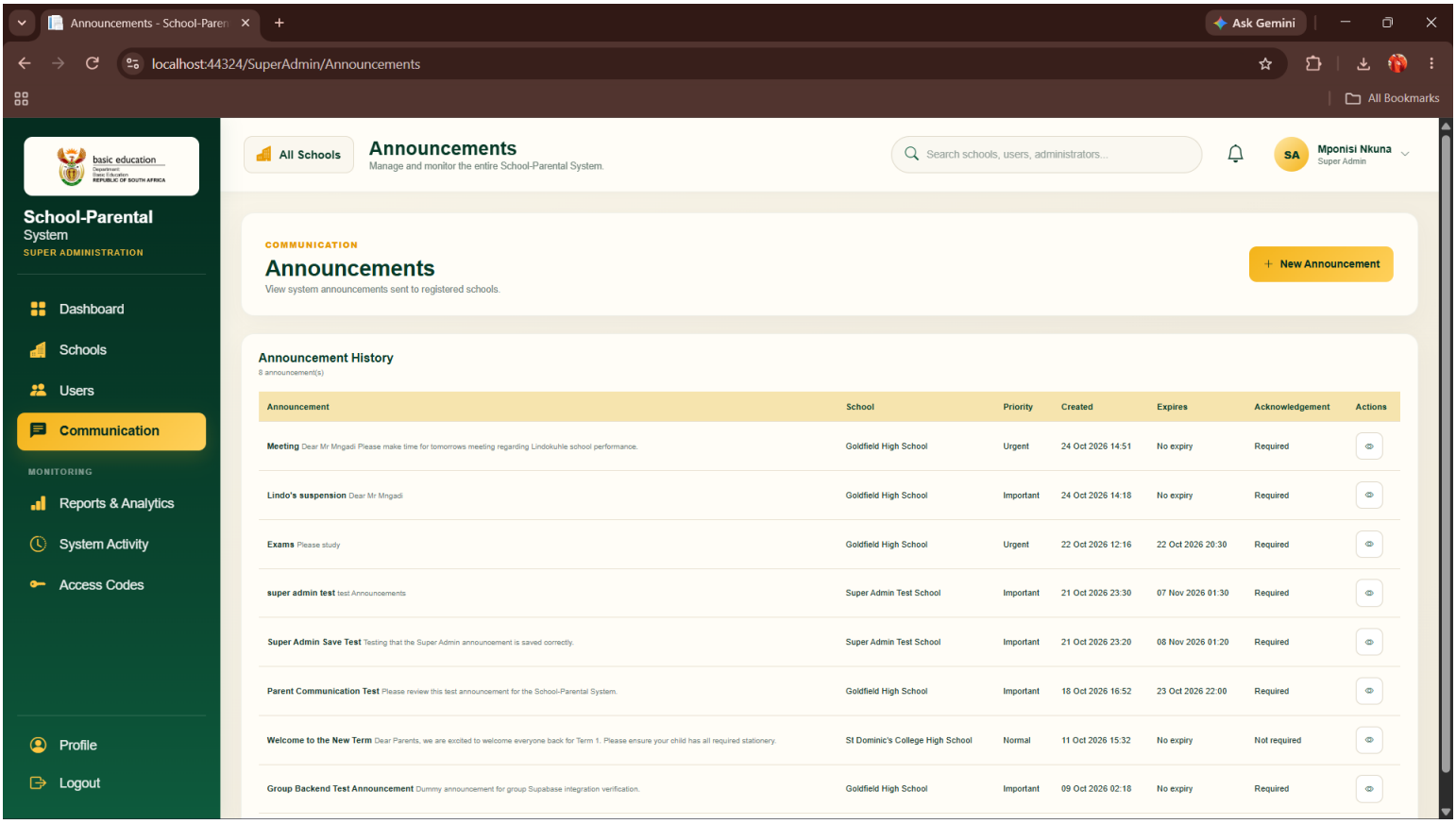


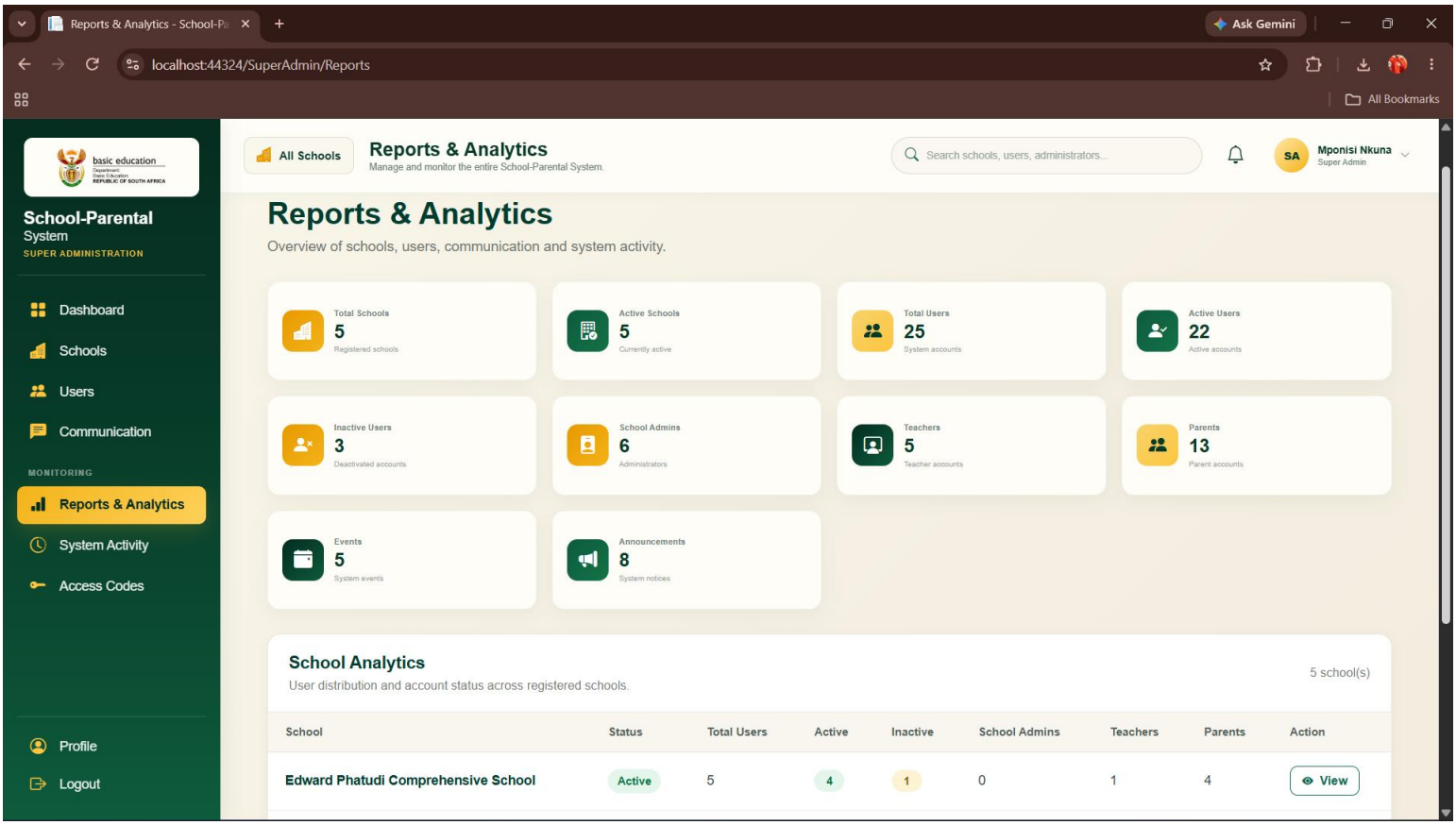


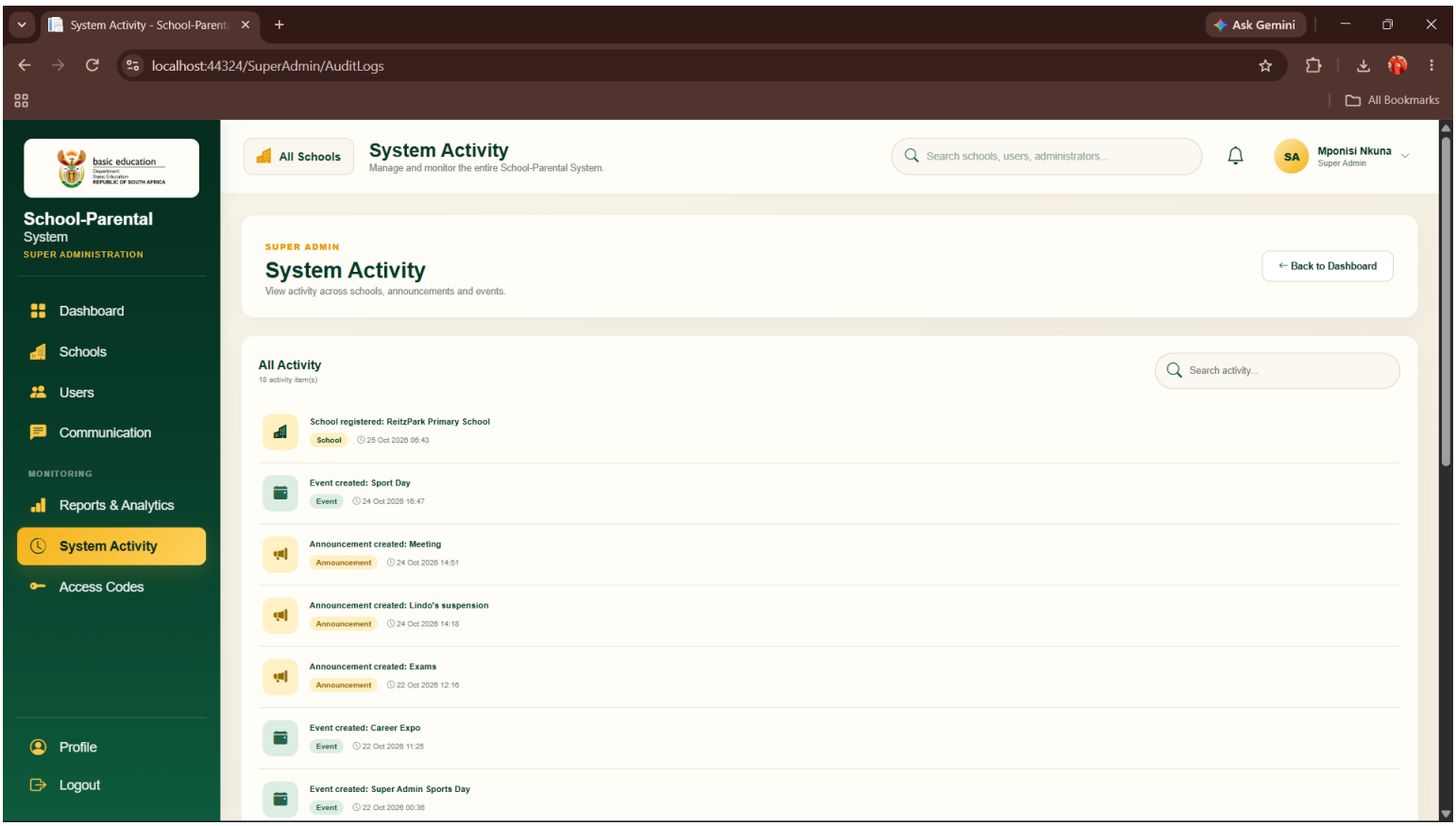

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MONITORING
Reports & Analytics
System Activity
Access Codes
Profile
Logout
All Schools
Dashboard
Manage and monitor the entire School-Parental System.
Search schools, users, administrators...
SA
Mponisi Nkuna
Super Admin
SYSTEM OVERVIEW
Welcome back, Mponisi Nkuna!
Here is an overview of the entire School-Parental System.
Total Schools 5 Registered schools
Total Learners 12 Active learners
Total Parents 10 Active parent accounts
Total Teachers 5 Active teachers
School Admins 6 Active administrators
Active Announcements 6 Current notices
School Enrollment
View All
Learner enrollment across all schools
5
3
2
2
0
Goldfield High School
Edward Phatudi Comprehensive School
St Dominic's College High School
ReitzPark Primary School
Super Admin Test School
Schools by Status
Current school registration status
5
Schools
Active 5 schools
Pending 0 schools
Inactive 0 schools
Upcoming Events
View All
Upcoming events across the system
OCT 28
Sport Day
10:47:00
Goldfield High School
NOV 05
Super Admin Sports Day
14:34:00
Super Admin Test School
Recent Activity
View All
School registered: ReitzPark Primary School
Quick Actions
Recent Schools
View All

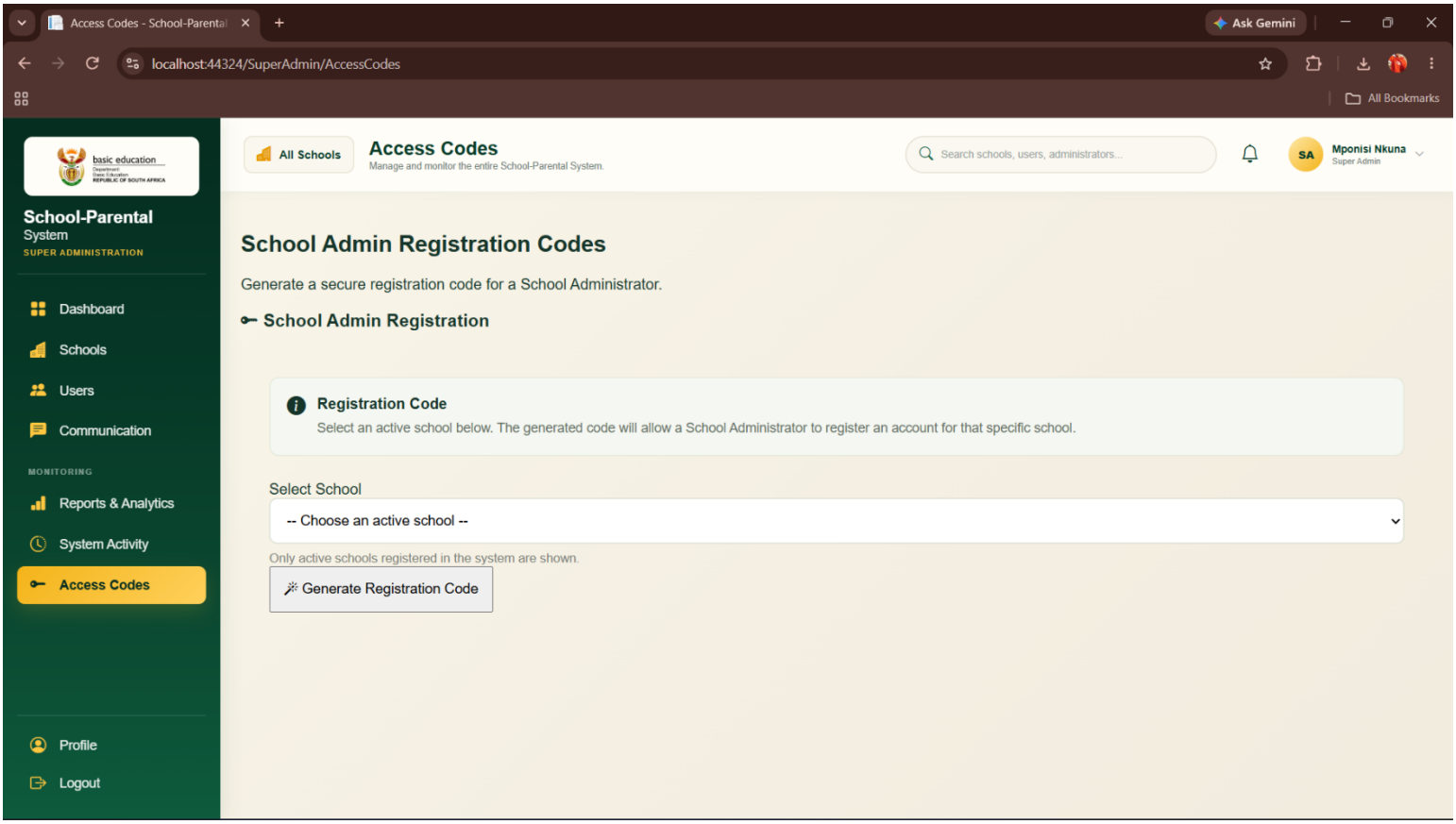

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School-Parental System
SUPER ADMINISTRATION
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Schools
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MONITORING
Reports & Analytics
System Activity
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Logout
All Schools
Schools
Manage and monitor the entire School-Parental System.
Search schools, users, administrators...
SA
Mponisi Nkuna
Super Admin
SCHOOL MANAGEMENT
All Schools
View and manage schools registered on the School-Parental System.
Add School
Total Schools 5
Active 5
Pending 0
Inactive 0
Registered Schools
5 school(s)
Search schools...
School
Code
Location
Type
Learners
Principal
Status
Actions
ReitzPark Primary School lehlogonolomoshoeu223@gmail.com RPS SAULSVILLE Free State Primary School 2 MR Maboe Active
Super Admin Test School nkunamponisi6@gmail.com SATS001 Welkom Free State Secondary School 0 Thabo Mabunda Active
Edward Phatudi Comprehensive School lehlogonolopontsho22@gmail.com EPCS SAULSVILLE Gauteng Combined School 3 Lehlogonolo Moshoeu Active
Goldfield High School asandajobe0312@gmail.com GF — Primary School 5 MR Maboe Active
St Dominic's College High School stdominics.test@brightwebcrafters.invalid — — — 2 — Active

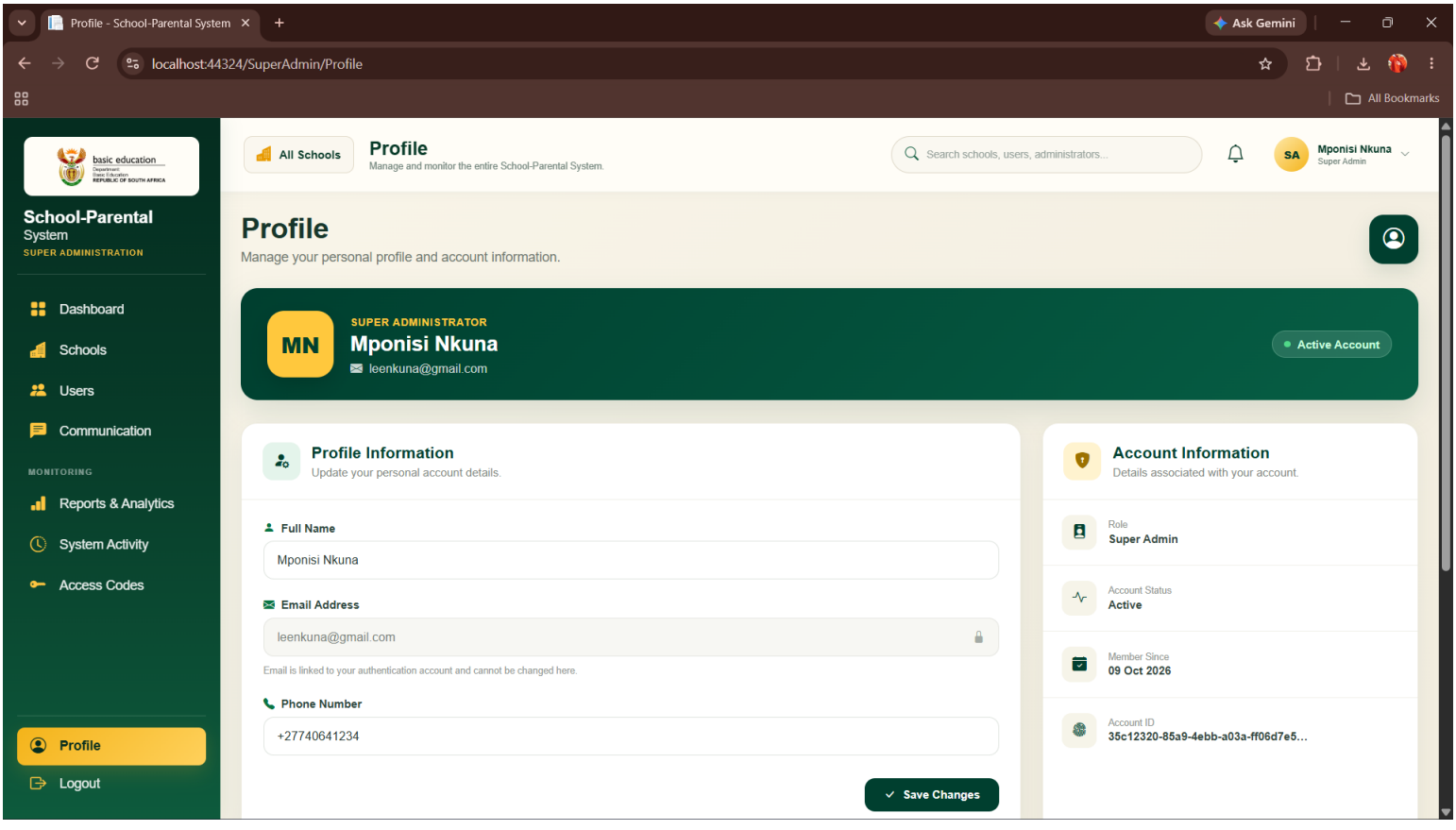

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School-Parental System
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Logout
All Schools
Manage Users
Manage and monitor the entire School-Parental System.
Search schools, users, administrators...
SA
Mponisi Nkuna
Super Admin
Manage Users
View and manage users across all schools.
Total Users 25 System accounts
Active 22 Active accounts
Inactive 3 Deactivated accounts
School Admins 6 School administrators
Teachers 5 Teacher accounts
Parents 13 Parent accounts
Search
Search name, email or phone...
Role
All Roles
School
All Schools
Status
All
Users
25 user(s)
Name
Email
Phone
Role
School
Status
Actions
A Asande Jobe asandajobe0312@gmail.com +27640741234 School Admin Goldfield High School Active View
B Ben Mabunda barnneyapee@gmail.com 0832107553 School Admin Super Admin Test School Active View
B Ben Mabunda brendleydelight@gmail.com 0633090205 School Admin Super Admin Test School Active View


Announcements - School-Paren
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School-Parental System
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Manage and monitor the entire School-Parental System.
Search schools, users, administrators...
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Mponisi Nkuna
Super Admin
COMMUNICATION
Announcements
View system announcements sent to registered schools.
New Announcement
Announcement History
8 announcement(s)
Announcement
School
Priority
Created
Expires
Acknowledgement
Actions
Meeting Dear Mr Mngadi Please make time for tomorrows meeting regarding Lindokuhle school performance. Goldfield High School Urgent 24 Oct 2026 14:51 No expiry Required
Lindo's suspension Dear Mr Mngadi Goldfield High School Important 24 Oct 2026 14:18 No expiry Required
Exams Please study Goldfield High School Urgent 22 Oct 2026 12:16 22 Oct 2026 20:30 Required
super admin test test Announcements Super Admin Test School Important 21 Oct 2026 23:30 07 Nov 2026 01:30 Required
Super Admin Save Test Testing that the Super Admin announcement is saved correctly. Super Admin Test School Important 21 Oct 2026 23:20 08 Nov 2026 01:20 Required
Parent Communication Test Please review this test announcement for the School-Parental System. Goldfield High School Important 18 Oct 2026 16:52 23 Oct 2026 22:00 Required
Welcome to the New Term Dear Parents, we are excited to welcome everyone back for Term 1. Please ensure your child has all required stationery. St Dominic's College High School Normal 11 Oct 2026 15:32 No expiry Not required
Group Backend Test Announcement Dummy announcement for group Supabase integration verification. Goldfield High School Important 09 Oct 2026 02:18 No expiry Required

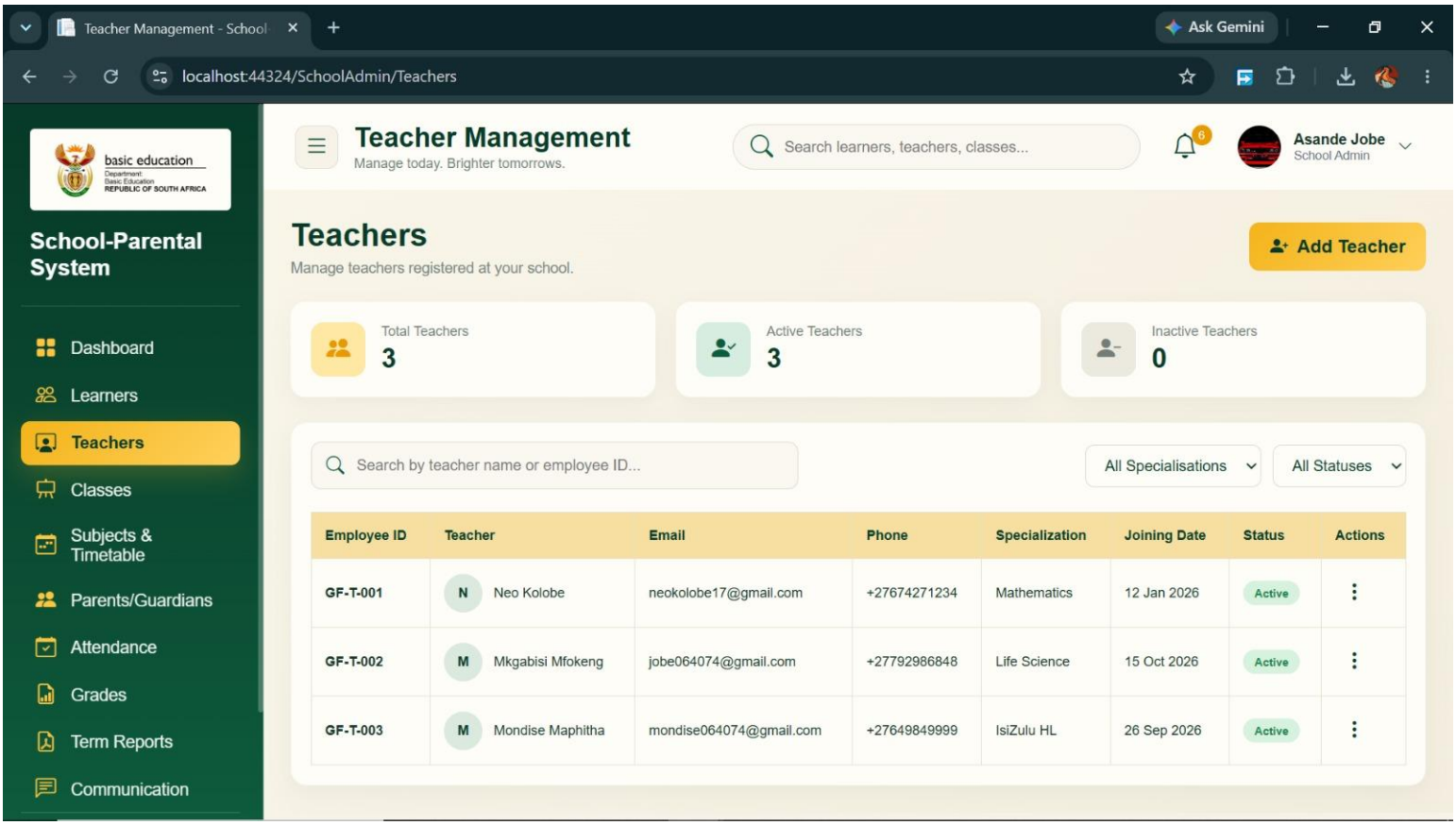

Reports & Analytics - School-Pa
Ask Gemini
localhost:44324/SuperAdmin/Reports
All Bookmarks
basic education
School-Parental System
SUPER ADMINISTRATION
Dashboard
Schools
Users
Communication
MONITORING
Reports & Analytics
System Activity
Access Codes
Profile
Logout
All Schools
Reports & Analytics
Manage and monitor the entire School-Parental System.
Search schools, users, administrators...
SA
Mponisi Nkuna
Super Admin
Reports & Analytics
Overview of schools, users, communication and system activity.
Total Schools 5 Registered schools
Active Schools 5 Currently active
Total Users 25 System accounts
Active Users 22 Active accounts
Inactive Users 3 Deactivated accounts
School Admins 6 Administrators
Teachers 5 Teacher accounts
Parents 13 Parent accounts
Events 5 System events
Announcements 8 System notices
School Analytics
User distribution and account status across registered schools.
5 school(s)
School
Status
Total Users
Active
Inactive
School Admins
Teachers
Parents
Action
Edward Phatudi Comprehensive School
Active
5
4
1
0
1
4
View

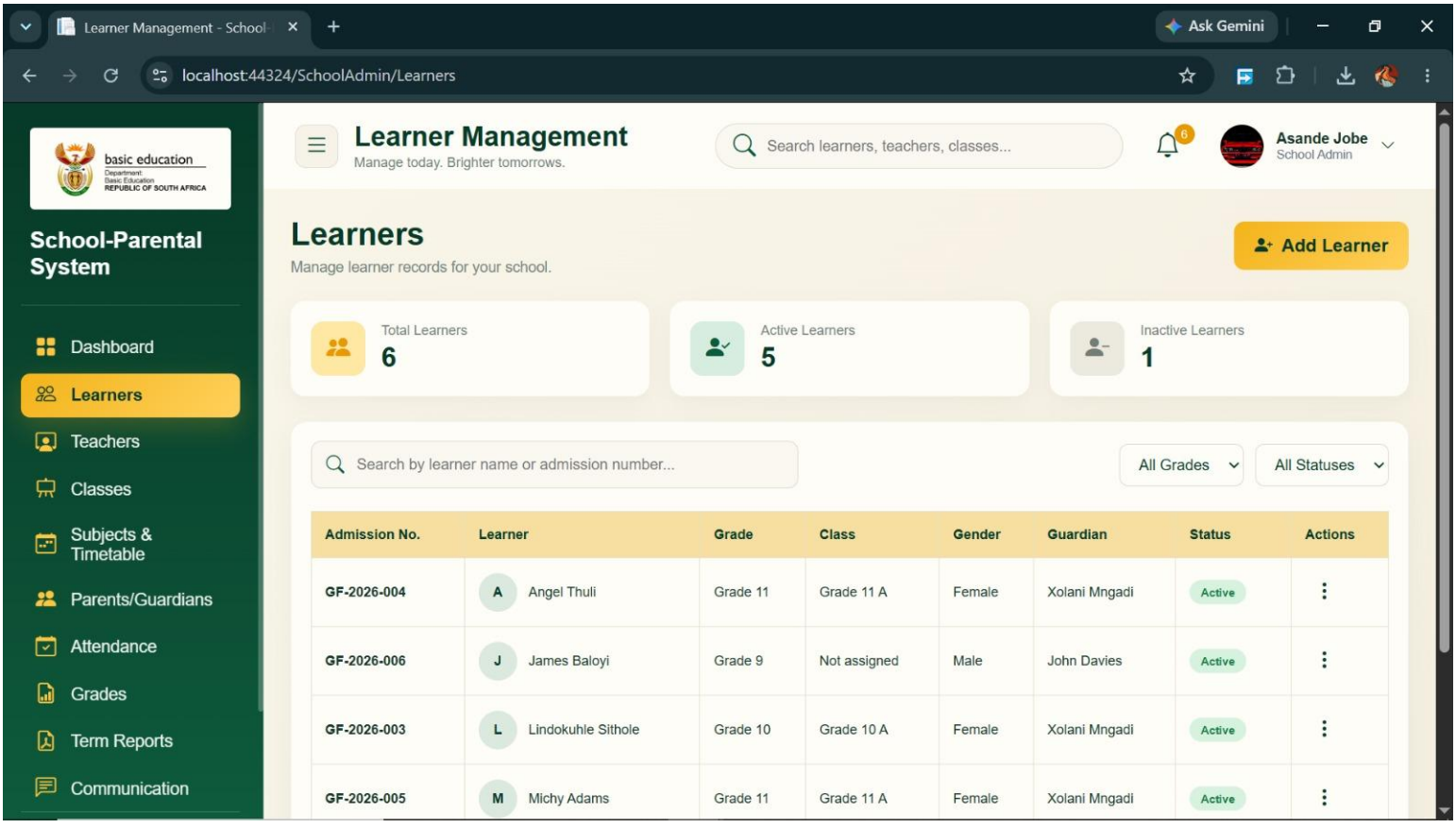

System Activity - School-Parent
Ask Gemini
localhost:44324/SuperAdmin/AuditLogs
All Bookmarks
basic education
School-Parental System
SUPER ADMINISTRATION
Dashboard
Schools
Users
Communication
MONITORING
Reports & Analytics
System Activity
Access Codes
Profile
Logout
All Schools
System Activity
Manage and monitor the entire School-Parental System.
Search schools, users, administrators...
SA
Mponisi Nkuna
Super Admin
SUPER ADMIN
System Activity
View activity across schools, announcements and events.
← Back to Dashboard
All Activity
18 activity item(s)
Search activity...
School registered: ReitzPark Primary School
School 25 Oct 2026 06:43
Event created: Sport Day
Event 24 Oct 2026 16:47
Announcement created: Meeting
Announcement 24 Oct 2026 14:51
Announcement created: Lindo's suspension
Announcement 24 Oct 2026 14:18
Announcement created: Exams
Announcement 22 Oct 2026 12:16
Event created: Career Expo
Event 22 Oct 2026 11:25
Event created: Super Admin Sports Day
Event 22 Oct 2026 00:36

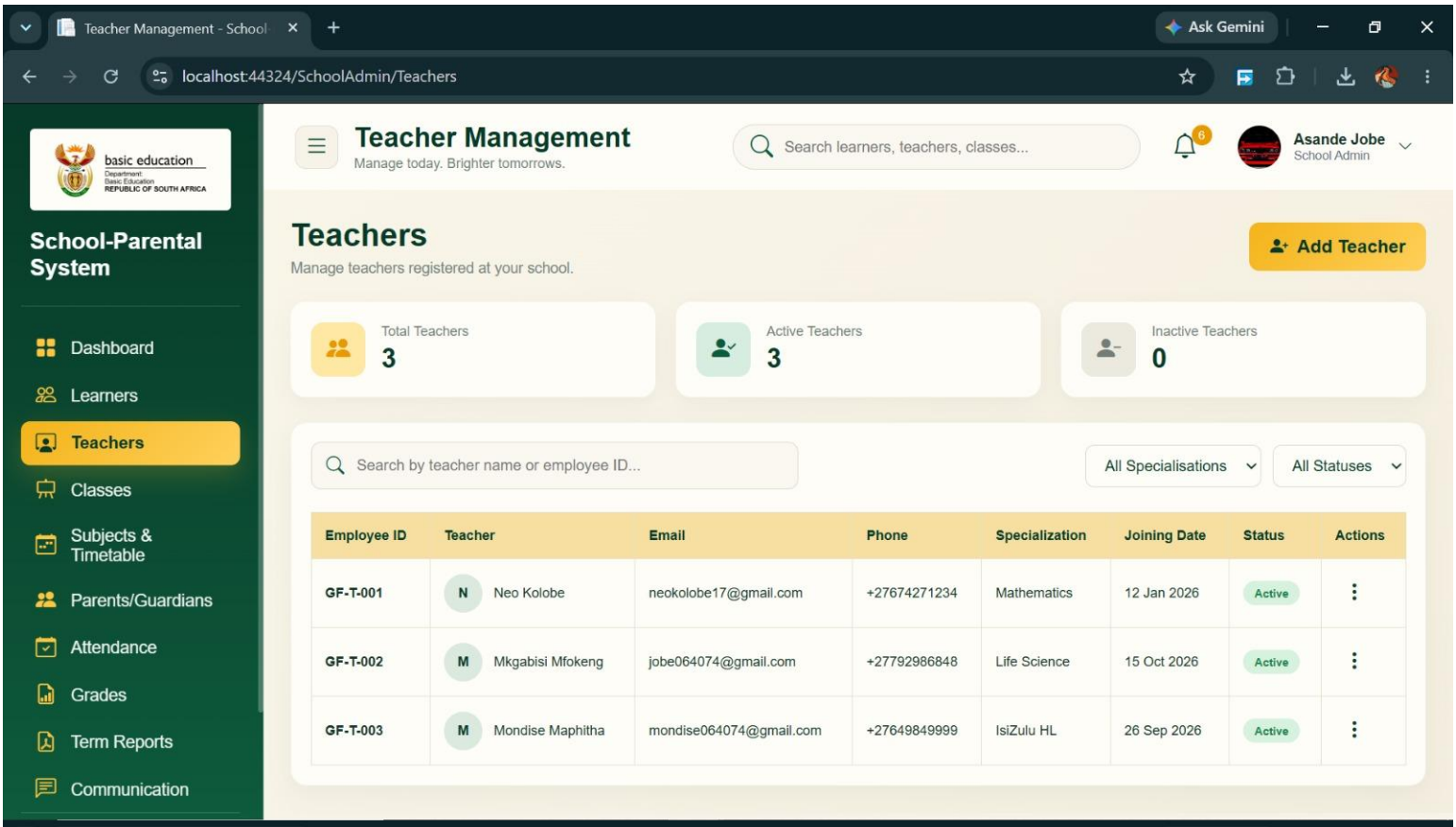

Access Codes - School-Parental
Ask Gemini
localhost:44324/SuperAdmin/AccessCodes
All Bookmarks
basic education
School-Parental System
SUPER ADMINISTRATION
Dashboard
Schools
Users
Communication
MONITORING
Reports & Analytics
System Activity
Access Codes
Profile
Logout
All Schools
Access Codes
Manage and monitor the entire School-Parental System.
Search schools, users, administrators...
SA
Mponisi Nkuna
Super Admin
School Admin Registration Codes
Generate a secure registration code for a School Administrator.
School Admin Registration
Registration Code
Select an active school below. The generated code will allow a School Administrator to register an account for that specific school.
Select School
-- Choose an active school --
Only active schools registered in the system are shown.
Generate Registration Code

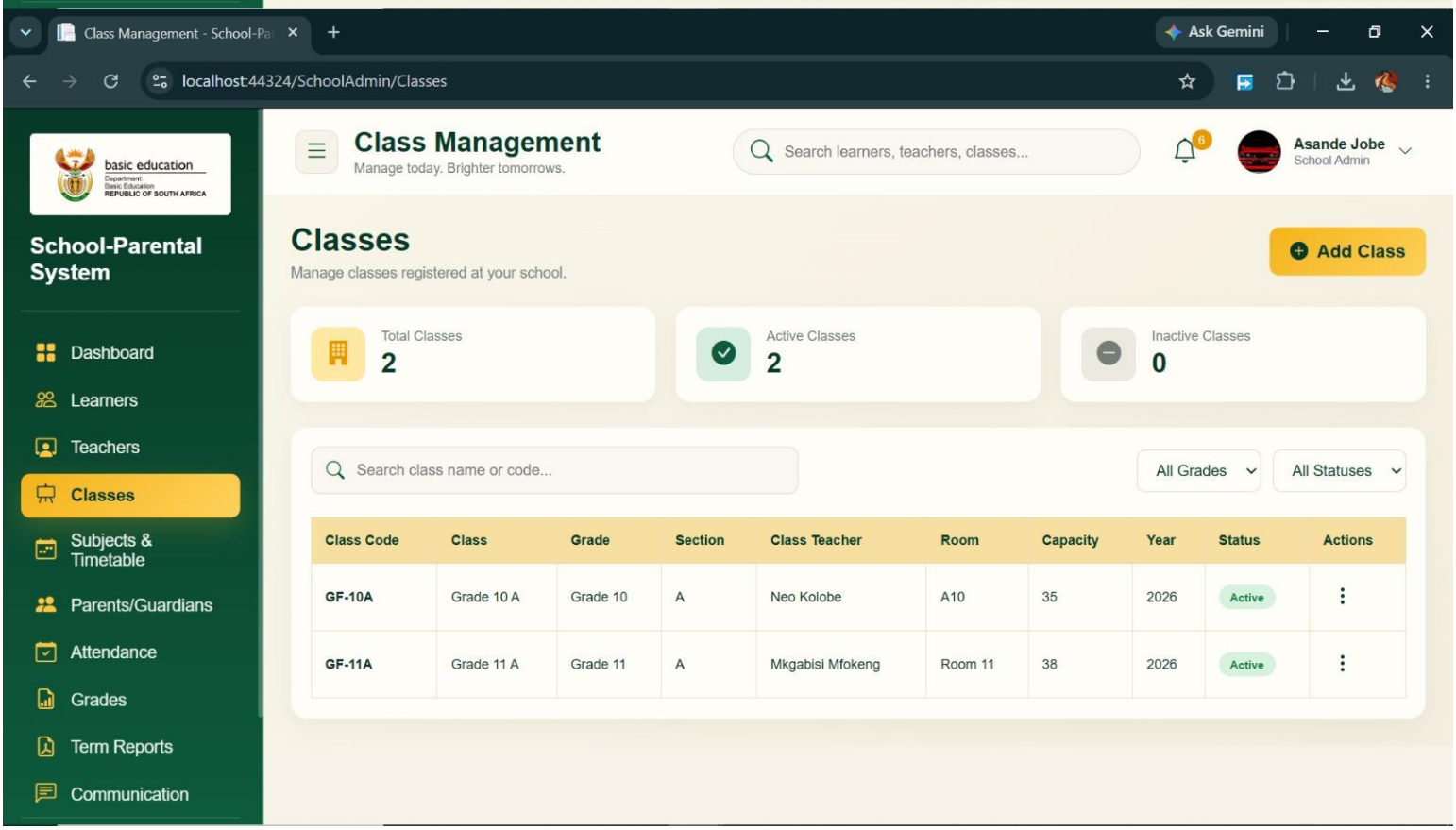

Profile - School-Parental System
Ask Gemini
localhost:44324/SuperAdmin/Profile
All Bookmarks
basic education
School-Parental System
SUPER ADMINISTRATION
Dashboard
Schools
Users
Communication
MONITORING
Reports & Analytics
System Activity
Access Codes
Profile
Logout
All Schools
Profile
Manage and monitor the entire School-Parental System.
Search schools, users, administrators...
SA
Mponisi Nkuna
Super Admin
Profile
Manage your personal profile and account information.
MN
SUPER ADMINISTRATOR
Mponisi Nkuna
leenkuna@gmail.com
Active Account
Profile Information
Update your personal account details.
Full Name
Mponisi Nkuna
Email Address
leenkuna@gmail.com
Email is linked to your authentication account and cannot be changed here.
Phone Number
+27740641234
Save Changes
Account Information
Details associated with your account.
Role
Super Admin
Account Status
Active
Member Since
09 Oct 2026
Account ID
35c12320-85a9-4ebb-a03a-ff06d7e5...

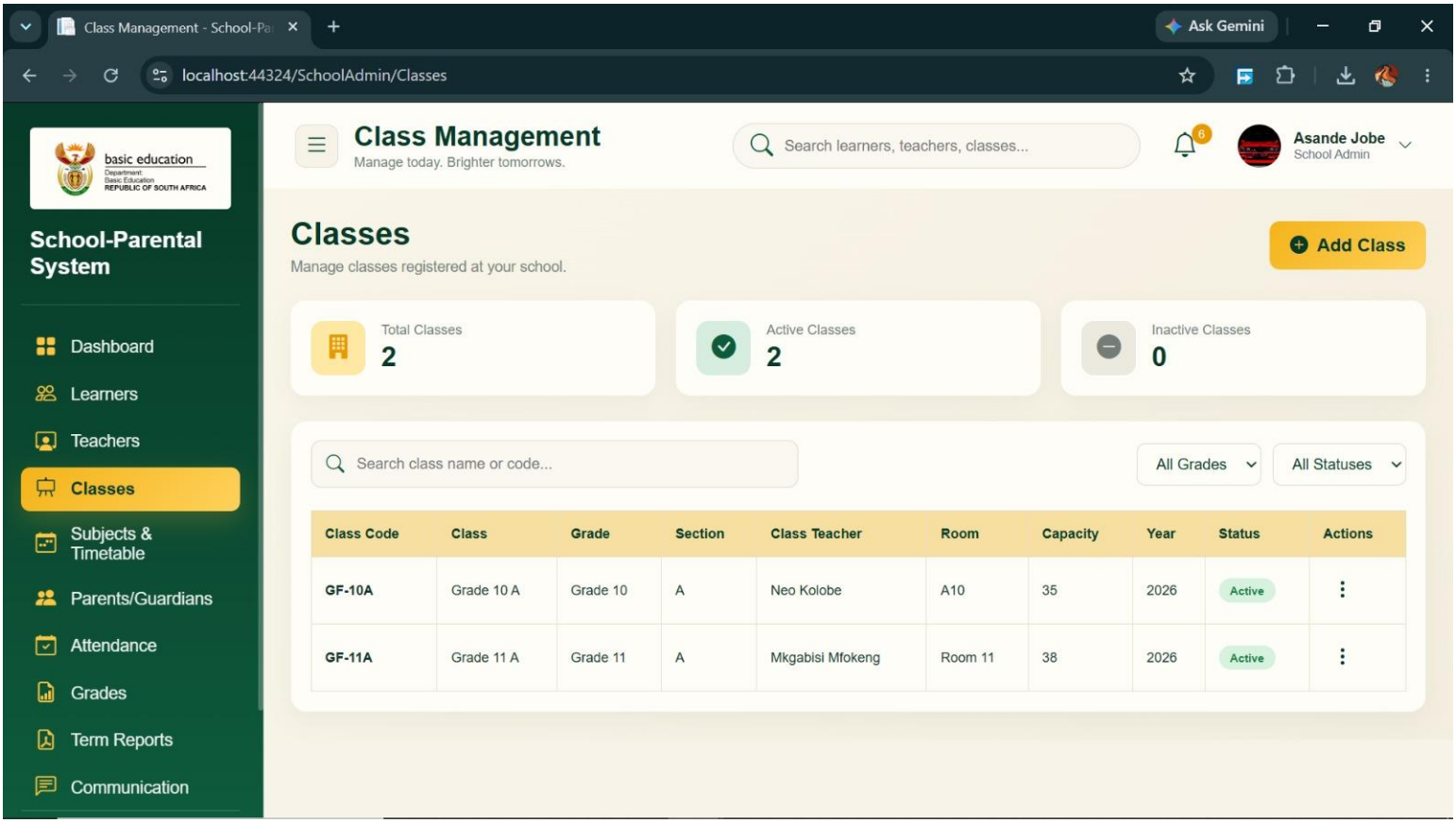
18.3.2 School Admin

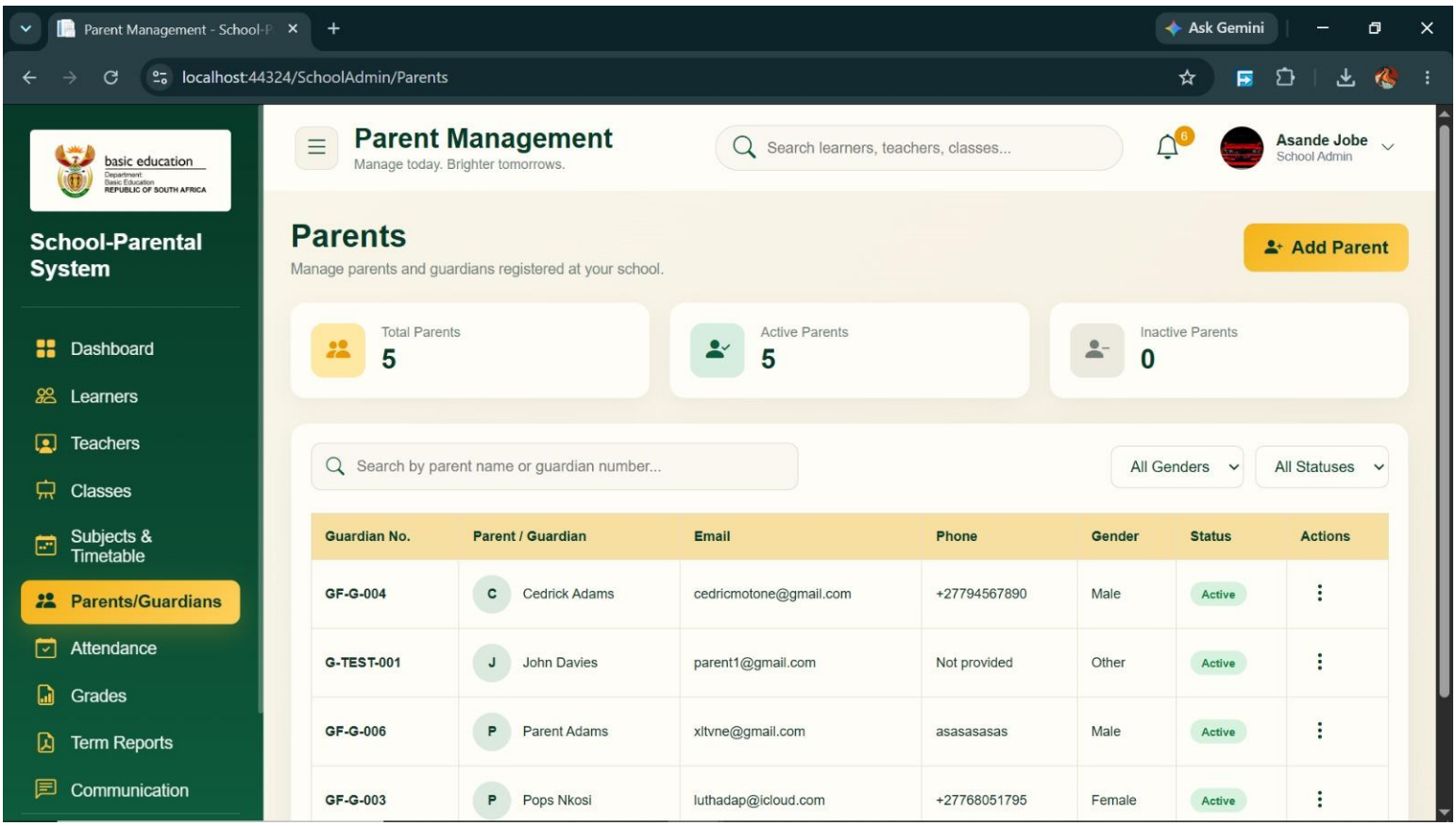


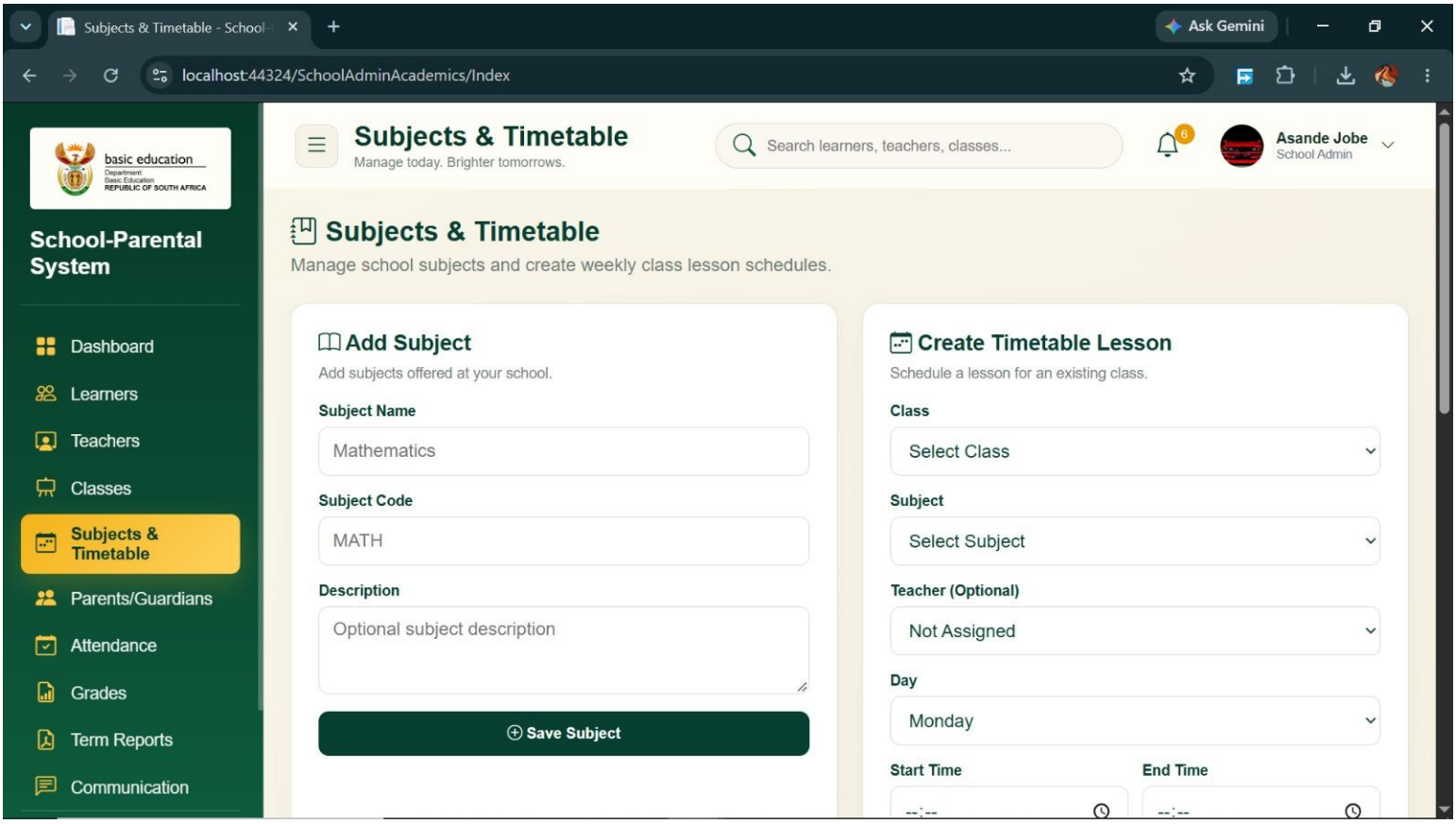


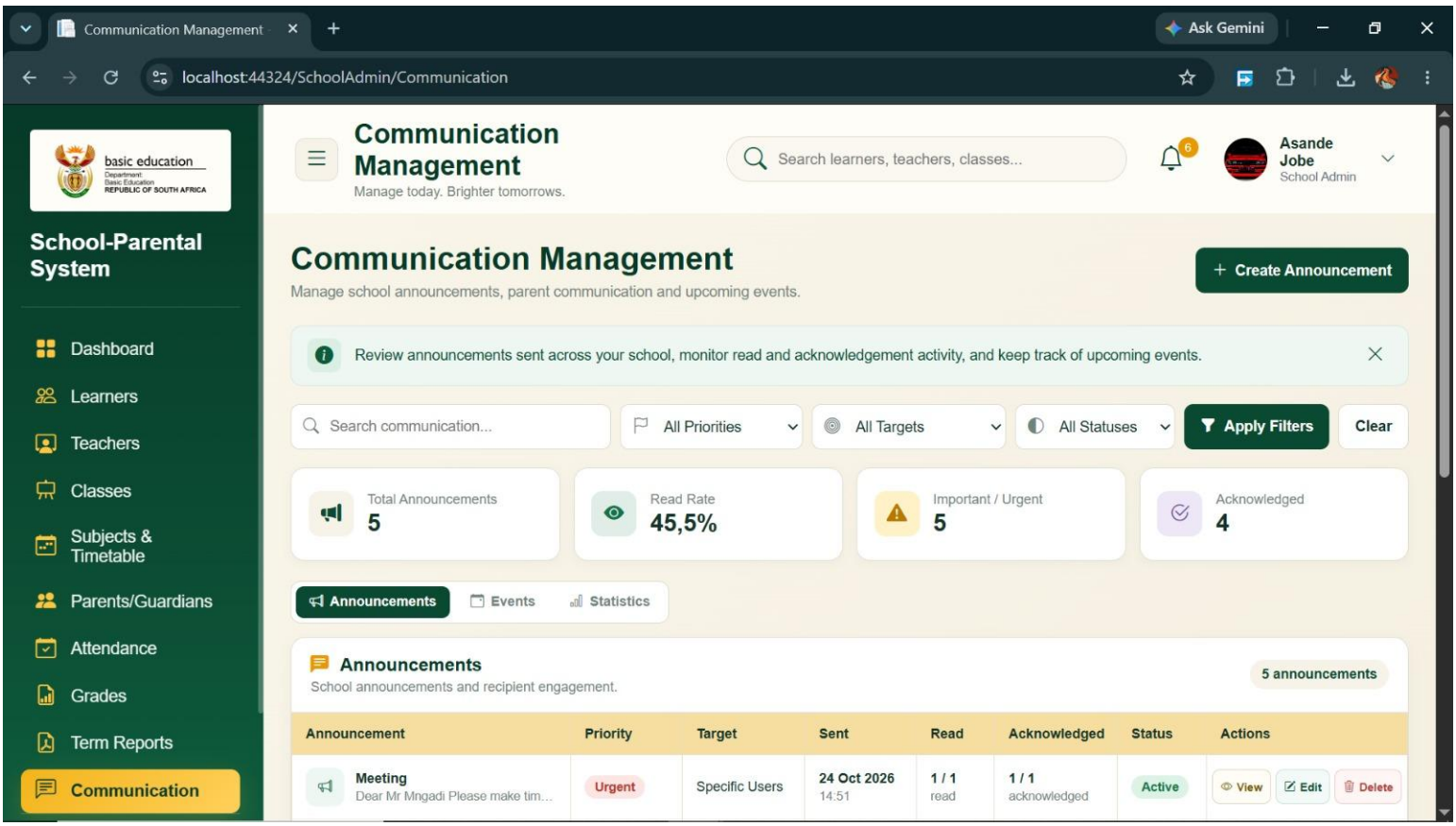

Teacher Management - School
Ask Gemini
localhost:44324/SchoolAdmin/Teachers
basic education
Department: Basic Education
REPUBLIC OF SOUTH AFRICA
School-Parental System
Dashboard
Learners
Teachers
Classes
Subjects & Timetable
Parents/Guardians
Attendance
Grades
Term Reports
Communication
Teacher Management
Manage today. Brighter tomorrows.
Search learners, teachers, classes...
Asande Jobe
School Admin
Teachers
Manage teachers registered at your school.
Add Teacher
Total Teachers
3
Active Teachers
3
Inactive Teachers
0
Search by teacher name or employee ID...
All Specialisations
All Statuses
Employee ID
Teacher
Email
Phone
Specialization
Joining Date
Status
Actions
GF-T-001
N
Neo Kolobe
neokolobe17@gmail.com
+27674271234
Mathematics
12 Jan 2026
Active
GF-T-002
M
Mkgabisi Mfokeng
jobe064074@gmail.com
+27792986848
Life Science
15 Oct 2026
Active
GF-T-003
M
Mondise Maphitha
mondise064074@gmail.com
+27649849999
IsiZulu HL
26 Sep 2026
Active


Class Management - School-Pa
Ask Gemini
localhost:44324/SchoolAdmin/Classes
basic education
Department: Basic Education
REPUBLIC OF SOUTH AFRICA
School-Parental System
Dashboard
Learners
Teachers
Classes
Subjects & Timetable
Parents/Guardians
Attendance
Grades
Term Reports
Communication
Class Management
Manage today. Brighter tomorrows.
Search learners, teachers, classes...
Asande Jobe
School Admin
Classes
Manage classes registered at your school.
Add Class
Total Classes
2
Active Classes
2
Inactive Classes
0
Search class name or code...
All Grades
All Statuses
Class Code
Class
Grade
Section
Class Teacher
Room
Capacity
Year
Status
Actions
GF-10A
Grade 10 A
Grade 10
A
Neo Kolobe
A10
35
2026
Active
GF-11A
Grade 11 A
Grade 11
A
Mkgabisi Mfokeng
Room 11
38
2026
Active


Class Management - School-Pa
Ask Gemini
localhost:44324/SchoolAdmin/Classes
basic education
Department: Basic Education
REPUBLIC OF SOUTH AFRICA
School-Parental System
Dashboard
Learners
Teachers
Classes
Subjects & Timetable
Parents/Guardians
Attendance
Grades
Term Reports
Communication
Class Management
Manage today. Brighter tomorrows.
Search learners, teachers, classes...
Asande Jobe
School Admin
Classes
Manage classes registered at your school.
Add Class
Total Classes
2
Active Classes
2
Inactive Classes
0
Search class name or code...
All Grades
All Statuses
Class Code
Class
Grade
Section
Class Teacher
Room
Capacity
Year
Status
Actions
GF-10A
Grade 10 A
Grade 10
A
Neo Kolobe
A10
35
2026
Active
GF-11A
Grade 11 A
Grade 11
A
Mkgabisi Mfokeng
Room 11
38
2026
Active


Parent Management - School-P
Ask Gemini
localhost:44324/SchoolAdmin/Parents
basic education
Department: Basic Education
REPUBLIC OF SOUTH AFRICA
School-Parental System
Dashboard
Learners
Teachers
Classes
Subjects & Timetable
Parents/Guardians
Attendance
Grades
Term Reports
Communication
Parent Management
Manage today. Brighter tomorrows.
Search learners, teachers, classes...
Asande Jobe
School Admin
Parents
Manage parents and guardians registered at your school.
Add Parent
Total Parents
5
Active Parents
5
Inactive Parents
0
Search by parent name or guardian number...
All Genders
All Statuses
Guardian No.
Parent / Guardian
Email
Phone
Gender
Status
Actions
GF-G-004
C
Cedrick Adams
cedricmotone@gmail.com
+27794567890
Male
Active
G-TEST-001
J
John Davies
parent1@gmail.com
Not provided
Other
Active
GF-G-006
P
Parent Adams
xltvne@gmail.com
asasasasas
Male
Active
GF-G-003
P
Pops Nkosi
luthadap@icloud.com
+27768051795
Female
Active


Subjects & Timetable - School
Ask Gemini
localhost:44324/SchoolAdminAcademics/Index
basic education
Department: Basic Education
REPUBLIC OF SOUTH AFRICA
School-Parental System
Dashboard
Learners
Teachers
Classes
Subjects & Timetable
Parents/Guardians
Attendance
Grades
Term Reports
Communication
Subjects & Timetable
Manage today. Brighter tomorrows.
Search learners, teachers, classes...
Asande Jobe
School Admin
Subjects & Timetable
Manage school subjects and create weekly class lesson schedules.
Add Subject
Add subjects offered at your school.
Subject Name
Mathematics
Subject Code
MATH
Description
Optional subject description
Save Subject
Create Timetable Lesson
Schedule a lesson for an existing class.
Class
Select Class
Subject
Select Subject
Teacher (Optional)
Not Assigned
Day
Monday
Start Time
End Time


Communication Management
Ask Gemini
localhost:44324/SchoolAdmin/Communication
basic education
Department: Basic Education
REPUBLIC OF SOUTH AFRICA
School-Parental System
Dashboard
Learners
Teachers
Classes
Subjects & Timetable
Parents/Guardians
Attendance
Grades
Term Reports
Communication
Communication Management
Manage today. Brighter tomorrows.
Search learners, teachers, classes...
Asande Jobe
School Admin
Communication Management
Manage school announcements, parent communication and upcoming events.
Create Announcement
Review announcements sent across your school, monitor read and acknowledgement activity, and keep track of upcoming events.
Search communication...
All Priorities
All Targets
All Statuses
Apply Filters
Clear
Total Announcements
5
Read Rate
45,5%
Important / Urgent
5
Acknowledged
4
Announcements
Events
Statistics
Announcements
School announcements and recipient engagement.
5 announcements
Announcement
Priority
Target
Sent
Read
Acknowledged
Status
Actions
Meeting
Dear Mr Mngadi Please make tim...
Urgent
Specific Users
24 Oct 2026
14:51
1 / 1
read
1 / 1
acknowledged
Active
View
Edit
Delete

Risk Register for School-Parental System

Prepar Bright Web Crafters **Date** 28 August 2026

Risk ID	Risk Statement	Category	Likelihood (1–3)	Impact (1–3)	Score	Level	Mitigation	Contingency	Risk Owner	Status	
R01	Unauthorized users may access confidential school or learner information.	Security	2	3	6	Medium	Enforce Supabase RLS, RBAC, HTTPS and JWT validation; test access permissions before release.	Revoke affected access or keys, reset credentials, investigate the exposure and correct security policies.	Security/Backend Lead – assign member	Monitoring	
R02	Incorrect school filtering may expose one school's data to another school.	Security / Data	2	3	6	Medium	Enforce school_id filtering through RLS and server-side validation; test each school context.	Temporarily disable affected access, correct RLS/filtering rules and re-audit exposed data.	Database/RLS Lead – assign member	Monitoring	
R03	Incorrect role enforcement may allow users to perform unauthorized functions.	Security / Access	2	3	6	Medium	Test every user role and permission; apply role-based middleware and server-side validation.	Correct the role or policy configuration, rollback affected permissions and re-test access.	Authentication Lead – assign member	Monitoring	
R04	Flutter, ASP.NET and Supabase integration may take longer than planned and delay development.	Technical / Schedule	3	3	9	High	Build and test a small end-to-end integration early; divide integration tasks and review progress weekly.	Simplify affected functionality, reassign development work and use the last stable integration approach.	Integration Lead – assign member	Open	
R05	Internet or network problems may interrupt access to the system or delay data synchronization.	Technical / Resource	2	2	4	Medium	Provide connection and sync-status messages, reduce unnecessary network requests and test unstable connections.	Retry synchronization when the connection returns and use temporary manual communication/processes where necessary.	Backend/Mobile Lead – assign member	Monitoring	
R06	Browser or device differences may cause screens or functions to behave incorrectly.	Technical / Quality	2	2	4	Medium	Test the web application on common browsers and the mobile application on available devices and screen sizes.	Correct affected screens/functions, release a patch or provide a temporary workaround.	Testing Lead – assign member	Open	
R07	Parent or guardian contact information may be incorrect or outdated, reducing communication reliability.	Operational / Data	3	2	6	Medium	Allow authorized school staff to maintain contact information and regularly verify parent/guardian details.	Verify the information with the learner or guardian and use an alternative communication method until updated.	Data Management Lead – assign member	Monitoring	
R08	Parents may not read or acknowledge important communication on time.	Operational / Communication	3	2	6	Medium	Use notifications, Read/Unread Tracking and Parent Acknowledgement for Important or Urgent announcements.	Identify non-responsive parents and follow up using telephone calls, printed notices or another approved method.	Communication Module Lead – assign member	Monitoring	
R09	Some parents may have limited access to smartphones, mobile data or reliable internet, reducing mobile-app adoption.	Operational / Resource	2	3	6	Medium	Keep the mobile app simple and data-conscious; avoid unnecessary data usage and support commonly available devices.	Continue using alternative communication methods such as telephone calls or printed letters for affected parents.	Mobile App Lead – assign member	Monitoring	
R10	The large number of proposed functions may delay prototype completion and leave insufficient time for testing.	Scope / Schedule	3	3	9	High	Prioritise core requirements, monitor the Microsoft Project schedule weekly and develop modules incrementally.	Simplify or postpone lower-priority functions so that core integration and testing can be completed.	Project Manager – assign member	Open	

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Priority	Risk ID	Risk	Score	Why Important?	Immediate Action
1	R04	Flutter, ASP.NET and Supabase integration may take longer than planned and delay development.	9	Failure or delay in integration affects both applications and the shared backend.	Build and test a small end-to-end integration immediately and monitor progress weekly.
2	R10	The large number of proposed functions may delay prototype completion and leave insufficient time for testing.	9	The project has a large scope and schedule pressure can reduce quality or leave modules incomplete.	Prioritise core functions and simplify/postpone lower-priority features if milestones slip.
3	R02	Incorrect school filtering may expose one school's data to another school.	6	Although tied with other medium risks, it has serious privacy and multi-school security consequences.	Test school_id filtering and Supabase RLS policies for every role and school context.

Central University of Technology, Free State

283 Koppie Alleen rd
Reitzpark,
Welkom,
9488

BRIGHT
WEB CRAFTERS

We are developing a centralized, web and mobile-based School Portal System to streamline school management. This system will provide a single platform for administrators, teachers, and parents to manage and access student information, including attendance, grades, and announcements, in real-time. By replacing manual, paper-based processes, our goal is to improve data accuracy, enhance parent-school communication, and support better academic decision-making for a more efficient educational environment.

Enter Your Initials and Surname:
J.B Setlhare

Enter School Name
Reitzpark Primary

What is your role at the school, and how long have you been in this position?
Administrator since 2021 to date

How would you describe your current comfort level with technology? (e.g., using computers, smartphones, apps)
Making life more easier

On a scale of 1-5, how satisfied are you with your current methods? Why?
5

What are the most common problems that you come across when you communicate with parents?
Not making app

How does the school currently inform parents or guardians about important matters such as learner attendance, academic performance, school announcements, meetings, and events? What problems, if any, do you experience with these methods?

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Parent meeting

What challenges do teachers or administrative staff experience when recording, updating, finding, or sharing learner information such as attendance, marks, classes, and reports?

No problem

When the school sends an important notice to a parent or guardian, how do you know whether they have actually received and read it? Is this currently a problem for the school?

I do not know if some parent have recieved the circular or whatsup ~~me~~

If you could improve or change anything about the way your school manages learner information and communicates with parents, teachers, and learners, what would you change and why?

parent to check their school bags to check if there is no circulars

What are the main tools or systems you currently use to manage student information? (e.g., paper registers, Excel, email, other software)

SAMS

Signature

School stamp

REITZPARK PRIMARY SCHOOL
BOX 220
WELKOM
9460
DEPT. OF EDUCATION

Central University of Technology, Free State

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Reitzpark,
Welkom,
9488

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Enter Your Initials and Surname:
Z. Horn

Enter School Name
Reitzpark Primary

What is your role at the school, and how long have you been in this position?
Educator 10 years

How would you describe your current comfort level with technology? (e.g., using computers, smartphones, apps)
Average

On a scale of 1-5, how satisfied are you with your current methods? Why?
1 -> Parents are not on whatsapp or Facebook. No data, no smartphones

What are the most common problems that you come across when you communicate with parents?
Not responding. Wrong numbers

How does the school currently inform parents or guardians about important matters such as learner attendance, academic performance, school announcements, meetings, and events? What problems, if any, do you experience with these methods?
Send notices on class group whatsapp and post on facebook.

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Welkom,
9488

BRIGHT
WEB CRAFTERS

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.....

What challenges do teachers or administrative staff experience when recording, updating, finding, or sharing learner information such as attendance, marks, classes, and reports?

No contact numbers, or wrong contact numbers. Learners coming late. Time !!

When the school sends an important notice to a parent or guardian, how do you know whether they have actually received and read it? Is this currently a problem for the school?

We don't know. Sometimes we will say respond with a thumbs up emoji. Not respecting private time of educators

If you could improve or change anything about the way your school manages learner information and communicates with parents, teachers, and learners, what would you change and why?

To update their contact information.

What are the main tools or systems you currently use to manage student information? (e.g., paper registers, Excel, email, other software)

Paper registers SA-SAMS Trusting parents that they will inform the educator.

Signature

School stamp

REITZPARK PRIMARY SCHOOL
P.O. BOX 220
WELKOM
9460
DEPT. OF EDUCATION

Central University of Technology, Free State

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Reitzpark,
Welkom,
9488

BRIGHT
WEB CRAFTERS

We are developing a centralized, web and mobile-based School Portal System to streamline school management. This system will provide a single platform for administrators, teachers, and parents to manage and access student information, including attendance, grades, and announcements, in real-time. By replacing manual, paper-based processes, our goal is to improve data accuracy, enhance parent-school communication, and support better academic decision-making for a more efficient educational environment.

Enter Your Initials and Surname:
R. I. Mocke

Enter School Name
Reitzpark Primary School

What is your role at the school, and how long have you been in this position?
Remedial Educator, 14 years

How would you describe your current comfort level with technology? (e.g., using computers, smartphones, apps)
Very Good

On a scale of 1-5, how satisfied are you with your current methods? Why?
3. Can improve, Maybe a more synchronized Method to reduce paper work.

What are the most common problems that you come across when you communicate with parents?
Not Responding, not attending meetings that are arranged.

How does the school currently inform parents or guardians about important matters such as learner attendance, academic performance, school announcements, meetings, and events? What problems, if any, do you experience with these methods?
Parent Meetings, Telephonically or send a letter with learners. - Letter does not get handed to parent, parents don't answer phone when they see it's the school.

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Reitzpark,
Welkom,
9488

BRIGHT WEB CRAFTERS

.....

.....

What challenges do teachers or administrative staff experience when recording, updating, finding, or sharing learner information such as attendance, marks, classes, and reports?

Time constraints, parent / learner information are not up to date.

.....

When the school sends an important notice to a parent or guardian, how do you know whether they have actually received and read it? Is this currently a problem for the school?

Yes its a problem, we just hope they do. - difficult to say, , with whatsapp more effective, but it causes problems because parents dont respect personal time.

If you could improve or change anything about the way your school manages learner information and communicates with parents, teachers, and learners, what would you change and why?

I would like to have access from my phone/laptop.

What are the main tools or systems you currently use to manage student information? (e.g., paper registers, Excel, email, other software)

Learner profiles, Register, SASAMS.

.....

.....

Signature

School stamp

REITZPARK PRIMARY SCHOOL
P.O. BOX 220
WELKOM
9460
DEPT. OF EDUCATION

Central University of Technology, Free State

283 Koppie Alleen rd
Reitzpark,
Welkom,
9488

BRIGHT
WEB CRAFTERS

We are developing a centralized, web and mobile-based School Portal System to streamline school management. This system will provide a single platform for administrators, teachers, and parents to manage and access student information, including attendance, grades, and announcements, in real-time. By replacing manual, paper-based processes, our goal is to improve data accuracy, enhance parent-school communication, and support better academic decision-making for a more efficient educational environment.

Enter Your Initials and Surname:
M. MOTJIE Z.

Enter School Name
LEBOHANG TECHNICAL SCHOOL

What is your role at the school, and how long have you been in this position?
ADMIN CLERK - 26 YEARS

How would you describe your current comfort level with technology? (e.g., using computers, smartphones, apps)
I EXPERIENCE NO PROBLEMS AS I AM USING IT DAILY AND WAS ONE OF THE SUBJECT

On a scale of 1-5, how satisfied are you with your current methods? Why?
5. WE LIVE IN A WORLD OF TECHNOLOGY, LIKE IT SO EASY WITH TECHNOLOGY

What are the most common problems that you come across when you communicate with parents?
HARD TO REACH PARENTS THEY DON'T UPDATE THEIR CELLPHONE NUMBER ON THE SCHOOL DATABASE.

How does the school currently inform parents or guardians about important matters such as learner attendance, academic performance, school announcements, meetings, and events? What problems, if any, do you experience with these methods?
ISSUING OF LETTERS TO THE PARENTS, GIVE THEM TO LEARNERS TO GIVE THEIR PARENTS. CLASS TEACHERS SHARE INFORMATION USING THEIR WHATSAPP PLATFORMS TO PARENTS.

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Reitzpark,
Welkom,
9488

BRIGHT
WEB CRAFTERS

What challenges do teachers or administrative staff experience when recording, updating, finding, or sharing learner information such as attendance, marks, classes, and reports?

THE ATTENDANCE BY PARENTS IS VERY POOR, WE EVEN INVITED PARENTS FOR ONE ON ONE SESSION WITH THEIR SUBJECT TEACHER, PARENTS THEY DON'T COME,

When the school sends an important notice to a parent or guardian, how do you know whether they have actually received and read it? Is this currently a problem for the school?

YES, ITS A CHALLENGE, LEARNERS THEY DON'T SHARE INFORMATION FROM SCHOOL WITH THEIR PARENTS, ESPECIALLY THOSE PARENTS WHO ARE NOT HAVING INFORMATION REPORTED ON SASAMS

If you could improve or change anything about the way your school manages learner information and communicates with parents, teachers, and learners, what would you change and why?

EMPHASISE THE ISSUE OF UPDATING PERSONAL INFORMATION FROM PARENTS, ESPECIALLY THE CELL NUMBERS SO THAT THEY CAN BE REACHED AS CAN AS THEY

What are the main tools or systems you currently use to manage student information? (e.g., paper registers, Excel, email, other software)

SASAMS

Signature .....

LENAKENG TECHNICAL SCHOOL
PRIVATE BAG X61
WELKOM, 9460
26 -08- 2026
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DEPT. OF EDUCATION

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9488

BRIGHT
WEB CRAFTERS

We are developing a centralized, web and mobile-based School Portal System to streamline school management. This system will provide a single platform for administrators, teachers, and parents to manage and access student information, including attendance, grades, and announcements, in real-time. By replacing manual, paper-based processes, our goal is to improve data accuracy, enhance parent-school communication, and support better academic decision-making for a more efficient educational environment.

Enter Your Initials and Surname:

P.V Ndweni

Enter School Name

Lenakeng Technical School

What is your role at the school, and how long have you been in this position?

Educator for 7 years

How would you describe your current comfort level with technology? (e.g., using computers, smartphones, apps)

Very comfortable with usage of all named above.

On a scale of 1-5, how satisfied are you with your current methods? Why?

3 because in most cases whereby devices are to be used I need WiFi and it's far.

What are the most common problems that you come across when you communicate with parents?

Parents do not show/put effort in terms of disciplining their kids, it's teachers walk alone.

How does the school currently inform parents or guardians about important matters such as learner attendance, academic performance, school announcements, meetings, and events? What problems, if any, do you experience with these methods?

Letters as well as whatsap message.

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What challenges do teachers or administrative staff experience when recording, updating, finding, or sharing learner information such as attendance, marks, classes, and reports?

The most challenge is having outdated information of learners/parents. None attendance of meetings by parents.

When the school sends an important notice to a parent or guardian, how do you know whether they have actually received and read it? Is this currently a problem for the school?

We have whatsap groups with parents per class and every information that should reach parents is also shared there.

If you could improve or change anything about the way your school manages learner information and communicates with parents, teachers, and learners, what would you change and why?

I would reccomend that parents should update learner/parent information every term.

What are the main tools or systems you currently use to manage student information? (e.g., paper registers, Excel, email, other software)

Educators have hard copy registers and the information is captured in the system by admin clerk.

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Enter Your Initials and Surname:

V.V. SMAIL

Enter School Name

Lonakeng Technical School

What is your role at the school, and how long have you been in this position?

Teacher, one year

How would you describe your current comfort level with technology? (e.g., using computers, smartphones, apps)

I am very comfortable using technology for teaching

On a scale of 1-5, how satisfied are you with your current methods? Why?

3, because my current methods allow me to perform my duties, but they are not always efficient

What are the most common problems that you come across when you communicate with parents?

Parents not receiving messages, changing or incorrect contacts details and parents not responding to messages

How does the school currently inform parents or guardians about important matters such as learner attendance, academic performance, school announcements, meetings, and events? What problems, if any, do you experience with these methods?

The school mainly uses letters, phone calls and WhatsApp channel

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What challenges do teachers or administrative staff experience when recording, updating, finding, or sharing learner information such as attendance, marks, classes, and reports?

The main challenge are manual capturing of data, duplication of information and difficulty of locating manually captured data.

When the school sends an important notice to a parent or guardian, how do you know whether they have actually received and read it? Is this currently a problem for the school?

Through WhatsApp, it is possible to know when the parents have read the messages as it has delivery and read indicators, but it is not always working for parents with limited connection. Yes

If you could improve or change anything about the way your school manages learner information and communicates with parents, teachers, and learners, what would you change and why?

I would introduce a free centralised web and mobile-based school portal where teachers and parents can access information

What are the main tools or systems you currently use to manage student information? (e.g., paper registers, Excel, email, other software)

Paper registers, files, Microsoft word and SASAMS

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Enter Your Initials and Surname:

SP Matoware

Enter School Name

Herateng Technical School

What is your role at the school, and how long have you been in this position?

An educator. I have been here for two years.

How would you describe your current comfort level with technology? (e.g., using computers, smartphones, apps)

I experience no problems at all as I did computer as one of my modules at CUT.

On a scale of 1-5, how satisfied are you with your current methods? Why?

5. Technology makes our lives easier. It is easy to type, edit, print tasks and also record marks.

What are the most common problems that you come across when you communicate with parents?

Parents are difficult to reach as they do not update their cellphone numbers, however when we do find them, they hardly know how to best handle their children

How does the school currently inform parents or guardians about important matters such as learner attendance, academic performance, school announcements, meetings, and events? What problems, if any, do you experience with these methods?

We issue out letters and give them to learners to pass on to their parents. We also have parents whatsapp groups, we send information directly to them.

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What challenges do teachers or administrative staff experience when recording, updating, finding, or sharing learner information such as attendance, marks, classes, and reports?

Parents hardly fetch their children's report cards and they never pick up their phones when we want to make a follow-up.

When the school sends an important notice to a parent or guardian, how do you know whether they have actually received and read it? Is this currently a problem for the school?

Parents usually (for those who keep contact) with teachers and report if they must be able to make it.

If you could improve or change anything about the way your school manages learner information and communicates with parents, teachers, and learners, what would you change and why?

I would make it a must for all parents to attend meetings, submit their cellphone numbers that actually work.

What are the main tools or systems you currently use to manage student information? (e.g., paper registers, Excel, email, other software)

paper registers are used to monitor learner attendance, this is how we are able to monitor their information.

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Enter Your Initials and Surname:

M.D. Goliath

Enter School Name

Thabong Primary School

What is your role at the school, and how long have you been in this position?

Administrator, 20 years.

How would you describe your current comfort level with technology? (e.g., using computers, smartphones, apps)

Computers

On a scale of 1-5, how satisfied are you with your current methods? Why?

5, because technology has made our work much easier

What are the most common problems that you come across when you communicate with parents?

Complaints and miscommunication.

How does the school currently inform parents or guardians about important matters such as learner attendance, academic performance, school announcements, meetings, and events? What problems, if any, do you experience with these methods?

Using letters and messages. Problems

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BRIGHT WEB CRAFTERS

is children not giving their parents letters

What challenges do teachers or administrative staff experience when recording, updating, finding, or sharing learner information such as attendance, marks, classes, and reports?

Sometimes mark lists are not complete and we don't receive atterdance registers on time

When the school sends an important notice to a parent or guardian, how do you know whether they have actually received and read it? Is this currently a problem for the school?

We issue them tear slips which they bring back to school as confirmation. No

If you could improve or change anything about the way your school manages learner information and communicates with parents, teachers, and learners, what would you change and why?

Use more effective communication methods like a group chat

What are the main tools or systems you currently use to manage student information? (e.g., paper registers, Excel, email, other software)

South African School Administration And Management System (SASAMS)

Signature

THABONG PRIMARY SCHOOL
Dept. of Education
Tel/Fax: 057 396 3146
School stamp
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PRINCIPAL
P.O. BOX 401 THABONG 9463

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Enter Your Initials and Surname:

T KHOARAI

Enter School Name

THABONG PRIMARY SCHOOL

What is your role at the school, and how long have you been in this position?

PL 1, TEACHER, 4 YEARS

How would you describe your current comfort level with technology? (e.g., using computers, smartphones, apps)

ADVANCED

On a scale of 1-5, how satisfied are you with your current methods? Why?

3, I AM NOT FULLY USING TECHNOLOGY IN MY CLASS TO TEACH.

What are the most common problems that you come across when you communicate with parents?

UNAVAILABLE TO COME TO MEETINGS AND NOT BEING ABLE TO READ THE TEXT ON TIME.

How does the school currently inform parents or guardians about important matters such as learner attendance, academic performance, school announcements, meetings, and events? What problems, if any, do you experience with these methods?

LETTERS AND MEETINGS

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What challenges do teachers or administrative staff experience when recording, updating, finding, or sharing learner information such as attendance, marks, classes, and reports?

PARENT CHANGE NUMBERS AND DO NOT INFORM THE SCHOOL OR THE TEACHER

When the school sends an important notice to a parent or guardian, how do you know whether they have actually received and read it? Is this currently a problem for the school?

WORD OF MOUTH FROM THE LEARNER, IT IS A PROBLEM BECAUSE SOME OF THE PARENTS DO NOT COME TO THE MEETINGS

If you could improve or change anything about the way your school manages learner information and communicates with parents, teachers, and learners, what would you change and why?

HAVE A FORM YEARLY TO UPDATE THE DETAILS OF THE PARENTS

What are the main tools or systems you currently use to manage student information? (e.g., paper registers, Excel, email, other software)

PAPER REGISTERS

Signature

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Dept. of Education
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No	Design Artefact	Tool Used	Original Source Link / Editable File
1	System Architecture Diagram	Draw.io	https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=_blank&layers=1&nav=1&dark=auto#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1ClYwvn3JjYWEF-mdmsHK64CJf6675N-G%26export%3Ddownload
2	Use Case Diagram	Draw.io	
3	Sequence Diagram 1	Draw.io	
4	Entity Relationship Diagram (ERD)	Data Modeler	
5	Flutter Mobile Wireframes	FIGMA	https://www.figma.com/proto/pG7mL3Oj2Rr37JAT9r7WJv/School-Parental-system?node-id=42-238&p=f&t=0WVSbhhXYOH6dG98-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=42%3A238
6	ASP.NET Web Interface Wireframes	FIGMA	https://www.figma.com/proto/lPObV3qt8TfARP7lGBJZqU/School-Parental-System?node-id=224-463&p=f&viewport=-107%2C84%2C0.19&t=7iRvLo52rvJGZx2o-1&scaling=contain&content-scaling=fixed&page-id=218%3A2&starting-point-node-id=281%3A2266

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BRIGHT WEB CRAFTERS EST. 2024

Questionnaire for our School Parental System that we developed to bridge the gap between schools , teacher and parents this questionnaire was made with a sole purpose to get feedback from the school to see your overall satisfaction of our system

Enter your initials and surname

W.M Landman

Is the system easy to learn and use without extensive training?

Yes very easy to use

Are the navigation and layout intuitive for your role?

Yes

Did you experience any errors while using the system?

No errors expreienced while using.

Is the system faster and more convenient than your previous method?

Yes - direct communication with staff/parents

How effective is the system in improving communication between parents and the school?

Very effective, safes time - Save paper and I do not have to contact parents on their cellphones

Are you happy with how the final product looks and works compared to what you imagined?

Very Impressed with what students are doing.

Out of 10 can you rate the system

10/10

Signature

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EST 2024
BRIGHT
WEB CRAFTERS

Questionnaire for our School Parental System that we developed to bridge the gap between schools , teacher and parents this questionnaire was made with a sole purpose to get feedback from the school to see your overall satisfaction of our system

Enter your initials and surname

R. Marks

Is the system easy to learn and use without extensive training?

Yes, very user friendly.

Are the navigation and layout intuitive for your role?

Yes, it will save a lot of time.

Did you experience any errors while using the system?

No, not at all.

Is the system faster and more convenient than your previous method?

Yes, definitly more effecient.

How effective is the system in improving communication between parents and the school?

Very, it saves time, and the proof of communication is on the system.

Are you happy with how the final product looks and works compared to what you imagined?

Yes, very impressed.

Out of 10 can you rate the system

10

Signature

REITZPARK PRIMARY SCHOOL
2026 -10- 23
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